Who Wins With Up-Zoning? Quasi-Experimental Evidence from California's Reforms for Accessory Dwelling Units*

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October 15, 2026

Abstract

Can up-zoning and permitting reforms for small units increase housing supply, and who captures the benefits? We study California's 2017 deregulation of accessory dwelling units (ADUs) through state preemption of local permitting. We exploit a feature of California's regulatory environment to isolate quasi-random variation in the geography of permitting reform: the California Coastal Zone. Leveraging a regulatory discontinuity at the Coastal Zone boundary, we find that permitting reforms increased ADU permitting activity by about 30%, although these effects are small in magnitude. However, the reform did increase land values by 7%. With a simple model, we show that the change in land values reflects the decreased permitting costs associated with the reform package. We also find that permitting reforms are likely not creating large price spillovers, and the increases in land values are associated with a 1.3% increase in property tax collections.

JEL codes: H71, R31, R38, R52

Keywords: Land Use, Permitting, Accessory Dwelling Units, Property Taxes

*Thank you to Judd Boomhower, Jeffrey Clemens, Julie Cullen, Gordon Dahl, and Seth Hill for their advice and guidance on this project. We would also like to thank Vincent Armentano, Prashant Bharadwaj, Jonathan Brownrigg, Leah Brooks, Richard Carson, Darwyyn Deyo, Itzik Fadlon, Arthur Gailes, Ed Glaeser, Steve Glazer, Roger Gordon, Idalyia Grigoryeva, Alec Hoover, John Juneau, Churn Ken Lee, Katherine Meckel, Ilia Murtazashvili, Paul Niehaus, Tobias Peter, Alicia Plemmons, Chris Severen, Danny Shoag, Alec Stapp, David Torregrosa, Stan Veuger, Heidi Williams, and Letian Yin as well as participants in the 2022-2023 Econ 250 Labor Sequence, the UCSD Public and Labor Workshop, the IHS Housing Papers Workshops, and the IHS Junior Fellows Program for their helpful comments. We would like to thank Arnold Ventures for funding the data purchases to support this research. Hoxie would also like to thank the Alfred P. Sloan Foundation, the Institute for Humane Studies, and the National Science Foundation for their fellowship support. The views expressed here in no way represent the views of our funders. All errors are our own.

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1 Introduction

Housing costs have been increasing as a share of household expenditures in cities for several decades, which has eroded the ability of workers to benefit from high productivity places (Ganong and Shoag, 2017; Legislative Analyst's Office, 2015; Paulin, 2018; Hoxie, Shoag and Veuger, 2023). This may have substantial macroeconomic consequences through decreased mobility to productive labor markets (Eckert, Ganapati and Walsh, 2020; Duranton and Puga, 2023). Economists have linked inelastic housing supply to higher housing costs, which suggests that regulatory reform may be a potential solution to housing affordability concerns (Glaeser, Gyourko and Saks, 2006; Saiz, 2010; Baum-Snow and Han, 2024; Bartik, Gupta and Milo, 2025). In most cases, regulations that govern the construction of new housing units are under local control, and local policymakers may maximize the welfare of current homeowners, perhaps at the expense of local renters, and without considering the cross-jurisdictional spillovers of their housing supply choices (Gordon, 1983; Fischel, 2001; Gyourko, Saiz and Summers, 2008). These considerations have led to increasing interest among state policymakers in circumscribing the powers of local government through state-mandated deregulation through up-zoning and permitting reform. Such reforms raise two questions that we address in this paper: how much additional housing supply can up-zoning and permitting reform deliver, and what is the incidence of the state-mandated reforms for stakeholders?

This paper quantifies the effects of the state of California's 2017 by-right preemption of local zoning and permitting authority for accessory dwelling units (ADUs). After 2017, most residential lots were up-zoned to allow one ADU, a small one-unit addition under 1,200 sq. ft., and if the building plans met state-set specifications, then a permit or denial report had to be issued within 120 days (Gray, 2024).[1] We exploit a unique regulatory discontinuity to identify the causal impacts of permitting reforms. Within the California Coastal Zone, the state-wide preemption of local regulatory authority was superseded by the California Coastal Act, passed in 1976. Permitting jurisdictions within this approximately one-kilometer coastal strip of land could not be forced to adopt permitting reforms for ADUs (California Coastal Commission, 2022*a*,*b*).[2] Therefore, we use a difference in discontinuities design to estimate the causal effects of these reforms (Grembi, Nannicini and Troiano, 2016; Butts, 2023; Cañedo Riedel and Hoxie, 2025).[3]

We measure the impacts of permitting reform on ADU permitting activity, land values, spillovers,

[1] Annual state-wide permitting of ADUs has increased almost 40-fold since 2016, as shown in Figure 1.

[2] A recent court case, Riddick v. City of Malibu (2025), suggests that the Coastal Commission was not allowing owners in the Coastal Zone to take advantage of the new ADU laws, but it also suggests that there may be heterogeneity across jurisdictions within the Coastal Zone about how stringent their regulations are towards ADUs. However, Monkkonen, Manville and Lens (2024) and Legislative Analyst's Office (2015) suggest that all coastal jurisdictions may be relatively stringent when it comes to housing policy, even compared to the rest of California.

[3] Before accessing any of the data from Cotality (2025), we pre-registered three of our main research designs in August 2025 (Cañedo Riedel and Hoxie, 2025). The designs presented in this paper are partially pre-registered as many features of the data were not known to us in advance. We deploy an adversarial compliance adjudication protocol using Opus 5 from Anthropic (2026). Our protocol builds on insights for auditing adherence to pre-analysis plans in non-experimental settings from Clemens, Mahajan and Sabia (2026) and Clemens and Mahajan (2026). We will identify important deviations from our pre-analysis plan in the methods section of this paper. We provide the artificial intelligence-written change log and compliance audit in Appendix B.

transactions, and property taxes using parcel-level property and transaction data from Cotality (2025) and permit data from California Department of Housing and Community Development (2025). We find that California's by-right permitting for ADUs increased permitting activity by about 30%, representing a small absolute increase in a setting where baseline permitting rates are low. Despite low take-up, we find that permitting reform increased residential land values by 7%. We show that this price increase reflects the *ability* to build an ADU, which is capitalized in the land and accrues to all parcels whether or not the owner decides to build an ADU.[4]

The Coastal Zone boundary difference in discontinuities design relies on a local parallel trends assumption: in the absence of permitting reforms in 2017, parcels just inside and just outside the Coastal Zone boundary would have remained on the same trends as before 2017. We find small level differences in parcel characteristics between parcels inside and outside the Coastal Zone in our pre-period, reflecting the Coastal Zone's regulatory effects (Severen and Plantinga, 2018; Kahn, Vaughn and Zasloff, 2010). However, our difference in discontinuities design can account for initial gaps from the pre-2017 discontinuity across the Coastal Zone by incorporating them directly into the estimator. We show all estimates on a mean squared error optimal bandwidth around the Coastal Zone boundary of 450 meters (Calonico, Cattaneo and Farrell, 2020). We find that local to the Coastal Zone boundary, parcels on either side were following similar trends in total sale prices per acre, land values per acre, and transaction volumes before the 2017 reforms.

Our main results for permitting activity and land values are robust in several ways. First, we present a range of specifications, including alternative running variable polynomials, additional controls, the inclusion of Coastal Zone segment by year fixed effects, and specifications using alternative bandwidths. While we present our pre-registered designs, the richness of the Cotality (2025) data within our analysis bandwidth allows us to extend our analysis to include a repeat sale model, which is common in the literature (Bailey, Muth and Nourse, 1963; Soltas and Gruber, 2026). We view this as an improvement over our pre-specified models. The repeat sales model allows us to include a parcel fixed effect to capture static, unobserved characteristics of each parcel. Our preferred specification is estimated on our repeat sale sample and includes both a parcel fixed effect and segment by year fixed effects.

To investigate spillovers from permitting reforms across the Coastal Zone boundary, we use a reference bin approach that compares changes in outcomes across 50-meter bins that span both sides of our baseline analysis sample to changes in a reference bin inside the Coastal Zone. We focus on two channels for spillovers. First, additional units may decrease nearby parcel values either by creating disamenities or by lowering the market value of similar parcels through supply channels. Second, nearby owners exposed to increased disamenities may elect to sell their parcels and move (Cullen and Levitt, 1999; Card, Mas and Rothstein, 2008). Comparing bins just inside the Coastal Zone to changes in our reference bin, we find no detectable evidence of either total price spillovers or transaction frequency spillovers.

[4]This option value may have been particularly valuable as demand for floor-space increased during the COVID-19 pandemic (Mondragon and Wieland, 2022).

To rationalize low permitting take-up and increased land values, we build a simple open city model with two permitting jurisdictions, property taxes, and endogenous construction of ADUs and land transactions.[5] This framework allows us to characterize our difference in discontinuities estimates using jumps in permitting activity and land values across the permitting jurisdictions within our model city. This model shows that our land value estimates are measuring differences in permitting costs in cases when there is no treatment heterogeneity by lot quality. Additionally, our model suggests that property tax shelter, the extent to which a lot is taxed on a below market basis, may depress transactions and, in turn, dampen ADU construction. These theoretical results motivate the following heterogeneity exercises.

In order to better understand how permitting reform increased land values, we explore treatment effect heterogeneity by lot quality for our permitting and land value results. We explore two versions of lot quality: excess square feet on a parcel available for an ADU and distance to the nearest central business district (CBD). Reflecting that lots with less excess space not covered by a structure may also be closer to CBDs, we find stronger heterogeneity for lot quality as proxied by distance to the CBD.[6] Our estimates of heterogeneity by lot quality suggest that the effects of permitting reform on permitting activity are larger when parcels are closer to CBDs, with a statistically significant interaction term. When we turn our attention to land values, we find that our preferred repeat sale estimates do not have a statistically significant interaction with CBD distance. This suggests that our land value results reflect the changes in permitting costs after ADU reform, rather than the full market option value that embeds the value of the ADU itself net of construction costs. This implies that permitting reforms for ADUs lowered permitting costs by 7% of land values.

Even though cities lose autonomy under state preemption, our finding that land values increased as a response suggests that cities may have seen a revenue gain. As properties transact and ADUs are built, the value of the ADU reforms enters the property tax base through two channels. First, if a lot transacts, then the reassessment after sale should reflect the value of permitting reform that was capitalized into the sale price. Second, if an ADU is built on a lot, that ADU's assessed value is added to the existing property tax base, without triggering a full reassessment (San Francisco Assessor Recorder's Office, 2025). We find that property tax collections increased by about 1.3% in our preferred specification. Although these results suggest that ADU reforms increased local revenues, whether or not these reforms resulted in a net fiscal gain depends on the composition of new residents and their preferences for, and use of, local services (Poterba, 1997; Hilber, 2017).

In our last exercise, we investigate to what extent newly permitted ADUs impacted local rents. We are limited to rental data from the American Community Survey, and our results are generally too imprecise to draw strong conclusions.[7] However, given the relatively small numbers of ADUs

[5] This model is similar to Turner, Haughwout and van der Klaauw (2014).

[6] Marantz, Elmendorf and Kim (2023) suggests that access to jobs is a key driver of ADU permitting.

[7] We use the extent to which a block group overlaps with the Coastal Zone as an instrument for changes in ADU permits per housing unit in 2016, and we measure percent changes in average gross rents between 2016 and 2024 (U.S. Census Bureau, 2016, 2024; IPUMS NHGIS, 2026). Our results are statistically insignificant, but our standard errors cannot rule out meaningful effects. Our first stage F-statistic is 17, and we include a number of controls for terrain and density, the area age structure of owners, nearby changes in job access, other permitting activity, and

permitted state-wide, it is unlikely that ADU reforms lowered rents in a meaningful way.

This paper primarily contributes to the literature on regulation, home prices, and housing supply. Several papers link land use restrictions to decreased housing supply and higher home prices (Glaeser, Gyourko and Saks, 2006; Bartik, Gupta and Milo, 2025; Turner, Haughwout and van der Klaauw, 2014). Moreover, Baum-Snow and Han (2024) find that about 40% of the supply response to price changes comes from small renovations, which could include ADUs. However, how effective decreasing local regulations are at lowering housing costs is currently being debated in the literature (Louie, Mondragon and Wieland, 2025; Duranton and Puga, 2023). Our findings help rationalize this divide. Although we find that permitting does increase as a result of reducing local discretion, we don't see a large shift in units as a share of all parcels. Therefore, it is unlikely that ADUs have meaningfully helped to depress rental prices in the broader market for small rentals, although they may reduce prices locally. In contrast, the increase in land values, created by decreased permitting costs for ADUs, does impact a meaningful number of parcels. It is likely that aggregate measures of home prices commonly used in macroeconomic studies capture the increased land values that can't be untangled from structure values.[8] Our results suggest that permitting and zoning may not be the only barriers to increasing housing supply, and other factors like incumbent preferences, construction costs, use restrictions, tax incentives, and other elements of the construction process may warrant additional focus in future research.[9]

Second, we contribute to an active literature on up-zoning and permitting costs. A survey from Freemark (2023) finds that up-zoning is often associated with land value increases, but per-unit cost reductions only materialize if units get built. D'Amico, Soltas and Wang (2026) find that permitting and construction timelines can vary widely across localities. The two closest papers to ours also study how land values change with permitting costs. Gold and Tarascina (2026) study ADU up-zoning and permitting reforms in Seattle, WA. Using a difference in differences approach, they find an increase in permitting activity and suggestive evidence of an option value of 1.6% of the sale price associated with lots exposed to reforms. Our estimates suggest a higher land value change from permitting reforms, of about 7% of land value (about 3% of the total sale price). The larger option value in our setting may reflect differences between Seattle, WA and California in the value of building an ADU, the initial and post-reform regulatory environments, and construction costs. Next, Soltas and Gruber (2026) find that developers will pay over 50% more for already permitted land of any type and an 11% increase in the sale price of land with an ADU permit in Los Angeles, CA.[10] Using their estimates as the total cost of permitting suggests that the 7% increase in land

access to amenities.

[8] For example, Ganong and Shoag (2017) and Hoxie, Shoag and Veuger (2023) use housing values from the American Community Survey, and Louie, Mondragon and Wieland (2025) use data from the S&P Cotality Case-Shiller US National Home Price Index and the American Community Survey. In these data sources land and structure values are not differentiated. However, it is also possible that increased housing production from reduced supply constraints could still be consistent with aggregate price increases in general equilibrium.

[9] For example, notice the large gaps between permitting and construction timelines for California cities compared to other jurisdictions in D'Amico, Soltas and Wang (2026).

[10] On a per acre basis, Soltas and Gruber (2026) find circumventing permitting is worth around $2 million per acre, whereas our estimates imply a benefit of about $500,000 per acre.

values we find as a result of the California up-zoning and permitting reforms removed about one third of the barriers to building an ADU by value.[11] Many of the papers in this literature focus their estimates within a single city or region. Our setting is unique in that the Coastal Zone extends along the California coast from Mexico to Oregon, offering a large and varied quasi-experimental setting to study permitting reform, which is rare in the literature. We also offer an identification improvement. In many applied papers, variation in regulation is chosen by the same jurisdictions whose outcomes are studied (Gold and Tarascina, 2026; Gyourko and McCulloch, 2024; Bartik, Gupta and Milo, 2025; Turner, Haughwout and van der Klaauw, 2014). In our setting, the state imposes reform on localities, and exempt segments of land were determined by legislation passed decades before the reforms that we study.

The economics literature around up-zoning has also focused on cross-plot spillovers from development. In some settings, negative spillovers from up-zoning can be substantial. Diamond and McQuade (2019) find negative spillovers for comparatively large low-income housing projects placed in higher-income neighborhoods. In a more residential setting, Gyourko and McCulloch (2024) find that 100 homes within 500 meters reduce total home prices by between 1% and 2%. Blanco and Sportiche (2026) find that smaller housing projects may increase home prices, whereas negative spillovers may be primarily features of larger development projects. In our setting, it is unlikely that any parcel is exposed to an increase in ADU density of the magnitudes in either Diamond and McQuade (2019) or Gyourko and McCulloch (2024), and we do not find evidence of large spillovers.[12]

There is a smaller literature linking regulations to land values. Grout, Jaeger and Plantinga (2011) use a regression discontinuity design to link Portland, Oregon's urban growth boundary to increases in land prices. They attribute these changes to the now limited supply of available land. Similarly, Cunningham (2007) finds that growth controls around Seattle, WA reduced the real option value of land, lowering land values and accelerating development. Turner, Haughwout and van der Klaauw (2014) also find that increased regulation lowers land values. Our findings suggest that the converse may also be true when regulations are relaxed. We show that in our setting deregulation for small units can increase land values and increase permitting, albeit modestly.[13]

This paper proceeds as follows. In Section 2, we discuss the institutional details surrounding California's ADU reforms in the context of our Coastal Zone border discontinuity design. We outline our parcel-level dataset and econometric methods in Sections 3 and 4, respectively. We discuss our partially pre-registered main results for ADU permitting, land value changes, and spillovers in prices

[11] To see this, take their 11% total sales price and capitalize it in the land using the average land share of value (56%). This converts their estimates into changes in land values. Then take the ratio of our land value changes from reform and their changes from completing permitting: $\frac{7}{20} = 0.35$.

[12] There is a small literature on spillovers for ADUs. Tanrisever (2025) estimates a cross-plot spillover from ADUs of about 3% in Los Angeles, CA. Davidoff, Pavlov and Somerville (2022) find an equally sized but insignificant spillover effect for all lots, but do find large spillovers for higher-value lots. Our results are not precise enough to reject spillovers of this magnitude. Gold and Tarascina (2026) also do not find evidence of significant cross-plot spillovers from ADUs in Seattle, WA.

[13] This literature is also related to the more standard capitalization literature of tax and spending policies, such as Oates (1969), Black (1999), Linden and Rockoff (2008), and Baylis and Boomhower (2023).

and transactions in Section 5. Then we build intuition on how homeowners respond to permitting reform for ADUs with an open city model that features endogenous ADU construction, property taxes, and a characterization of our difference in discontinuities estimator in Section 6. This simple model motivates our discussion in Section 7, which explores heterogeneity in our main estimates as well as the fiscal and rental impacts of permitting reform. Finally, we offer brief concluding remarks in Section 8.

2 Housing Policy and ADUs in California

Since at least 1926, cities have had the power to regulate local housing construction through zoning (Village of Euclid v. Ambler Realty Co., 1926). However, the exact powers that cities possess have largely been determined by the states (Davidson et al., 2020). In 2016, the State of California passed the first of several laws that aimed to encourage the construction of ADUs by removing local regulatory barriers enforced through zoning, permitting backlogs, parking requirements, and other pre-construction hurdles (Dillon and Khouri, 2016; Minott, 2023; Gray, 2024).

Prior to 2016, building an ADU in a particular jurisdiction was largely under local control. Although the State passed legislation aimed at enabling ADU permitting in 2002, local governments were still able to use the discretion inherent to the permitting process, as well as a myriad of additional requirements, as a *de facto* veto of the attempted state-wide reforms.[14] This left cities with the power to impose additional site requirements and inspections, impact fees, parking requirements, and other cost-increasing measures on proposed ADUs (Brinig and Garnett, 2013).[15]

Starting in 2016, the State has passed a series of reforms that stripped local jurisdictions of their power to impose barriers to ADU construction (Gray, 2024; Minott, 2023). Assembly Bill (AB) 2299 and Senate Bill (SB) 1069 required "by-right" permitting of ADUs on most residential lots with a maximum processing time of 120 days. This meant that if an ADU met state-set requirements and was in a residentially zoned area, then the ADU must be approved. The reforms required jurisdictions to allow ADUs on lots that already have single-family homes, and removed a number of local requirements such as street access, new utility hookups, and large setbacks. The reforms also reduced existing parking requirements for ADUs near transit or ADUs built as part of a primary residence (Minott, 2023; County of Humboldt, 2017). These reforms took effect in 2017.

Continuing its efforts in the 2017 legislative session, the state passed SB 229 and AB 494. These bills clarified that local jurisdictions cannot ban homeowners from renting out ADUs separately from the attached residence, that ADUs can be part of proposed single-family housing projects, and limited parking requirements to only one additional space per-unit. Then in 2019, the legislature continued its reforms to ADUs with three new bills, AB 68, AB 881, and SB 13.[16] Together, these pieces of legislation removed minimum lot sizes, capped setbacks, eliminated floor area ratios,

[14] Moreover, a second bill in 2003 was proposed to address local roadblocks, but Governor Schwarzenegger vetoed the bill since it encroached on local authority (Brinig and Garnett, 2013).

[15] For a more complete discussion of the regulatory environment in the 150 largest California cities surrounding ADUs, please see Brinig and Garnett (2013).

[16] These bills also expanded the reforms to Junior ADUs, which are smaller than regular ADUs (Minott, 2023).

prohibited owner-occupancy requirements, expanded garage conversions, and allowed ADUs in multi-family areas. These additional limitations on local governments were paired with increased enforcement tools through the California Department of Housing and Community Development. The state also limited the ability of homeowners associations (HOAs) to restrict ADU rentals in AB 670 (2019) and AB 3182 (2021), clarified that selling ADUs separately from the main house is legal in AB 587 (2019), and required permitting jurisdictions to communicate existing grants and financing programs to encourage ADU construction in AB 671 (2019). Finally, in 2022 the state clarified that all ADUs up to 16 feet high are permitted, further limited the setback restrictions, and required localities to enumerate to applicants why their permits were denied (AB 2221 and SB 897) (Minott, 2023; Gray, 2024).

After implementing the full set of reforms for ADUs, a typical residential lot in California is entitled to build a 1,200 square-foot ADU and a 500 square-foot Junior ADU with only a 4 foot setback on the side and rear. Moreover, existing detached structures can be converted into ADUs of up to 1,200 square feet. These projects must be approved within 60 days or issued a denial report so they can reapply (California Department of Housing & Community Development, 2026).

However, the set of ADU reforms could not be enforced for land within the California Coastal Zone, a roughly one-kilometer strip of land along California's entire coastline.[17] The California Coastal Zone's land use has been governed by the Coastal Act and the California Coastal Commission since 1976 (California Coastal Commission, 2022*a*). The Coastal Act prohibits the state from imposing the reforms to ADUs that "increase the intensity of" land use on local jurisdictions within the Coastal Zone, such as up-zoning and by-right permitting, which are key features of the state's ADU reforms (California Coastal Commission, 2022*b*). Instead, within the Coastal Zone, new development must receive a special coastal development permit, and any of the state-wide reforms may only be implemented by jurisdictions updating their local coastal plans voluntarily (California Coastal Commission, 2022*b*; Severen and Plantinga, 2018; Kahn, Vaughn and Zasloff, 2010).[18] In practice, the Coastal Zone is governed by local coastal plans, which cover 145 segments spanning the state's coastline from Mexico to Oregon. The local jurisdiction (e.g., the City of San Diego, UC San Diego, or the County of San Diego) that overlaps with the Coastal Zone will file a plan that details how the land will be managed in accordance with the Coastal Act.[19] These plans must be approved by the Coastal Commission, and can periodically be updated.[20] In Figure 2, we show how the Coastal Zone boundary (in blue) overlaps with real estate transactions (dots), incorporated cities (orange), and unincorporated land (white) in San Diego County.

There are two additional features of California housing policy that will play an important role in

[17] The Coastal Zone can extend from 1,000 yards inland to several miles. Some of this variation can be seen in Figure 2.

[18] As of this 2022 memo, only 5 cities have voluntarily adopted parts of the reforms to ADUs. These include Santa Cruz, Santa Barbara, Encinitas, San Mateo, and Pacifica (California Coastal Commission, 2022*b*).

[19] In most cases, cities like San Diego have portions of land within the Coastal Zone and portions outside of the Coastal Zone. In this case, the local coastal plan is only applicable in the fraction of land within the city that overlaps with the Coastal Zone.

[20] A recent court case showed that the Coastal Commission has been requiring coastal development permits for ADUs (Riddick v. City of Malibu, 2025).

how we evaluate the impacts of permitting reforms. First, before the ADU reforms, housing regulations were relatively restrictive in coastal California. Monkkonen, Manville and Lens (2024) collect data from the 2014 California Housing Element's Regional Housing Needs Assessment (RHNA) process, in which the state and regional governments set housing targets for housing unit growth over the next 8 years for each city and county. Then, the cities respond to these targets by reporting zoned capacity figures that reflect their assessments of how many new units they can accommodate. Following Monkkonen, Manville and Lens (2024), we view the ratio of zoned capacity to existing housing units as a reflection of both regulatory stringency in a jurisdiction and that jurisdiction's stance towards additional housing. More (less) zoned capacity per-unit would reflect both a lighter (more stringent) regulatory environment and more (less) favorable local attitudes towards new units.[21] In Appendix Figure A.1a we compare zoned capacities for all jurisdictions in California and jurisdictions within the coastal counties of California. The median jurisdiction's zoned capacity per housing unit statewide reflects zoning for a 13% expansion in units over 8 years. Appendix Figure A.1b shows the zoned capacity per-unit for each parcel in our analysis sample. Within 450 meters of the Coastal Zone boundary, the median parcel sits in a jurisdiction that only zoned for an 8% increase in units over 8 years, which is reflected in the compressed distribution in Appendix Figure A.1b. These figures suggest that coastal areas were in general relatively restrictive in their policy attitudes towards new units, so ADU reforms are likely to meaningfully change the regulatory environment in this setting.[22]

Second, California's property tax policies likely interact with the effectiveness of the ADU reform package. Proposition 13 allows properties to be assessed at market value after they are purchased, and then subsequent assessments, on which property taxes are applied, are capped at 2% growth each year. A property will be reassessed above this cap in one of two ways. First, if the property is sold, then a full reassessment takes place, and the property is assessed at or near market value. Second, if major improvements are made on the property, then a partial reassessment takes place on the improvements alone. If an improvement like an ADU is built, then the net value of the ADU is added to the baseline assessed value, but the full land and unchanged structure values are not reassessed (San Francisco Assessor Recorder's Office, 2025). This results in most properties being taxed on an assessed value below their market value, which can create lock-in effects (Wasi and White, 2005). However, in participating counties, owners over the age of 55 can transfer their below-market assessed value to a new primary residence through Propositions 60 and 90 (Ferreira, 2010).[23] We will leverage these features of California's property tax system to assess how property

[21] In the text, Monkkonen, Manville and Lens (2024) refer to this measure as unbuilt capacity. We also extend this measure to include the unincorporated areas and non-reporting cities by applying the residual RHNA target units to these areas, distributed in proportion to their populations.

[22] Even if a city zones for units, these units may not be delivered. Glaeser and Gyourko (2025) show that Los Angeles, San Diego, and San Francisco also had some of the lowest housing stock growth between 2010 and 2020, adding about half a percent per year. This also suggests that realized housing construction is likely below what a city is zoned to accommodate.

[23] The benefit applies to primary residences and may be used one time. Participating counties with Coastal Zone overlap are Los Angeles, Orange, San Diego, San Mateo, Ventura. Other participating counties are: Alameda, Riverside, San Bernardino, Santa Clara, and Tuolumne (California State Board of Equalization, 2025*b*).

taxes may mediate the impacts of the ADU reforms.

3 Data

Our dataset construction, variable definitions, and econometric designs were pre-registered before we began our data work with the Cotality (2025) parcel data.[24] As a result of unforeseen features of our dataset, we have deviated in a limited number of ways from our original plan. At the end of the methods section, we will identify deviations from our pre-analysis plan as appropriate. In addition, we provide a full artificial intelligence enabled audit of our compliance, following a novel adversarial protocol, in Appendix B.

3.1 Dataset Construction

To investigate the impacts of permitting reforms for accessory dwelling units, we build two datasets that link parcel-level property characteristics from assessment records, real estate sales data, and building permits. The first dataset is a balanced parcel panel that has one observation for each parcel in each year. The second dataset uses transaction records, which may have multiple observations or no observations for a parcel in a given year. Both datasets are constructed for the 15 coastal counties of California from 2010 to 2024, as data are available.[25] Our final dataset contains just over 4 million residential parcels and over 2 million transactions between 2010 and 2024.

The Cotality (2025) (formerly CoreLogic) property data create the backbone of our balanced parcel-year dataset. Cotality (2025) collects and standardizes county assessor records, which are used to levy property taxes. These records include information on assessed values for the structure, building and lot characteristics, and location identifiers including coordinates and a street address. Using property identifiers, we create a balanced panel of about 4 million unique properties between 2010 and 2024. This requires us to summarize building and lot characteristics when a parcel has multiple assessments within a given year, as well as to expand our dataset to fill in gaps between assessments.[26]

We prune these data, expanding on a process outlined in Frank et al. (2025). First, we limit our dataset to parcels that are ever listed as having a land use or property use that is for small residential uses.[27] Additionally, the parcel must have an assessed value above $1,000 (real) USD, have a room-to-unit ratio within the 1st and 99th percentiles, and have an internally consistent implied age. The impacts of each sample restriction are tracked in Appendix Table A.2.[28] The

[24] The 2025 pre-analysis plan can be accessed on OSF.

[25] From north to south, the 15 coastal counties of California are: Del Norte, Humboldt, Mendocino, Sonoma, Marin, San Francisco, San Mateo, Santa Cruz, Monterey, San Luis Obispo, Santa Barbara, Ventura, Los Angeles, Orange, and San Diego.

[26] To expand the data, we take property characteristics from the last available assessment and carry them forward through the panel.

[27] This includes parcels marked as single-family residences, duplexes, or with land uses marked as residential uses between 1 and 10 units. The vast majority, over 90%, of our final dataset consists of single-family homes, as shown in Appendix Table A.1.

[28] We explore how these sample restrictions interact with our research design in Appendix Table A.3. These

final parcel-year sample has just over 60 million parcel-year observations between 2010 and 2024. We use the balanced panel dataset to evaluate the share of parcels that permit an ADU, that transact, or how much property tax each parcel pays in a given calendar year.

In addition to the property records from Cotality, we incorporate data on building permits from the State of California's housing element progress reports (California Department of Housing and Community Development, 2025). As part of the post-2016[29] housing reforms, the State of California began requiring local jurisdictions (cities and counties) to report on the number of land use entitlements, building permits, and certificates of occupancy they issue each reporting year, by project, starting in 2018.[30] This state-wide permitting dataset is then linkable to our property and transaction data using either an assessor parcel number (APN) or matching the reported addresses directly.[31] One limitation is that we do not have parcel-level permit data available to us before 2018.[32]

Figure 1 compares the California Department of Housing and Community Development (2025) post-2018 state-wide counts of ADU building permits that we attached to our parcel dataset with a prior vintage of the same time series.[33] The combined series for state-wide ADUs from 2017 to 2024, using the deprecated CA DHCD series (blue) for 2017 and the current housing element (orange) after 2017, shows about 176,000 permits issued in total post-reform. Pre-reform, the deprecated CA DHCD series shows only about 4,000 permits issued between 2012 and 2016. These series suggest that the overwhelming majority of ADU permits issued between 2012 and 2024 are captured by the detailed post-2018 permit data from California Department of Housing and Community Development (2025), shown as an orange dashed line.[34] Although we do not have any pre-reform permits to match, Figure 1 suggests that ADU permitting before the reforms for ADUs was generally negligible.

For analyzing the impacts of permitting reforms on land values and sale prices, we use a dataset based on the Cotality (2025) transaction records. These transaction data also include information

restrictions differentially remove observations from inside the Coastal Zone in levels, but these differences appear to be stable over time in our balanced panel dataset.

[29]Throughout the paper, pre-2016 refers to 2010 to 2016, and post-2016 refers to 2017 to 2024.

[30]We are unable to account for illegal ADU construction, which may be a meaningful share of ADUs constructed pre-reform. Jo et al. (2025) use satellite data and a machine learning model to detect unpermitted ADUs in San Jose, CA, between 2016 and 2020. Their estimates suggest that informal units may outnumber formal units.

[31]To do this matching, we build a hierarchical matching algorithm that prioritizes matching by APN and then uses the available address information. After we separate all of the listed addresses associated with a project, we attach a permit to all addresses associated with it first using the APN, then an exact match on the address, then on the Google location key from the Google Maps Platform (2026), and finally on previously listed APNs. We describe this process in Appendix C.

[32]Another issue is that we can only measure permitting. We attempted to measure construction of ADUs using the Cotality (2025) data, but it was clear that we could only capture the effects of likely construction events in general rather than specifically the construction of ADUs.

[33]The prior vintage time series comes from the California Department of Housing and Community Development's (CA DHCD) Housing Element, but we were unable to locate the original official source. The same time series of state-wide ADU permits going back to 2012 is corroborated by Gray (2024), Minott (2023), and Peterson (2021).

[34]There are slight discrepancies between the two series. The detailed data from California Department of Housing and Community Development (2025) include unincorporated permits, and it is unclear whether or not these areas are included in the older series.

on the real property being sold, the date of the sale, an arm's-length transaction indicator, and the nominal sale price. We attach building characteristics, assessment information, location, land use categories, and other parcel-level information to these transaction records from the balanced parcel panel using the unique property ID provided by Cotality. Next, we trim the sample to include only residential properties and arms-length transactions as well as properties with assessed values greater than \$1,000 (real) USD, (real) sale prices between the 1st and 99th percentiles, plausible transaction volumes within a single year (99th percentile and above are trimmed), room-to-unit ratios between the 1st and 99th percentiles, (real) land values per acre greater than \$1 USD, and an internally consistent implied age of the structure based on the date of sale and the building date. We describe how these restrictions impact our final sample of 2 million transactions in Table A.2.[35]

3.2 Outcome Variable Construction

Our first outcome variable is a dummy for whether or not a parcel received an ADU permit in a particular year, which is constructed on our balanced parcel panel. This variable takes a value of 1 if the parcel has an associated ADU permit in a particular calendar year, and a value of zero otherwise. We only use this variable starting in 2018, as shown in Table 1. In the full county sample, only about 0.3% of parcels had an ADU permit in a given year. Although this is a flow measure, it still reflects the fact that ADU permitting, even with reforms, appears to be relatively rare. However, Table 1 does suggest that ADU permits are likely a large share of building permits overall. For legibility in tables, we scale our ADU permit dummy to reflect permits per 1,000 parcel-years.

Our second outcome variable that we build on our balanced panel dataset is a transaction dummy. We construct a dummy variable using both the panel dataset and transaction records from Cotality (2025) that takes a value of 1 if a parcel has an arms-length transaction in a given calendar year, and 0 otherwise. We will primarily use this variable to investigate transaction spillovers that may result from disamenities associated with ADUs.

We now turn to outcome variables constructed on our transaction dataset. Since land values are not measured directly in our transaction data, we multiply the transaction sale price by the land share of total value from the first available assessment after the transaction to create our measure of (real) land values.[36] We then take the total land price and divide it by the parcel land acreage

[35] We also show how these restrictions interact with the Coastal Zone boundary on our analysis sample in Appendix Table A.3. These restrictions drop more transactions inside the Coastal Zone than outside, and this relationship strengthens after 2017. However, the magnitude of these changes is small: an additional 1% of transactions get dropped on top of a base rate of over 80% of transactions being trimmed. The vast majority of transactions are dropped because they are not registered as arms-length, meaning both parties have a prior relationship such as family ties.

[36] Typically, assessments are updated after arms-length transactions in California, so this is the best available measure of the land share of total value available to us. Moreover, assessors are supposed to estimate the land value by removing the structure value from the total sale price, similar to an approximated hedonic approach (California State Board of Equalization, 2025*a*). We view combining the transaction price and assessed land value share as an improvement over using the assessed land value directly, which can have issues in panel settings (Ahlfeldt, Koster and Vu, 2025; Ihlanfeldt and Martinez-Vazquez, 1986).

to get the (real) land value per acre.[37] As shown in Table 2, before the 2017 permitting reforms, transactions in the full 15 county sample averaged $3.1 million per acre (2025 USD), compared to $4.5 million per acre (2025 USD) after the reforms.

How does our measure of land values per acre, built from market transactions and the first post-sale assessed ratio of land and total value, compare to the literature? One common method is to use vacant parcels to estimate land values (Albouy, Ehrlich and Shin, 2018; Nichols, Oliner and Mulhall, 2013; Haughwout, Orr and Bedoll, 2008). The second method is to try to estimate the value of the land by accounting for the cost of replacing any structures on the land and removing it from the total sale price (Davis and Palumbo, 2008; Davis et al., 2017). In our setting, using teardowns or vacant lots is difficult because there are very few instances of suitable parcels in coastal California and near the Coastal Zone boundary. Moreover, estimating the land value under a built parcel requires a form of spatial interpolation that would be ill-suited for measuring the effects of a regulatory environment that changes discretely. When a property is being assessed, and comparable land sales are not available, an assessor will use a "land residual" approach that estimates the value of the structures and assigns the remaining value to the land itself (California State Board of Equalization, 2025*a*). As an alternative to our preferred measure of land values built from the total sales price and assessed land value ratio, we will present estimates that reflect land values by using time-updating hedonics to remove the value of current structures from the total sales price.[38] As a consistency check, we compare our preferred land value measure, derived from sale and assessment data, to an alternative measure from Davis et al. (2021) directly in Appendix Figure A.2. Davis et al. (2021) provide a national set of estimates of land values built using appraisal data on new homes. To make our measures comparable, we show correlations of average residential land values per acre for each of the 15 coastal counties in all available years (2012-2024). Each dot represents a county-year, with land value estimates from Davis et al. (2021) on the vertical axis, and land value calculations from our transaction and assessment data on the horizontal axis.[39] A simple linear fit shows an R-squared of 0.9 between the two measures, and a coefficient of about 1.1. Although our measure does not perfectly fit the equality line, these results suggest that these two measures of land values are relatively similar, at least at the county-year level.

To capture the impacts of spillovers from increased density across the Coastal Zone boundary, we use the (real) total sale price per acre. We construct sale prices per acre as the transaction sale price divided by the parcel land acreage from its matched assessment records. In Table 2, we can see that total sale prices increased from on average $5.6 million per acre before the reforms to almost $8 million per acre between 2017 and 2024.[40] The typical total sale price for a home in our 450m sample is between about $2 million and $3 million, but there is substantial variation given

[37] We deflate all monetary variables by the personal consumption expenditures chain-type price index to convert values to 2025 USD (U.S. Bureau of Economic Analysis, 2025).

[38] We do not have access to construction data to improve on this method, which is why it will be used as robustness.

[39] Davis et al. (2021) have continued to update this series post-publication.

[40] Note that if a 3,500 square-foot home in La Jolla, CA, is on a 0.2 acre (8,700 square-foot) lot and sells for $5 million, then its price per acre is $25 million.

that our sample spans the entire state.

3.3 Construction of the Running Variable and Other Covariates

The core of the regression discontinuity and difference in discontinuities designs is the running variable. Using each parcel's coordinates, we construct our running variable as distance to the Coastal Zone boundary in meters using shapefiles from the California Coastal Commission (2014).[41] We then center our running variable at 0 so that values for the distance to the Coastal Zone are less than 0 when a parcel is within the Coastal Zone, and greater than 0 when a parcel is outside the Coastal Zone. We will use this running variable to define the bandwidth for our final analysis sample, as well as to assign parcels to distance bins from the Coastal Zone boundary. We selected our analysis bandwidth by applying the mean squared error optimal bandwidth selection criterion to a regression discontinuity in land values per acre estimated on pre-period data (2010 to 2016) (Calonico, Cattaneo and Farrell, 2020).[42] This gives us an optimal bandwidth of 450 meters from the boundary. We also calculate each parcel's nearest Coastal Zone boundary segment, which corresponds to the 145 local coastal plans that govern each Coastal Zone segment.

We also build several covariates, which we use both to ensure the robustness of our estimates and to characterize pre-period differences across the Coastal Zone boundary. First, we construct a set of fixed effects that are meant to capture hedonic characteristics of the parcel. These fixed effects are a set of interacted bins that capture the parcel's build year, land use category, number of bedrooms, number of bathrooms, and building square footage before the permitting reforms. We construct fixed effects in a similar style to Yagan (2019), which include the full interaction of the following groups, with bins for missing values in each category. The bins are: above or below county median effective age (and a missing bin), land use category (single-family, other, or missing), number of bedrooms (0-2, 3-4, 5 and up, and missing), number of bathrooms ($[0, 2)$, $[2, 4)$, $[4, \text{inf})$, and missing), and within county tertiles of building square footage (and a missing bin). In most cases, we include versions of these fixed effects constructed before our 2017 post-ADU reform period.[43] However, to accompany the alternative land value measure robustness exercise we allow these bins to update throughout the panel. We also use parcel coordinates to attach a rich set of spatial data, such as distance to the nearest CBD from Holian (2019),[44] distance to the nearest coastline from California Department of Fish and Wildlife (2026), and elevation from U.S. Geological Survey (2026) as well as data from the American Community Survey (ACS) attached by block group (U.S. Census Bureau, 2016).[45]

We also build additional variables to explore heterogeneity in our estimates. First, the decision to permit and build an ADU is likely a function of lot quality. We identify two major determinants

[41] To be included in our sample, a property must have a valid identifier and a valid coordinate. If a parcel coordinate is unavailable, we use the block centroid when possible.

[42] We used the rdrobust package in R (Calonico et al., 2017).

[43] We use the most recent building features available between 2010 and 2016 to construct our pre-2016 static hedonic fixed effects set.

[44] We choose the Google Earth points, but all choices are highly correlated (Holian, 2019).

[45] We also use jurisdictional shapefiles from California Office of Emergency Services (2015).

of ADU-relevant lot quality, although in principle there could be many: the amount of unbuilt "excess" space on the lot where an ADU could be placed and the distance of the lot from the nearest central business district. A lot with additional area not covered by existing structures will be better able to accommodate an ADU without eliminating the existing yard or open space compared to a more space-constrained lot. Lots with more excess space can also accommodate larger ADUs. Additionally, lots closer to the central business district in standard urban models, such as Alonso (1964), Muth (1969), and Mills (1967), will have greater access to jobs and likely command higher rents. Therefore, ADUs closer to the city center (CBD) will likely be more valuable to the owner than ADUs built towards the periphery. We will include excess lot square footage and distance to the CBD as two binary variables. The first for excess lot square footage takes a value of 1 when a parcel has above median excess square feet available for an ADU and 0 otherwise. The second variable takes a value of 1 when the lot is above the median distance to the nearest CBD, and 0 otherwise. In turn, these two factors should also interact with the magnitude of the land value jump at the boundary.

Next, we will explore how property taxes interact with permitting reforms. We will use two variables to evaluate the role of property taxes in transmitting or muting the effects of permitting reforms. The first is to use the estimated property tax shelter for each parcel in 2016. We construct each parcel's property tax shelter as a function of an estimated market value from a hedonic price regression and the imputed assessed value under Proposition 13 in 2016. The formula for constructing our measure of property tax shelter is:

$$PropTaxShelter = \frac{M - A}{M} \tag{1}$$

where M is the estimated market value in 2016 and A is the assessed value in 2016 for each parcel.[46] The second variable that we use to explore the role of property taxes in our estimates is a parcel's block-group-level exposure to Propositions 60 and 90. We construct this variable using the share of homeowners above the age of 55 in 2016, attached to each parcel by block group (U.S. Census Bureau, 2016).[47] Since property tax shelter is likely not independent of build costs, as more tenured owners may accrue large tax shelters, face more difficulties completing construction projects on their lots, or have changing preferences for density due to age, these variables likely provide imperfect evidence of the role that property taxes play in the effectiveness of ADU reforms.

3.4 Summary Statistics

In Tables 1 and 2, we present means and standard deviations for variables in our 450-meter bandwidth analysis sample. Columns 1 through 4 present summary statistics for the sub-sample of parcels inside the Coastal Zone from 2010 to 2016, inside the Coastal Zone from 2017 to 2024, outside the Coastal Zone from 2010 to 2016, and outside the Coastal Zone from 2017 to 2024. Across

[46] We describe the construction of the property tax shelter variable, including the underlying hedonic estimates, in Appendix C.

[47] In 2021, this benefit was extended statewide as part of Proposition 19.

all columns, around 90% of parcels are single-family parcels. Buildings on these parcels tend to be relatively old, with build years around 1970.[48] The houses on these parcels tend to have around 3 bedrooms, 2.5 bathrooms, and have a building square footage of about 2,000 square feet.[49]

We graphically show the location of our final 450-meter bandwidth sample in Figure 3. We present the density of our Coastal Zone sample by latitude as a solid blue line. Our sample ranges from San Diego in the south, just north of 32 degrees latitude, to the Oregon border at almost 42 degrees latitude. The majority of our analysis sample is located in Southern California, with a large coastal sample in both San Diego and Los Angeles. As we move north, there is a meaningful sample around Santa Cruz and San Francisco, and then a final set of parcels in Eureka. The density for the remaining (non-Coastal) parts of the 15 counties is shown as an orange dashed line. Even though the ocean-adjacent parts of California include some of the most desirable real estate in the United States, the wide range of this sample will allow us to evaluate the impacts of ADUs in a number of different communities. We will present our methodology for doing so in the following section.

4 Empirical Methods

We can build intuition around our empirical designs using the map of San Diego in Figure 2. This figure displays the Coastal Zone, and its segments, in blue, incorporated cities in orange, and 2016 residential real estate transactions from the Cotality (2025) data as dots. Throughout our analysis period within the Coastal Zone (the blue region), ADUs are subject to a stable, albeit stringent, regulatory environment. However, outside the Coastal Zone (in the orange and white regions), parcels benefit from the suite of up-zoning and permitting reforms for ADUs that took effect in 2017 and after. The core of our spatial difference in discontinuities design is a comparison of changes for parcels just outside the Coastal Zone which benefit from the ADU reform package to changes for parcels just inside the Coastal Zone that are subject to land use rules under their local coastal plans.[50] Critically, this stark change in the regulatory environment represented by the

[48] Effective build year is the year in which the parcel had its currently assessed features constructed. So, if a new extension to the house was added, such as an ADU, the effective build year would update.

[49] Full sample statistics, including means, standard deviations, minima, and maxima can be found for the full county sample in Appendix Table A.1. It is important to note that the sample is mostly, but not entirely, single-family units. We intentionally exclude large apartment or condominium multi-family parcels as these types of units would not have their highest economic use increased by the ability to build an ADU. However, given that we have a dataset of tax parcels, rather than lots, it is the case that several commonly owned units may be combined in a single parcel, which can skew the extremes of the distribution for characteristics such as bedrooms, bathrooms, etc. We have tried to remove these parcels from our sample.

[50] There are more pieces of legislation that have been passed by the State of California to address housing affordability. These measures include the ADU reforms, as well as rezoning efforts surrounding transit-dense housing, infill housing, and lot splitting among other reforms. To our knowledge, only the ADU reforms and the SB 35 environmental permitting waivers for infill housing conflict with the Coastal Act directly. SB 35 does limit discretionary permit review for developments that qualify as infill. There have been about 160 SB 35 projects since its enactment in 2017. These projects are generally much larger projects, with 90% of SB 35 projects containing more than 10 units, and 74% of projects containing more than 50 units. Using SB 35 in a smaller residentially zoned setting would be difficult due to the cost of assembling an economically viable tract of land to build on. This largely puts SB 35 projects farther away from the small residentially zoned lots that are the focus of this paper (Manji and Finnigan,

Coastal Zone boundary cuts directly through neighborhoods, which we can see in Figure 2. We formalize this intuition in the remainder of this section.

4.1 A Regression Discontinuity for Permitting Activity

To evaluate the impacts of permitting reform on ADU permitting activity, we will leverage the discrete change in the regulatory environment at the Coastal Zone boundary in a regression discontinuity design. We will estimate the following specification for 2018 to 2024 using our balanced parcel dataset:

$$Y_{ict} = \alpha + \beta_1 OutCZ_i + \beta_2 DistCZ_i + \beta_3 (OutCZ_i \cdot DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \sigma_{s,t} + \varepsilon_{ict} \quad (2)$$

where Y_{ict} is an indicator variable taking a value of 1 if parcel i in county c received an ADU permit in year t. $OutCZ_i$ is an indicator taking a value of 0 if lot i is within the Coastal Zone, and therefore has stable regulations, or a value of 1 if lot i is outside the Coastal Zone and therefore will receive the state-wide permitting reform package in 2017. $DistCZ_i$ is lot i's distance from the closest point on the Coastal Zone boundary in meters, which we also interact with $OutCZ_i$ to allow the slope to vary across the boundary.[51] In order to control for pre-reform characteristics, we include a set of hedonic fixed effects using pre-2016 parcel characteristics, $\theta_{(i)}$. We also include a set of county (λ_c) fixed effects. The county fixed effects will control for constant characteristics of county assessors' offices as well as other static characteristics of the county. In our preferred specifications, we include a set of Coastal Zone border-segment by year fixed effects ($\sigma_{s,t}$) to ensure that all comparisons are within local segments of the Coastal Zone boundary. We also present specifications using a year fixed effect (τ_t), but controlling for border segments that may experience different real estate market conditions or other shocks is likely important with such a varied spatial setting. ε_{ict} is an error term.

The main assumption for our spatial regression discontinuity model is that permitting activity across the Coastal Zone boundary would be the same absent the difference in the regulatory environments. Explicitly, we are assuming that parcels inside the Coastal Zone boundary and parcels just outside the Coastal Zone boundary would have had similar ADU permitting levels if reforms had not taken effect in 2017. Since Figure 1 suggests that pre-reform permitting activity was negligible, we will proceed under the assumption that pre-reform ADU permitting activity was essentially zero on both sides of the boundary and therefore there is no pre-2017 discontinuity in ADU permitting to account for. Additionally, the sharp regression discontinuity design demands that the regulatory environment jumps discretely at the Coastal Zone boundary, whereas other relevant features for building ADUs besides regulation change smoothly across the boundary. Severen and Plantinga (2018) and Kahn, Vaughn and Zasloff (2010) both show that land just inside

2023).

[51] We will use first-degree and second-degree polynomials, each interacted with the $OutCZ$ dummy to allow for slope changes across the boundary.

and just outside of the Coastal Zone is similar in terms of physical characteristics and amenities. Critically, we are identifying the effects of a package of reforms, as we will be unable to separate the impacts of any individual element of the post-2017 up-zoning and permitting reform package.

We explore how covariates evolve across the Coastal Zone boundary in Figure 4a. In this figure, we show estimates of Equation 2 within our analysis sample for the pre-reform years, 2010 to 2016. In orange circles we show standardized coefficients using just county and year fixed effects, and in hollow blue squares we replace the year fixed effect with our preferred segment by year fixed effect. We can see that for sale price per acre and land values per acre, values are not distinguishable from zero across the Coastal Zone boundary.[52] We also see pre-period balance in transactions. However, several covariates jump across the Coastal Zone boundary before permitting reform, and they are statistically distinguishable from zero. When we undo the scaling on these estimates, however, we find that they are small differences in practical terms. Compared to inside the Coastal Zone, parcels outside the Coastal Zone have houses on them that are about a year newer, have 0.04 fewer bathrooms, are 26 square feet smaller, sit 1.4 meters lower in elevation, are 250 meters closer to the nearest CBD, and are about 30 meters closer to the coastline.[53] These baseline differences motivate the inclusion of our set of pre-2016 hedonic fixed effects and our repeat sales model, which both address static differences in lots that may interact with our treatment.

Throughout our empirical estimates, we will use our 450 meter analysis bandwidth. All specifications also include triangle weights, which give additional weight to parcels closer to the Coastal Zone boundary. In our main tables and figures, we will present parcel clustered standard errors. In practice, at every bandwidth these standard errors are almost identical to the more conservative Coastal Zone segment clustered standard errors. We present our pooled and preferred specifications using parcel clustered, segment clustered, and spatial correlation robust standard errors from Conley (1999) when we explore the bandwidth sensitivity of our estimates.

4.2 A Difference in Discontinuities Design for Land Value Changes

We use a difference in discontinuities estimator to show how permitting reform changes at the Coastal Zone boundary impact land values. Since we have land values for the full panel starting in 2010, we use our transaction data to estimate models both with and without a parcel fixed effect. We estimate the following equation using our transaction sample from 2010 to 2024:

$$\begin{aligned} ln(Y)_{ict} =& \alpha + \beta_1(OutCZ_i \cdot Post_t) + \beta_2(DistCZ_i \cdot Post_t) + \beta_3(DistCZ_i \cdot OutCZ_i \cdot Post_t) \\ &+ \beta_4 OutCZ_i + \beta_5(DistCZ_i) + \beta_6(DistCZ_i \cdot OutCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \sigma_{s,t} + \varepsilon_{ict} \end{aligned} \tag{3}$$

where $ln(Y)_{ict}$ is the log of the real land value per acre for parcel i in county c sold in year t. A difference in discontinuities model stacks two regression discontinuities in one design. The first

[52] We can see this visually in the pre-2016 series of the binned scatter plots for land values and total sales prices in Appendix Figures A.3 and A.4, respectively.

[53] The shape of the Coastal Zone and the coastline can diverge substantially. Several bays, coastal inlets, and other features that are classified as coastline extend inland past the Coastal Zone boundary, such as the San Francisco Bay. Moreover, the Coastal Zone itself can extend inland several miles at points, as shown around San Diego in Figure 2.

regression discontinuity, parameterized by β_4, β_5, and β_6, captures the pre-2016 level effects of Coastal Zone regulations, which Figure 4a and prior work suggest are likely present (Severen and Plantinga, 2018; Kahn, Vaughn and Zasloff, 2010).[54] Then, a second set of regression discontinuity parameters is interacted with a post-treatment dummy, $Post_t$, to capture the changing effects of regulation at the Coastal Zone boundary after permitting reform is enacted in 2017. Our parameter of interest is β_1, the coefficient on the interaction term $OutCZ_i \cdot Post_t$. Just as with Equation 2, we include a set of segment by year fixed effects ($\sigma_{s,t}$) that keep the comparisons local to each of the 145 segments of the Coastal Zone boundary. We also present estimates with only a year fixed effect (τ_t), but these estimates may suffer from biases generated from comparisons along distant parts of the Coastal Zone boundary that may experience different economic shocks. Including a parcel fixed effect limits us to a sample of parcels that transact both before and after 2017. The segment by year fixed effects also help control for features of the policy and economic environment that may vary over our sample and impact both sides of the Coastal Zone boundary. We have a sufficient sample of these repeat sales parcels in our dataset to include a parcel fixed effect. Therefore, we present a repeat sales model that includes a parcel fixed effect to control for static characteristics of the parcel, similar to Soltas and Gruber (2026). A repeat sales model also removes our pre-reform regression discontinuity terms, which get absorbed by the fixed effect. Our preferred specification will use a repeat sales model with a set of segment by year fixed effects.

This difference in discontinuity design, developed by Grembi, Nannicini and Troiano (2016), compares changes in land values on parcels just inside the Coastal Zone boundary to changes in land values on parcels just outside the Coastal Zone boundary before and after the reform. The resulting estimator relies on a similar logic to difference in differences and regression discontinuity designs. The core assumption is that, local to the cutoff, land values would have evolved similarly in the absence of the post-2016 reforms to ADUs. However, unlike the regression discontinuity design, stable differences in the characteristics of parcels across the Coastal Zone boundary are accounted for through differencing. In Figure 4, we show level differences in Panel A, and pre-trends compared to our base year, 2016, in Panel B. As with the level differences discussed above, our trends in total sale prices per acre, land values per acre, and transactions shift towards zero when we use segment by year fixed effects instead of the year fixed effect in our pooled estimates. However, for all three outcomes (sales price per acre, land price per acre, and transactions), Figure 4b supports the argument that our local parallel trends assumption likely holds near the Coastal Zone boundary.[55]

The second assumption for our difference in discontinuities estimator is that the regulatory environment changes sharply at the Coastal Zone boundary after 2017. Our regression discontinuity estimates for differences in permitting rates for ADUs at the boundary will serve as evidence that

[54] Note that Severen and Plantinga (2018) focus on multi-family units, and Kahn, Vaughn and Zasloff (2010) use price data from census tracts and Los Angeles property transactions from 2008. So, our baseline balance may not align with either study exactly.

[55] Butts (2023) formalizes the local parallel trends assumption into a continuity assumption of potential outcomes across the cutoff. He also notes that we need to assume the land cannot sort in response to our reforms differentially across the boundary.

the permitting reforms had sufficient bite to justify our design.[56] Finally, we are also assuming that the pre-existing level differences created by Coastal Zone regulations can be accounted for by including our pre-period discontinuity terms (Butts, 2023).

4.3 Detecting Spillovers Across the Coastal Zone Boundary

To detect spillovers local to the boundary, we will use a more flexible binned approach following the designs in Hornbeck and Keniston (2017) and Severen and Plantinga (2018). Parcels just inside the Coastal Zone should be unable to take advantage of the new ADU law, but they will be exposed to any spillovers associated with the new construction. We can estimate a flexible specification as follows for 2010 to 2024:

$$ln(Y)_{itbs} = \alpha + \sum_{b \neq 1} \beta_b (I\{Bin = b\}_i \cdot Post_t) + \theta'_{(i)}\gamma + \phi_b + \sigma_{s,t} + \varepsilon_{itbs} \tag{4}$$

where $ln(Y)_{itbs}$ is the log of the real total sale price (land and structures) per acre on parcel i, in county c, in year t, in Coastal Zone distance bin b, and Coastal Zone border segment s. Here, we will construct a series of dummies for 50 meter Coastal Zone distance bins, $b \in \{1, ..., B\}$, with an omitted bin 400 to 450 meters within the Coastal Zone as a comparison group.[57] Our set of estimates, β_b, will capture how prices evolve across a set of bins relative to the comparison bin. We also include segment by year fixed effects, $\sigma_{s,t}$, in our presented estimates. If spillovers are present, then prices should change just on the inside of the Coastal Zone, reflecting either positive or negative spillovers from the permitting reforms outside the Coastal Zone.

Motivated by our theoretical model, we also use this design to show how transactions evolve across the Coastal Zone boundary. We show spillover estimates that replace the log of the real total sale price per acre with a dummy indicating if the parcel had an arms-length transaction in a given year. In these transaction specifications, we also include a parcel fixed effect.

4.4 Heterogeneity

We include heterogeneity variables following Calonico et al. (2025). This requires fully interacting our regression discontinuity variables with the heterogeneity variable of interest. Including heterogeneity gives the modified regression discontinuity equation:

$$\begin{aligned} Y_{ict} =& \alpha + \eta_1(H_i \cdot OutCZ_i) + \eta_2(H_i \cdot DistCZ_i) + \eta_3(H_i \cdot OutCZ_i \cdot DistCZ_i) + \eta_4 H_i \\ &+ \delta_1 OutCZ_i + \delta_2 DistCZ_i + \delta_3(OutCZ_i \cdot DistCZ_i) + \lambda_c + \tau_t + \varepsilon_{ict} \end{aligned} \tag{5}$$

where H_i is our heterogeneity variable of interest. Our new object of interest is the conditional

[56] We also find that our regression discontinuity design doesn't detect changes in larger permitted units, such as multi-family units impacted by SB 35.

[57] Our selection of the 400 to 450 meters bin as the reference bin is based on our mean squared error optimal bandwidth choice, but in principle we could use any bin as a reference bin.

average treatment effect, or CATE. We can write the CATE as a combination of estimates of η_1 and δ_1 at different levels of H_i:

$$CATE(H_i) = (\hat{\eta_1} \cdot H_i) + \hat{\delta_1} \tag{6}$$

where H_i is once again our heterogeneity variable. Calonico et al. (2025) note that with a continuous H_i, we are required to assume that at the boundary the CATE is linear in H_i. This exercise changes the coefficient of our baseline discontinuity, and it will generally be true that $\delta_1 \neq \beta_1$ from estimates of Equation 2. Rather, δ_1 reflects the treatment effect of permitting reform across the Coastal Zone boundary in the special case when $H_i = 0$. We extend the method to our difference in discontinuity estimates by fully interacting the specification with our heterogeneity variable of interest. We will show heterogeneity estimates by lot quality, as measured by excess lot square footage and distance to the CBD, by property tax shelter, and by block-group exposure to Propositions 60 and 90. When presenting estimates for heterogeneity by exposure, we break this exercise up into estimates stopping before COVID-19 and estimates on the full dataset. We do this because COVID-19 likely decreased the willingness of older adults to move.[58]

4.5 Pre-Registration

Our pre-registration plan specified a set of choices around variable construction and estimating versions of our permitting regression discontinuity design (Equation 2), land value difference in discontinuities design (Equation 3), and total price spillovers design (Equation 4) (Cañedo Riedel and Hoxie, 2025). However, we have deviated from our initial plan in several ways. We will discuss the most important deviations here, and leave the remaining deviations for our artificial intelligence-enabled compliance audit in Appendix B.

Initially, we planned to measure ADU permitting, ADU construction, land values per acre, and total prices per acre. For ADU permitting, we initially planned to use a difference in discontinuities design and collect locally issued permits before 2018. However, Figure 1 suggests that 95% of all state-wide permits for ADUs were issued in 2018 and afterwards. Given that no state-wide dataset is publicly available before 2018, and collecting permits for all permitting jurisdictions within the coastal counties directly was unlikely to yield a substantial number of ADU permits, we decided to only use permits available in California Department of Housing and Community Development (2025). We also hoped to supplement permitting activity with a measure of ADU construction derived from the Cotality (2025) property characteristics. However, the building characteristics available to us are not detailed enough to separate construction related to an ADU from construction in general. We initially planned to measure land values per acre using the assessed land value directly. However, assessed values are subject to non-classical measurement error, and in panel

[58] Foster, Fiorio and Ellis (2024) use US Postal Service change of address data to show that mobility among older natives declined in the US in 2020 and 2021 relative to 2019, likely in response to the COVID-19 pandemic.

settings the assessed value alone can be slow to update (Ahlfeldt, Koster and Vu, 2025; Ihlanfeldt and Martinez-Vazquez, 1986; Chen and Cohen, 2024). In California after a transaction, assessors are instructed to assign values to structures and land separately, whereas owners are only taxed on the total value (California State Board of Equalization, 2025*a*). Therefore, we elect to multiply the total sales price by the ratio of assessed land to total assessed value using the first assessment after a sale to measure land prices. We initially expected anticipation effects to show up in our data as the first set of laws were passed in the third and fourth quarters of 2016. These anticipation effects don't appear in our data and we therefore use the enactment year, 2017, as the start of our post period. Our final measurement deviation from our pre-analysis plan is the addition of transaction rates as an outcome variable in our spillovers design and our difference in discontinuities estimates for property tax collections. These additions were motivated by our theoretical model, presented in Section 6.[59]

We also made minor deviations in our methods. First, instead of estimating a difference in discontinuities for permitting, we estimate a regression discontinuity since all of our measured permitting is post-reform. Implicitly, we are assuming that baseline differences in ADU permitting are negligible across the boundary, and Figure 1 suggests there was not much permitting activity to detect. Second, across our specifications we have elected to include several features that are common for regression discontinuities and other boundary designs. These include the use of triangle weights and mean squared error optimal bandwidth selection (Cunningham, 2021; Calonico, Cattaneo and Farrell, 2020). Finally, we did not pre-specify a repeat sale model because we were unsure about whether or not we would have a large enough sample of parcels that were both local to the Coastal Zone boundary and that transacted before and after the 2017 reforms. We view these deviations as either changes necessitated by our available data or improvements over our pre-specified designs.

5 Main Results

In this section, we will present our estimates of the impacts of ADU reforms on permitting activity, land values, and spillovers in both total sale prices and transactions. The first three subsections present each of these estimates in turn. Then, in the final subsection, we explore robustness in our main permitting and land value results.

5.1 Permitting

First, we will explore differences in permitting activity for ADUs across the Coastal Zone boundary in response to 2017 regulatory changes graphically in Figure 5. We display permits in bins by their distance from the Coastal Zone boundary within our 450-meter analysis sample, where negative values (in blue) show locations within the Coastal Zone that did not receive permitting reforms and positive values (in orange) show locations that were impacted by the state permitting reforms.

[59] Our heterogeneity analysis, analysis on rental impacts, and supporting figures and tables were also not pre-registered.

Each bin covers 25 meters, and the Coastal Zone boundary is marked by a line at 0.[60] Just inside the Coastal Zone, about 1.9 ADUs per 1,000 parcel-years are being permitted, whereas just outside the Coastal Zone more than 2.4 ADUs are being permitted. The sharp difference in the regulatory environment across the Coastal Zone boundary post-reform is directly reflected in the visual discontinuity in permitting activity.

In Table 3, we use our post-2018 balanced parcel dataset and regression discontinuity design to show how reforms outside the Coastal Zone impacted permitting. Our outcome variable is a dummy taking a value of 1 if a parcel in a given year has any ADU permit associated with it, and 0 otherwise. For display, we present coefficients scaled to be ADU permits per 1,000 parcel-years. In Column 1, we include our pooled specification, which only uses the first-degree polynomial, county fixed effects, and year fixed effects. In Column 2, we add in our pre-2016 hedonic fixed effects to capture differences in parcel characteristics. Columns 3 and 4 include segment by year fixed effects to keep our comparisons local to each Coastal Zone segment. Across our specifications, these coefficients suggest an increase in ADU permitting of about 30% when comparing parcels just outside the Coastal Zone to parcels just inside the Coastal Zone. These results are qualitatively consistent across alternative specifications. Every estimate is significant at least at the 1% level, and the point estimates range from 0.53 (Column 3) to 0.629 (Column 2) ADU permits per 1000 parcel-years. Our preferred specification is shown in Column 3, which includes a first-degree polynomial and segment by year fixed effects. Cumulatively, over the 7 years represented by these data, about 0.5% of parcels outside the Coastal Zone will get an ADU permit that wouldn't have been pulled absent these reforms.

These results suggest a meaningful change in the regulatory environment for ADUs occurs when moving from the Coastal Zone to the rest of the state. However, if we were to take the direct change in permitting activity as the only impacts of the post-2016 permitting reforms, then we would have to conclude that these impacts were relatively limited, as a very small fraction of parcels have elected to build an ADU. Extrapolating from these estimates, about 60,000 of the 176,000 statewide ADUs were generated by the up-zoning and permitting reforms that jump at the Coastal Zone boundary. This means a very small portion of lots ever build an ADU, and the natural question is what might explain the low take-up.

5.2 Land Values

On smaller residential lots, the ability to add an ADU increases the highest economic use of that lot, reflecting the ADU's potential value to the owner. This option value should capitalize in the land

[60] We show an alternative scatter plot in Appendix Figure A.5a. Here, the vertical axis shows counts of permits issued per parcel-year. We can see that in terms of volume, ADU permitting activity jumps when moving from the stringent Coastal Zone to the recently reformed area just outside of the Coastal Zone. In terms of raw counts of permits, activity more than doubles, but the density of parcels also increases across the boundary. In Sub-Figure A.5b, we also show how ADU permitting changes as a share of all building permits. In the more lenient non-Coastal Zone areas, ADU permitting is about 65% of all building permits issued, whereas inside the Coastal Zone ADU permitting is only about 45% of all permits issued. Marantz, Elmendorf and Kim (2023) also find that ADUs were a large share of overall permitting in California.

regardless of the amount of permitting activity. We explore how the post-2016 permitting reforms impacted land values using our transaction dataset and a difference in discontinuities design in Table 4.

Column 1 of Table 4 shows a specification for the impact of permitting reforms on land values without segment by year fixed effects. This column only uses a first-degree polynomial in distance to the Coastal Zone boundary, a county fixed effect, and a year fixed effect. In Columns 2 and 4, we add our pre-2016 hedonic fixed effects, which control for baseline characteristics of the parcel. Then in Columns 3, 4, and 6 we include segment by year fixed effects instead of year fixed effects to keep estimates local to the 145 local coastal segments. Finally, in the last two columns we include a parcel fixed effect and produce estimates for the repeat sale sample. Our preferred specification is the repeat sale model with segment by year fixed effects shown in Column 6. The estimates suggest that by-right permitting reform increased the value of land by about 7%. Our estimates range from 7% (Column 6) to 14% (Column 1), and all estimates are significant at the 1% level. The inclusion of segment by year fixed effects reduces the magnitudes of our estimates substantially. Intuitively, models without segment by year fixed effects are capturing variation both across Coastal Zone segments and within Coastal Zone segments. Recall from Figure 3 that our analysis sample includes a wide range of communities from San Diego to Crescent City, which may have local time-varying factors that interact with the value of the ability to build an ADU. In contrast, estimates that include segment by year fixed effects are only leveraging variation within a particular segment. As we can see in Figure 4a, limiting our comparisons to just within-segment variation reduces baseline differences in levels across parcels. Our repeat sale model takes this a step further and accounts for static differences across parcels with a parcel fixed effect. Comparing Column 5 to Column 1, we can see that static parcel characteristics, such as location, drive a large amount of the variation in these estimates. We can also see this by comparing Columns 4 and 6, where Column 4 controls for static parcel characteristics with a set of hedonic fixed effects and Column 6 controls for these characteristics with a parcel fixed effect. Both Column 4 and Column 6 show changes in land values of about 7%.

We can show this change in land values visually in Figure 6. The graph shows within-bin changes in land values for 25 meter distance bins from the Coastal Zone boundary. These within-parcel changes are calculated between 2010 and 2016 and between 2017 and 2024 on our repeat sale sample. The vertical axis shows the log difference of the real land value per acre in each bin between the pre-reform and post-reform periods. The effect of the permitting reforms can be seen as the jump in land value per acre changes just before and just after crossing the Coastal Zone boundary. The roughly 7% jump in land value changes visually corroborates our estimates in Table 4.[61] We also show year-by-year event studies from our preferred estimates that use a repeat sale model with segment by year fixed effects in Figure 7. The pooled pre-trend estimate is centered around zero, and shows a slightly positive and insignificant point estimate.[62] After the reforms

[61] The un-differenced version of this figure is available as Figure A.3.

[62] Note that Figure 4b doesn't use the repeat sale sample.

in 2017, the year-by-year estimates are too imprecise to say that any one year had a particularly large impact on land values, which might have suggested parts of the post-2017 reform package that were effective at raising land values. However, the event study does appear to suggest that several years after the onset of the COVID-19 pandemic saw larger increases in land values across the border. This might reflect increased demand for floor-space, which was common during the work-from-home era (Mondragon and Wieland, 2022). Alternatively, it may suggest that increased state-level enforcement of ADU reforms after 2019 may have had an impact on land values (Minott, 2023).

These results are consistent with previous findings from Ihlanfeldt (2007). He finds that increased land use regulations increase structure prices and decrease land prices. Our findings are the mirror image of Ihlanfeldt (2007); we find that decreased land use regulations increase land prices. Gold and Tarascina (2026) also find an option value of between 1.6% and 4% of the total sale price from permitting reforms for ADUs in Seattle, WA using a within-city difference in differences estimator. Although our estimates suggest that landowners who received deregulation for ADUs outside the Coastal Zone likely benefit from a higher value for their land, disamenities from increased density may negatively impact their total sale prices as well as their neighbors'.

5.3 Spillovers

The public concern with up-zoning is typically a discussion about spillovers. Neighbor A may use her property in ways that negatively affect the property value or utility of Neighbor B. Disamenities from density could arise from ADU construction through decreasing street parking, increasing utilization of local services, building something of low aesthetic value, or simply competing down rental rates for units. To investigate these spillovers, we first show how total sale prices, which include both land and structure values, change on a per acre basis across the Coastal Zone boundary.

We present our estimates of Equation 4 in Figure 8. This figure presents coefficients that capture how the (log of) real total sale prices per acre changed before and after the 2016 reform legislation. All coefficients are relative differences from changes in a reference bin. We use the bin farthest inside the Coastal Zone boundary and within our 450-meter bandwidth sample, $[-450, -400)$ meters. The horizontal axis shows the bin's (median) distance from the Coastal Zone boundary, with negative values in blue representing transactions inside the Coastal Zone that did not receive the permitting reforms after 2016. Estimates for bins outside the Coastal Zone that received permitting reforms are shown in orange. If spillovers are present, they should be shown in the coefficients just inside the Coastal Zone.

In Figure 8a we find no evidence of price spillovers into the Coastal Zone from permitting reforms. The four estimates just inside the Coastal Zone ($[-200, 0)$ meters) have indistinguishable changes in total sales prices per acre compared to our reference bin. By contrast, a strong jump in estimates, of between 4 and 9%, can be seen after crossing the Coastal Zone boundary. These are broadly consistent with our estimates for land value of a 7% increase in Table 4. Additionally, the estimates outside the Coastal Zone from $[0, 450]$ meters also capture cross-parcel spillovers,

suggesting that the net effect on price reflects mostly a change in land values rather than a decrease in structure values.[63] These results suggest that the option value likely dominates concerns over negative spillovers in this setting. Our findings are not consistent with Davidoff, Pavlov and Somerville (2022), who find evidence of negative cross-plot spillovers. However, Gold and Tarascina (2026) do not find cross-plot spillovers from ADU construction, which is similar to our results.

We next show how transaction volumes change across the boundary in Figure 8b. Here, we again find no evidence of spillovers. On the reform side of the boundary ($[0, 450]$ meters), we see transaction rate changes that appear similar to those in our reference bin. These results suggest that ADU reforms may not be associated with negative disamenities that induce owners to sell and exit. The magnitudes of the point estimates (-0.2 percentage points) are also small in a setting where about 4% of parcels transact at arm's-length each year.

5.4 Robustness

In addition to the robustness presented in our main tables, we provide several additional exercises to further probe our results. First, we explore how bandwidth selection and choices around the appropriate standard errors impact our results for both estimates of Equations 2 and 3 for ADU permits and land values, respectively. We present these alternative choices in Figure A.6, with ADU permit results sensitivity shown in Sub-Figures A.6a and A.6b and land value results sensitivity shown in Sub-Figures A.6c and A.6d. Panels A and C show our pooled estimates and Panels B and D show our preferred estimates. For both sets of permitting estimates, with and without county by year fixed effects, point estimates are remarkably stable at 0.6.[64] For land value estimates in Panels C and D, we see a slightly different pattern. In Panel C, point estimates are steadily increasing as the bandwidth gets closer to the boundary, with estimates from the 100-meter bandwidth reaching almost a 20% increase in land values. However, in Panel D using our repeat sale sample and segment by year fixed effects, we see much more stable estimates around 7%.

In all panels of Figure A.6, we show standard errors three ways. First, we show our preferred parcel-clustered standard errors. Then, we show segment-clustered standard errors in orange. Since the Coastal Zone is administered by 145 local coastal programs, if these segments differed substantially in their pre-reform treatment of ADUs, this would vary our treatment along the boundary and warrant clustering (Abadie et al., 2023). In all specifications, these segment-clustered standard errors are almost identical to the parcel-clustered standard errors. Finally, we also show spatial-correlation robust standard errors from Conley (1999) with a 200 meter cutoff in light blue. These standard errors are larger in our repeat sales model, and render estimates from bandwidths of 350 meters and smaller insignificant. Across all three standard error choices, our 450-meter bandwidth remains significant.

[63] We corroborate this with difference in discontinuities estimates using total prices per acre in Appendix Table A.4.

[64] We use county by year fixed effects here because segment by year fixed effects drop a number of segments without enough variation in permitting with bandwidths below 200 meters from the boundary. At 450 meters, the county by year fixed effects and segment by year fixed effects give almost the exact same point estimate.

Next, we probe the land value results in Table 4 in three additional ways. In Appendix Table A.5, we show estimates that use a second-degree polynomial. In general, these results increase both the point estimate and the standard errors associated with permitting reform. Even so, these estimates are qualitatively similar to those presented in Table 4.[65] Second, we explore trying to isolate changes in land values using updating hedonics in Appendix Table A.7. We estimate a modified version of Equation 3 where we use the log of the real total sale price per acre on the left-hand side and allow the binned hedonic fixed effects to update as characteristics of the structures on the property change over time. This exercise mimics the types of adjustments done by assessors when assigning land values in settings when structure values are known and comparable land sales are scarce (California State Board of Equalization, 2025*a*). This alternative approach again produces qualitatively similar estimates to those in Table 4 that are all significant at the 1% level. Finally, in Appendix Table A.8, we show that our results are robust to dropping parcels around the five cities that voluntarily updated their local coastal programs to implement the ADU reforms on both sides of the boundary (California Coastal Commission, 2022*b*).[66]

6 Theoretical Framework

Our main results show that even with low permitting activity, a large number of parcels can benefit from up-zoning through increased land values with few measurable spillovers. To rationalize these results and motivate heterogeneity, we build an open city model. A few relevant papers also use open city models to characterize estimators of discrete regulatory effects in regression discontinuity settings. Turner, Haughwout and van der Klaauw (2014) organize their model as two adjacent open linear cities and decompose the effects of regulatory boundaries on land prices into an own-lot and external effect that changes with distance to the boundary. Similarly, Gyourko and McCulloch (2024) characterize the effects of discrete changes in minimum lot sizes on home prices. In both of these models, density exposure decays as lots on the high-minimum-lot-size side of the border get farther from the denser jurisdiction. Finally, Severen and Plantinga (2018) build a model of how multi-family unit values and rents responded to the Coastal Zone's permitting regime. In their model, the sign of the price effect of regulation local to the boundary hinges on whether or not the effect of limiting construction by neighbors outweighs the increased compliance costs from regulation, assuming amenities are similar across the border.[67] Our model abstracts away from the running variable, which can be embedded in our variable "lot quality," but it endogenizes the lot owners' construction and sale decisions to characterize the sale price and creates an explicit "option value" from deregulation. This option value is capitalized in land above a certain lot quality, and incumbent landowners and entrants may value the ability to build an ADU differently.[68]

[65] We also show robustness to polynomial choice for our regression discontinuity results for permitting in Appendix Table A.6.

[66] These cities are Santa Cruz, Santa Barbara, Encinitas, San Mateo, and Pacifica.

[67] Severen and Plantinga (2018) note that if regulation reduced supply, then prices would increase for all lots. It is, in practice, hard to separate the aesthetic effect of neighbor's construction from the supply effects.

[68] Alternatively, we show how ADU reforms fit into a more traditional real options model in Appendix D.

Additionally, we show how California's property tax system can distort the choice to sell a parcel, the decision to build an ADU, and the value of constructing an ADU.

6.1 An Open City Model in Coastal California

Our model is built to resemble the cities in California that straddle the Coastal Zone boundary. We construct a single city with two sub-jurisdictions with different permitting costs, to mirror sections of the city within the Coastal Zone and sections of the city outside the Coastal Zone. We denote permitting costs, κ_m, for each jurisdiction as $m \in \{L, R\}$, where L represents the stringent Coastal Zone and R represents the reformed side of the boundary outside the Coastal Zone:

$$\kappa_L > \kappa_R \geq 0 \tag{7}$$

where permitting costs are always non-negative and strictly larger on the left side of the city.

The city starts with a mass of incumbent landowners, evenly dispersed across differentiated lots on the left and right sides of the city. We assume that each lot has one owner and is entitled to build a single ADU, abstracting away from existing structures. No lot starts with an ADU. Lots differ in their overall quality, $l \sim F_l$, which will determine the flow amenity that an owner gets from residing on the lot, $v(l)$, and whether or not the lot is suitable for building an ADU of flow value $rG(l)$ (r is a discount rate). Incumbent owners each face an ADU construction cost, $k \sim F_k$, with a minimum cost of k_{min}. The costs to the owner of building an ADU include the costs of building the ADU (labor, materials, etc.) as well as costs to the owner like hassle costs of contracting, partial losses of their lot values during construction, and other idiosyncratic preferences over the process of building an ADU. Together, the construction costs, k, and permitting costs, κ_m, give the total one-time costs to the owner of building an ADU.

Owners who construct an ADU get utility from it. The net present value of building an ADU, $G(l)$, is a function of lot quality. Here, the value of an ADU is net in the sense that it includes changes that an ADU might induce to the value of the rest of the lot, such as taking away yard space or impacting existing structure values. Following California's property tax rules, when an owner builds an ADU, she pays the property tax rate, τ, on the addition. The flow tax value paid on an owner-built ADU is $\tau G(l)$. Therefore, the value to the owner of the ability to build an ADU, with permitting costs κ_m, on lot quality l, and facing construction costs k, can be written as:

$$\begin{aligned} OwnOpt_m(l,k) &= max\{(1-\frac{\tau}{r})G(l) - k - \kappa_m, 0\} \\ &= max\{\bar{k}_m(l) - k, 0\} \end{aligned} \tag{8}$$

where $OwnOpt_m(l,k)$ is the owner's option value of an ADU. Notice that an owner only values this right positively when $OwnOpt_m(l,k)$ is greater than 0, which implies that construction costs are low enough such that $\bar{k}_m(l) > k$.[69] Therefore, the owner will only build an ADU if her construction

[69] We also need to assume that $\tau < r$ so $(1 - \frac{\tau}{r}) > 0$.

costs are below the threshold, $\bar{k}_m(l)$.

We can now express the incumbent owners' welfare from remaining on their lot. In addition to the flow amenity value of the unbuilt lot, $v(l)$, the owners have idiosyncratic attachments, $\alpha \sim F_\alpha$, to remaining on their particular lot. These attachments can be thought of as a distaste for moving or an attachment to their particular location. The owner also benefits from a public service, $u(g)$, which is provided by the city and financed through the head tax, h, and property taxes.[70] Following Proposition 13, owners benefit from a below-market property tax assessment base, $B \sim F_B$, and the base is taxed at the state-set rate τ. We model the below-market base as a random draw, but in practice the gap between the market value and assessed value of a property grows with owner tenure when market values increase more than the 2% cap. The incumbent owner also has a distaste for new ADUs, which we model as $D = \phi N$, where θ is a preference parameter for density, and ϕ is a parameter that converts ADU construction, N, into density D. With these features, we can write the owner's welfare, $W(l, \alpha, B, k, m)$ as:

$$W(l, \alpha, B, k, m) = \underbrace{\frac{v(l) + \alpha + u(g) - h - \theta D - \tau B}{r}}_{\text{Value of living on the incumbent's lot}} + OwnOpt_m(l, k) \tag{9}$$

which is the present value of living on the lot to the incumbent owner plus the owner's option value of building an ADU. Embedded in this equation, we are assuming that distaste for density is not side specific, but impacts all incumbent owners equally. Similarly, the public services and head tax apply to all incumbents equally, and they are taken as given by the owners. Notice that the equal value of ADU disamenities, D, across the boundary is consistent with our empirical findings of limited price and transaction spillovers.

6.2 A Market for Land and ADUs

So far, we have set up the initial conditions of our open city model. In this subsection, we will set up a market and introduce entrants and firms that can purchase lots from owners. Outside of the city, there is a deep pool of potential entrants that bid competitively for lots. The utility of living outside the city is given by O, and is valued by incumbents and entrants equally. Within each side of the city, entrants are identical, differing only by their side-specific attachment α^e_m.

We will start by solving for the price of a lot where the option to build an ADU is zero. This can be thought of as a lot pre-ADU reform, or simply a lot with a low ADU-quality, l, such that $OwnOpt_m(l, k) = 0$ at any cost k. Entrants value a lot within the city as follows:

$$W^e = \frac{v(l) + \alpha^e_m + u(g) - h - \theta D - \tau P_{0,m}}{r} \tag{10}$$

where α^e_m is the attachment an entrant places on a lot on side m. In our model, α^e_m can be thought of as pre-reform price differences across the boundary. The entrant also places the same value

[70] The function $u(\cdot)$ is assumed to be well-behaved, such that $u'(\cdot) > 0$, $u''(\cdot) < 0$, and Inada conditions are respected.

on the public service as the incumbent, although there is some evidence from the capitalization literature that this may need to be relaxed (Hilber, 2017). Notice that the entrant pays property tax on the sale price of the lot, $P_{0,m}$, when it is purchased. Here, $P_{0,m}$ is the baseline price of the lot without the ability to build an ADU.

Entrants will buy a lot in the city if the value of the lot, less the price of the lot, is at least as valuable as the outside option. However, here we assume that entrants cannot build ADUs themselves. To solve for the baseline price, we can set the entrant's value from Equation 10 equal to the price of the lot and the value of the outside option and solve for the baseline price.

$$\begin{aligned} &\frac{v(l) + \alpha_m^e + u(g) - h - \theta D - \tau P_{0,m}}{r} = P_{0,m} + O \\ \implies P_{0,m} = &\underbrace{\frac{v(l) + \alpha_m^e + u(g) - h - rO}{r + \tau}}_{V_m(l)} \underbrace{- \frac{\theta D}{r + \tau}}_{\text{Density Discount}} \end{aligned} \tag{11}$$

We can then break up this price into two parts, which are convenient for presentation purposes. The left-hand piece is the market value of the lot without any density spillovers from ADU construction, $V_m(l)$. Then, the right-hand piece is the amount by which density from additional ADUs decreases the market price. These two pieces make up the market price of the incumbent's lots, which will be directly comparable to their assessed property tax bases, with $B \leq V_m(l)$.

For simplicity, we will assume that a perfectly competitive firm purchases the lot and builds the ADU at the lowest cost available, k_{min}, and with no profit.[71] When an owner sells a lot, if the lot is in the buildable lot quality set for side m, $Q_m = \{l : \frac{rG}{r+\tau} > k_{min} + \kappa_m\}$, then it is purchased by a construction firm that builds an ADU.[72] This simplicity allows us to pin down our market price for the right to build an ADU:

$$\Omega_m(l) = max\{\frac{r}{r + \tau} G(l) - k_{min} - \kappa_m, 0\} \tag{12}$$

where $\Omega_m(l)$ is the market option value. If the lot is not within the buildable set, $l \notin Q_m$, the market option value is zero. When the competitive firm buys an unbuilt lot from an incumbent, the incumbent gets the baseline price plus the market option:

$$P_m(l) = P_{0,m}(l) + \Omega_m(l). \tag{13}$$

Notice that when the lot has a quality outside the buildable set then the baseline price is equal to the price the firm would pay for the lot, $P_m(l) = P_{0,m}(l) + 0$. Here, $\Omega_m(l)$ capitalizes in the land, and $P_m(l)$ gives the market price of the land including the market value of the ADU build option and the preexisting structures on the lot. Since we are abstracting away from preexisting structures, $P_m(l)$ corresponds to the land value per acre.

[71] This is equivalent to a free entry condition.

[72] Note that $\kappa_R < \kappa_L \implies Q_L \subset Q_R$.

Lots with a firm-constructed ADU are sold to the pool of entrants.[73] Since the entrant will pay property tax on the full value of the ADU, and on the lot value, the price of the lot with a constructed ADU is given by:

$$P_m^{Built}(l) = P_{0,m}(l) + \frac{rG}{r+\tau} \tag{14}$$

where $\frac{r}{r+\tau}$ reflects the present value of an ADU, net of taxes.[74]

Entrants can buy either low-quality lots ($l \notin Q_m$) that a firm cannot profitably build on or higher- quality lots that can support an ADU ($l \in Q_m$). The deep pool of entrants bids competitively in both markets, which drives the prices to where the entrants are indifferent between the outside option O. Therefore, $P_{0,m}(l)$ and $P_m^{Built}(l)$ give us an entrant's indifference curve across lot types. The sold lots within the buildable set Q_m will all be purchased by a firm, built to include an ADU, and then sold to an entrant. The remaining lots that transact are sold to entrants without an ADU.

We now construct the incumbent owner's decision to stay or sell her lot. If the incumbent decides to stay, she can build an ADU as described above. The incumbent is indifferent between selling the lot and staying when her welfare from staying, W, is equal to the price of the lot plus the outside option, $P_m(l) + O$. Rearranging, we can express this in terms of an attachment threshold:

$$\alpha_m^*(l,k,B) = \alpha_m^e - \underbrace{\tau(V_m(l) - B)}_{\text{Prop-13 Lock-In}} + \underbrace{\frac{\tau}{r+\tau}\theta D}_{\text{Density Push-out}} + \underbrace{r[\Omega_m(l) - OwnOpt_m(l,k)]}_{\text{Option Wedge} \geq 0} \tag{15}$$

where $\alpha_m^*(l,k,B)$ is the cutoff attachment at which incumbents with $\alpha > \alpha_m^*(l,k,B)$ stay, and incumbents with $\alpha < \alpha_m^*(l,k,B)$ sell. $\alpha_m^*(l,k,B)$ has four main components. First, higher entrant demand, α_m^e, will induce more owners to sell. Second, owners with more property tax shelter, $\tau(V_m(l) - B)$, are less likely to transact. This lock-in effect has been shown empirically in previous work by Wasi and White (2005) and Ferreira (2010). Third, additional density will push more owners to sell. Finally, when there is a gap between the market value of the option and the owner's option of building an ADU, the incumbent will be incentivized to sell and realize that value. Since the market can build at lower costs than the owners, on buildable lots this wedge is greater than zero. The gap between the market and owner's options is greatest when $k > \bar{k}_m(l)$ and only the market can build a viable ADU on the lot. For unbuildable lots, this wedge is equal to 0 and the indifference condition doesn't depend on build costs, k.[75]

In this model, there are two channels through which ADUs are constructed. In the first channel, the owner builds, which requires both a high enough attachment $\alpha > \alpha_m^*$ so the owner doesn't sell

[73] We assume that the firm holds the lots with built ADUs only momentarily and therefore they do not flow into the property rolls until the entrant buys the lot and ADU bundle.

[74] Here, the assessment rules under Proposition 13 create a wedge between the value of a purchased ADU to the entrant and the value of a constructed ADU to the owner,$(\frac{r}{r+\tau}) - (1 - \frac{\tau}{r}) = \frac{\tau^2}{r(r+\tau)} > 0$.

[75] We show this threshold graphically in Appendix Figure A.7. For lots with buildable quality, the function $\alpha_m^*(l,k,B)$ is kinked.

her lot and a low enough build cost to benefit from the final ADU $k \leq \bar{k}_m$. The second channel is through market construction, where a firm builds an ADU after the incumbent owner sells the lot. We can then express the total number of ADUs built in permitting jurisdiction m as:

$$N_m = \iiint \mathbb{1}\{l \in Q_m\} \Big[\underbrace{(1 - F_\alpha(\alpha_m^*))\mathbb{1}\{k \leq \bar{k}_m\}}_{\text{Owner stays and builds}} + \underbrace{F_\alpha(\alpha_m^*)}_{\text{Market Builds}} \Big] dF_l dF_k dF_B \tag{16}$$

where N_m is the number of built ADUs on side m, and $N = N_L + N_R$. First, *ceteris paribus*, lower permitting costs should induce more ADU construction through both channels. Second, transactions drive ADU construction since the market can build at k_{min}. This is a simplification. Empirically, more than half of ADUs constructed are built by incumbent owners, as shown in Appendix Figure A.8. However, transactions are likely important for building, as lots that build an ADU are most likely to transact just before a permit is pulled. Appendix Figure A.9 shows that in the year before an ADU permit is pulled, about 15% of lots that pulled an ADU permit transacted, compared to the average transaction probability for a lot that receives an ADU permit of about 6%. The data suggest that a third channel, entrants building on newly purchased lots, is likely empirically important, although more complex than our exposition here.

This model is set up as a Stackelberg game. First, nature assigns owners, characterized by attachment (α), their ADU build costs (k), and their below market assessed values (B) to lots of various qualities (l). Next, the city selects the head tax, h, to provide the government service g subject to a balanced budget that includes head tax and property tax revenues. With perfect foresight, the city chooses h to maximize welfare, which is equivalent to maximizing incumbent welfare since the outside option pins down entrant welfare. Once the city selects h, incumbents and entrants trade parcels and build ADUs until an equilibrium is reached. Critically, all variables involving the city are independent of the permitting side m, and will not drive differences across the boundary. Therefore, we show the full solution to the city's problem in Appendix E.

6.3 Differences Across the Permitting Boundary

In this subsection, we will characterize our regression discontinuity estimates for permitting activity and our difference in discontinuities estimates for land values. In this simple model, we have built in the assumptions that before permitting reform, lots did not build ADUs and there was no valuable option. Therefore, we will show both estimates as simple differences across the permitting boundary. Our estimates for permitting activity are:

$$\beta^{Permits} = 2(N_R - N_L) \tag{17}$$

where $\beta^{Permits}$ corresponds to our regression discontinuity.[76] The difference in building activity, $N_R - N_L$, is largely mediated by the transaction cutoff, α_m^*. When we compare α_m^* across the

[76] Recall that each side has $\frac{1}{2}$ a measure of lots. In practice, this will not be the case and the estimates of $\beta^{Permits}$ will be the difference in average permitting rates across the two regions with differing numbers of lots on each side.

boundary, $\alpha_L^* < \alpha_R^*$ holding l, k, and B constant, we see that the market option wedge is larger when permitting costs are lower. This will increase ADUs through the market channel. The owner option channel is larger as well, as $\bar{k}_L < \bar{k}_R$ when permitting costs are higher on the un-reformed side of the boundary ($\kappa_L > \kappa_R$). Finally, the buildable set of lots includes more lot qualities on the reformed side of the boundary with lower permitting costs. However, several forces can also mute take-up generally. For example, low lot qualities, high building costs, high incumbent attachments, and high incumbent tax shelter can all depress overall ADU take-up in the model while still allowing for a discontinuity in building activity.

We now turn to our difference in discontinuities estimate for land values. In our model, the corresponding land values to our empirical estimates are $P_m(l)$. Since pre-existing differences in prices are differenced out of our difference in discontinuities estimator, our difference in land values will be the average difference in market options across the quality distribution on both sides.[77] We can write the difference in discontinuities estimator as a piecewise function in lot quality:

$$\begin{aligned} \beta^{Land} &= \Big[\Omega_R(l) - \Omega_L(l)\Big] \\ &= \begin{cases} 0, & \text{if } l \leq l_R, \\ \Big[\dfrac{r}{r+\tau}G(l) - k_{min} - \kappa_R\Big], & \text{if } l_R < l \leq l_L, \\ \Big[\kappa_L - \kappa_R\Big], & \text{if } l > l_L. \end{cases} \end{aligned} \tag{18}$$

where β^{Land} has three distinct cases.[78] In the first case, where no lot has a high enough quality to build even on the reformed side, the difference in land values should be 0. In the second case, when lot qualities are such that building is only viable on the reformed side of the boundary, our empirical estimates will reflect the full market option value of an ADU. Finally, in the third case when ADU construction is viable on either side of the boundary, the value of the ADU itself and construction costs cancel out. Critically, we can separate cases 2 and 3 by whether or not our estimates show heterogeneity in lot quality.[79] We show how β^{Land} changes with lot quality graphically in Appendix Figure A.10.

Since the estimates of β^{Land} reflect the functional form of the market option, $\Omega_m(l)$, our model shows that the land values do not require a built ADU to show up in land prices. Simultaneously, ADU construction can be depressed by property tax shelters, high construction costs to the owners, and high owner attachment, while the market value of the option is pinned down by the firm that builds at $k_{min} < k$ and embedded in every parcel that transacts.

[77] In our model, pre-existing differences would arise from α_m^*, which can be seen as a side-specific demand shifter.

[78] Note that the pre-reform baseline difference is $\frac{\alpha_R^e - \alpha_L^e}{r+\tau}$, which our difference in discontinuities estimate accounts for directly.

[79] We write our model in levels, and estimate our difference in discontinuities in logs. In logs, we write: $\beta^{Land} = ln(1 + \frac{\Omega_R}{P_{0,R}}) - ln(1 + \frac{\Omega_L}{P_{0,L}}) \approx \frac{\Omega_R}{P_{0,R}} - \frac{\Omega_L}{P_{0,L}}$. We find no detectable level difference in land values at the boundary. In this case, $\frac{\kappa_L - \kappa_R}{P_0(l)}$ requires us to assume that permitting costs scale with lot quality, for example: $\kappa_m = c_m P_0(l)$. For simplicity, we assumed that permitting costs were a fixed dollar amount, but in our estimates they likely reflect carrying costs which scale with lot value.

6.4 Motivating Heterogeneity

Our model motivates a few heterogeneity exercises for both our permitting results and our land value results. Our theoretical representations of our permitting and land value estimates suggest that we may see heterogeneity by lot quality. For permits, $\beta^{Permits}$ changes ambiguously with lot quality, as N_m has components that vary with lot quality that push ADU construction in different directions. For example, tax shelter, $V_m(l) - B$, is increasing in lot quality and drives down transaction rates, and in turn construction, whereas the market option $\Omega_m(l)$ pulls in the opposite direction.[80]

Equation 18 suggests that heterogeneity in lot quality can help us characterize our difference in discontinuities estimates. If we find heterogeneity in lot quality, then our estimates are in the middle case, and we are capturing the full market option value. On the other hand, if we do not find heterogeneity by lot quality, then our estimates capture only the local differences in permitting costs across the boundary, changing the interpretation of our estimates.

Our model also suggests that the property tax shelter of each parcel may be important for understanding why ADU construction was so low. If parcels are sheltered by assessed values below market, then they should transact less. However, the property tax shelter shouldn't show heterogeneity in transactions in our model, as it enters additively in Equation 15 and doesn't differ by side. However, it may depress transactions generally, and therefore shut down the market building channel for ADU construction. This should decrease the number of ADUs as incumbent owners build less frequently than the market in this model.

7 Discussion

7.1 Heterogeneity

7.1.1 Does Lot Quality Matter?

We investigate how lot quality, as measured by a parcel's distance to the nearest CBD, and suitability for an ADU, as measured by excess land on the lot, may influence the impacts of permitting reforms. Our theoretical model in Section 6 suggests that lots of higher quality for an ADU should build them at higher rates. Additionally, the piecewise nature of Equation 18 implies that if we are capturing the full option value for just the reform side of the boundary, then our difference in discontinuities estimates should increase with lot quality. If we do not see heterogeneity by lot quality, then our estimates are capturing differences in permitting costs across the Coastal Zone boundary.

Our conditional average treatment effect estimates by lot quality are presented in Table 5. We present estimates in odd columns for our pooled specification interacted with the heterogeneity

[80] To clarify, Equation 17 has regions by lot quality. When lot quality is only high enough to build on one side, the estimate gives $N_R(l)$, which has an ambiguous derivative in lot quality. In the case where building can happen on both sides, our estimates will decrease, since $N_R > N_R - N_L$, but the derivative with respect to lot quality for $N_R - N_L$ is also ambiguous.

variable of interest following Equation 5. Then, in the even columns we present our preferred specification with segment by year fixed effects. Panel A Column 2 presents our preferred heterogeneity specification showing how ADU reform impacts on permitting activity vary by excess square footage on the lot. The coefficient on the Coastal Zone boundary dummy is interpreted here as the treatment effect of permitting reform on lots with below 4,226 sq. ft. of available space on their lot for an ADU. The coefficient is significant at the 5% level and comparable to our main estimates in Table 3, 0.618. Although our coefficient on the interaction of the Coastal Zone dummy and the excess lot square footage dummy suggests that lots with additional space permit additional ADUs, the coefficient itself is insignificant. the coefficient itself is insignificant. The insignificant interaction likely reflects the fact that the majority of lots in our sample are suitable for an ADU, according to Table 1.

Estimates in Panel A Column 4 of Table 5 suggest that lots within 26 kilometers (16.2 miles) of a CBD have larger treatment effects from permitting reform. Our coefficient on the Coastal Zone boundary dummy suggests that permitting reform within 26 kilometers of a CBD increased permitting activity by almost 1 unit per thousand parcel years. These effects are significant at the 0.1% level and are almost 50% larger than our preferred estimates in Table 3. Moreover, the interaction term coefficient suggests that for parcels more than 26 kilometers from a CBD, the impacts of permitting reform were just 30% of our main estimates. This suggests that a large component of lot quality for an ADU is related to access to urban agglomerations. Moreover, if smaller lots are in general located near CBDs, then Table 5 suggests that a lot's relationship to urban agglomerations may be a more important measure of quality than unbuilt area on the lot, which is abundant in our sample.

We present our estimates of heterogeneity in treatment for land values by lot quality in Panel B of Table 5. In Columns 1 and 2 we show difference in discontinuities with heterogeneity using the above and below median excess lot square footage dummy. In Columns 3 and 4 we show estimates that use above- and below-median dummies for distance to the nearest CBD. Odd columns use our pooled specification without parcel fixed effects and segment by year fixed effects, and even columns use our preferred specification with a parcel fixed effect and segment by year fixed effects.[81] Our preferred estimates in Columns 2 and 4 for the coefficient on the Coastal Zone boundary dummy are positive and significant at the 5% level. The coefficients are also consistent in magnitude with our main estimates, showing point estimates around 6%. The interaction coefficients for both excess lot square footage and distance to the CBD show null effects. The null heterogeneity estimates and theoretical model suggest that our repeat sale estimates with segment by year fixed effects are likely detecting mostly differences in permitting costs across the boundary rather than full market option value, consistent with the third case in Equation 18.[82]

[81] Since the sample varies between Panels A and B, the medians used to build the binary heterogeneity variables change with the sample.

[82] Moreover, in the overwhelming majority of Coastal Zone segments there is ADU permitting on both sides of the boundary.

7.1.2 Do Property Tax Shelters Mitigate the Impacts of Permitting Reform?

Our second set of heterogeneity exercises unpacks how property taxes interact with ADU permitting through their impacts on transactions. The owner's stay-or-sell indifference condition, Equation 15 in Section 6, suggests that additional property tax shelter should decrease the likelihood that a parcel transacts equally on both sides of the boundary, ceteris paribus. However, both our model and Appendix Figure A.9 suggest that ADU permitting typically occurs around a transaction event. Across all parcels, we see that additional property tax shelter is correlated with both lower transaction rates and lower permitting rates for ADUs in Figure 9.

We display our heterogeneity estimates exploring how property tax shelter influences the effects of permitting reform on transactions in Table 6 Panel A. Columns 1 and 2 interact the Coastal Zone boundary dummy with the 2016 estimated property tax shelter for each parcel. Our preferred estimate is in Column 2, which includes segment by year fixed effects.[83] The coefficient on the Coastal Zone boundary dummy suggests that parcels that were assessed at market value in 2016 saw no difference in transaction volumes compared to similar parcels inside the Coastal Zone. This effect is not significant at traditional levels and is small in magnitude. The interaction term suggests that as tax shelter increases, transaction volume differences barely move. In the entire sample, about 4% of parcels transact each year. These estimates support the functional form of Equation 15, which suggests that property tax shelter decreases transactions independently of permitting reforms.

Table 6 Panel B shows heterogeneity estimates for how the property tax interacts with treatment effects for permitting activity. In Column 1 we show our pooled regression discontinuity design fully interacted with the property tax shelter in 2016. In Column 2 we show our preferred specification that adds segment by year fixed effects. The coefficient for the Coastal Zone border dummy here captures the effect of permitting reform on ADU permitting for parcels assessed at market value (no shelter). The estimates show that these parcels with no tax shelter in 2016 build ADUs at higher rates between 2018 and 2024. The coefficient is larger than our main estimates and significant at the 5% level. The coefficient on the interaction term suggests that parcels with more tax shelter permitted fewer ADUs, and the estimate is of a meaningful magnitude, but the coefficient is not significant at traditional levels. These estimates suggest that property tax shelter may decrease ADU permitting, and mute the impacts of permitting reform to an extent. However, it is difficult to disentangle property tax shelter from preferences that reduce mobility and ADU construction that likely correlate with sufficient amounts of tenure to accrue large amounts of tax shelter on a parcel.

We show our estimates of how property tax base mobility for owners over 55 impacted the effects of permitting reform on transaction rates in Panel A of Table 6. Columns 3 and 4 show pre-COVID estimates, and Columns 5 and 6 show estimates for all available years, including the COVID years. Column 4 shows our preferred pre-COVID heterogeneity specification which uses segment by year fixed effects. These estimates suggest that a parcel in a block group with no homeowners above

[83] Note that for transactions we can now always include a parcel fixed effect.

the age of 55 would be about 1.3 percentage points less likely to sell their home or transact the parcel compared to a similar parcel within the Coastal Zone. The coefficient is significant at the 5% level.[84] The interaction term of the share of owners above 55 and the Coastal Zone boundary and post dummy shows a strong upward slope in transaction rates with block group age before COVID. The coefficient implies that for every additional 10 percentage point increase in the share of area owners over the age of 55, the transaction rate increases by 0.3 percentage points. This coefficient is significant at the 5% level. However, the full sample estimates from 2010 to 2024 in Column 6 reflect an overall increase in transactions with additional exposure to Propositions 60 and 90, but the interaction term becomes insignificant. Together, these results suggest that allowing wider property tax base transfer rights could have increased transactions post-reform.

Our estimates in Panel B of Table 6 will shed light on whether or not access to property tax mobility impacted the effects of ADU reforms on permitting activity. Across specifications, pre-COVID and on the full sample, in Columns 3 through 6, we see that the coefficient on the interaction term between the share of owners over 55 and the Coastal Zone dummy is negative and insignificant, showing no detectable heterogeneity in treatment by exposure to Propositions 60 and 90. These estimates could be consistent with differing preferences over density and ADUs across age groups, rather than property tax mobility, as influencing transactions and permitting. Taken together, the evidence in Panels A and B shows that parcels in block groups with more owners over 55 may have transacted more, but this likely didn't lead to additional ADU permits as a result.

Overall, the evidence that the property tax lowered ADU permitting by depressing transaction volumes is only suggestive. The best evidence comes from the heterogeneity using the property tax shelter itself. These estimates support the conclusion that property tax shelter alone did not meaningfully impact the effects of permitting reform on transactions, but the least sheltered properties did permit ADUs at higher rates. This pattern is supported in the aggregate by Figure 9.

7.2 Why Was Permitting So Low?

We have explored treatment heterogeneity in permitting by distance to the CBD and property tax shelter. Although it might be the case that low take-up is the result of a lack of suitable lots or that Proposition 13 is depressing ADU construction, the presence of option value in the land points to alternative mechanisms.

The literature on option value in housing development typically finds that option value is associated with low rates of development. Cunningham (2007) studies growth controls around Seattle and finds that when the uses for lots are constrained, the option value of waiting to develop decreases and more owners develop their land. In a low-demand setting, Brooks, Rose and Veuger (2024) find that owners of lots in Washington, DC destroyed by the riots of 1968 allowed these lots to remain vacant for decades. They rationalize this finding with an option story: owners delayed development until uncertainty over future prices decreased. In contrast, Hornbeck and Keniston

[84] It should be noted that there are no block groups with zero owners above the age of 55 in our dataset.

(2017) find that large redevelopments can have positive spillovers that incentivize redevelopment of nearby parcels. There are also several other papers that view real estate development from an options perspective (Titman, 1985; McMillen and O'Sullivan, 2013; Yao and Pretorius, 2014).

Our theoretical model in Section 6 and a simpler real option model we show in Appendix D both point to take-up as a function of the expected benefits from an ADU and the costs of building an ADU. Permitting reform directly lowers one type of cost: the cost, in time or money, of getting the permission to build an ADU. However, permitting reform cannot address construction costs, which can be substantial. Industry quotes for a new ADU range from $200 to $300 per square foot (Henson, 2023). Construction costs have been rising generally, but we are unable to directly incorporate them into our analysis in this paper (Gyourko and Saiz, 2006; Brooks and Liscow, 2023; Eriksen and Orlando, 2024).

A second channel that may dampen construction and increase option value is uncertainty about future rents. In the open city model this would be equivalent to assigning a distribution to how ADU value might scale with lot quality. In the traditional option model in Appendix D, uncertainty is modeled as a distribution across a high state and a low state in the future. Increased uncertainty over the future value of an ADU should decrease development and increase the value of waiting to invest.

There are several reasons why landowners in California may believe that their ability to benefit from their ADU is uncertain. First, California in 2019 passed state-wide rent control, and several cities also have their own forms of rent control, which may lower the supply of ADUs for rent (Asquith and Reed, 2021; Diamond, McQuade and Qian, 2019). Second, a number of jurisdictions also impose restrictions on short-term rentals, which might reduce the number of ADUs permitted as a potential source of rental income is restricted (Koster, van Ommeren and Volkhausen, 2021; Seiler, Siebert and Yang, 2024; Bibler, Teltser and Tremblay, 2025). We will leave investigating the role of rental income uncertainty on ADU construction for future work.

7.3 Fiscal Impacts

Although the permitting reforms may not have put an ADU in every backyard, they did increase the value of these backyards, which could have significant fiscal impacts through property taxes. If the land values across the state have increased, then jurisdictions that rely on the property tax may see revenue gains through two channels under Proposition 13. First, as ADUs get built, assessments on the new property will add the value of the ADU to the property tax base without reassessing the whole parcel. Second, when a parcel transacts, the entire parcel gets reassessed, land and structures. Both of these channels are impacted by ADU permitting reforms, as shown in our theoretical model and in the empirical results above. The overlap of these channels makes a simple back-of-the-envelope estimate of the impacts of permitting reforms on property tax collections non-trivial.

We estimate the impact of ADU reforms on property tax collections in Table 7. Here, we estimate Equation 3 after replacing the outcome variable with the log of real property taxes paid

per acre. Our preferred specification is in Column 6, which includes segment by year fixed effects and a parcel fixed effect. Our estimates suggest that permitting reforms for ADUs increased property tax revenues per acre by 1.3%, and our estimates are significant at the 1% level and qualitatively robust to alternative specifications.

Our results are broadly consistent with previous work on the relationship between home values and property tax collections. Lutz (2008) finds an elasticity of property tax receipts to total sales prices of 0.4.[85] In order to compare our setting directly to Lutz (2008), we estimate our difference in discontinuities models for total sale prices per acre in Table A.4. Our preferred estimates in Column 6 show an increase in total sales prices per acre of 3%, which is consistent with our land value estimates of 7%. If we take the ratio of our property tax estimates in Table 7 and our total sales price estimates in Table A.4, we get an implied elasticity of about 0.4, matching Lutz (2008).[86] Our results show that state preemption policies can impact local budgets through the link between property values and property tax collections, even in a setting with high levels of property tax shelter.

However, from these revenue increases alone, we cannot draw conclusions about the overall impact of ADU reforms on local governments' fiscal health. The net fiscal impacts could be positive if the composition of residents by demand for local services (police, fire, education, etc.) remains constant and the resident population stays constant. However, our estimates suggest that parcels in older areas, at least before COVID, were more likely to transact in response to permitting reform. This may suggest that these areas are getting younger, which may come with a shift in preferences for spending and associated changes in local expenditures that may shift the fiscal gain in revenues into a net-neutral fiscal impact or even a fiscal loss (Poterba, 1997; Hilber, 2017).

7.4 Rents

Basic economic models and empirical evidence suggest that more elastic housing supply should reduce rents (Glaeser, Gyourko and Saks, 2006; Molloy, 2020; Molloy, Nathanson and Paciorek, 2022). Naturally, since our estimates suggest that permitting reform for ADUs led to additional permitting, the open question is whether or not permitting reforms reduced rents. We use an instrumental variables approach based on the Coastal Zone boundary and block group data on gross rent from the ACS 5-year samples from 2016 and 2024 to probe this question (U.S. Census Bureau, 2016, 2024). We use the share of block group area that overlaps with the Coastal Zone as an instrument for the cumulative change in ADU permits from 2018 to 2024 per housing unit in 2016. We then use this instrument to show how changes in ADU permits impacted percent changes in block group gross rents between 2016 and 2024. Our first stage F-statistic for our preferred specification with controls using block groups within 2,000 meters of the Coastal Zone boundary is 17. We show details about these estimates in Appendix F.

[85] However, the relationship between home prices and tax revenues may not be symmetric for both positive and negative changes in home prices (Doerner and Ihlanfeldt, 2011).

[86] Our estimates in Appendix Table A.4 also suggest that on the reform side of the boundary, price increases from land values likely outweighed any negative spillovers from constructed ADUs.

Our estimates in Appendix F suggest that permitting reform for ADUs had no detectable impact on area rents. For the average block group in our sample, rents increased between 2016 and 2024 by 28%. Our point estimate suggests that a one standard deviation increase in ADU permits (16.5 permits per 1,000 units in 2016) decreased rent changes by 1.2 percentage points. These results are imprecise and not statistically significant at traditional levels. These estimates are only suggestive, as the low uptake of ADU permitting as described above is likely the cause of the imprecision of these, and related, estimates.[87]

The evidence presented here is not sufficient to conclude that ADU reforms didn't lower rents. However, given the low take-up associated with ADU permitting, about 176,000 units statewide since 2017, it is unlikely that ADUs had a large effect on rents at the market level. However, Appendix Figure A.11 suggests that there are some areas with quite concentrated clusters of ADU permitting. It might be the case that ADUs meaningfully decreased rents in jurisdictions where ADUs made up a significant portion of the supply of units. Outside of our Coastal Zone boundary analysis sample, the San Fernando Valley and Bakersfield appear to have enough ADU permitting as a share of existing pre-reform units to see meaningful rent declines.

8 Conclusion

Who benefits from permitting reforms for ADUs? We deployed a plausibly exogenous boundary discontinuity that captures discrete changes in where permitting reforms applied. Our results suggest that reforms increased permitting activity by about 30%, local to our boundary. However, the baseline permitting rate is so low that only about 1 in 250 parcels was induced to permit an ADU. Even with low take-up, the ADU reforms increased land values by 7%. This increased land value applied to every reformed parcel, whether the owner applied for an ADU permit or not. On average, the increase was about $500,000 per acre ($100,000 on a 0.2 acre lot) and it accrued to incumbent landowners, who could either exercise their option to build an ADU or monetize the benefits by selling their parcel. Local and county governments also saw about 1.3% higher property tax collections, although the full net fiscal impact depends on several factors outside the scope of this paper. We do not find evidence of significant negative spillovers from permitting reform, which is consistent with our finding of low overall take-up. Our results do not support the finding that permitting reforms for ADUs lowered rents, but these estimates are imprecise.

Our simple open city model characterizes the increase in land values with permitting reform as a step function in lot quality. In an environment where it is only economically viable to build on one side of the boundary, our difference in discontinuities estimates should capture the full market option of building an ADU and exhibit heterogeneity by lot quality. However, if it is viable to build on both the low permitting cost side and high permitting cost side of the boundary, then our estimates will capture the difference in permitting costs and not show heterogeneity in lot quality. Our results suggest that we are in the latter case, and the value of permitting reform for ADUs was

[87] Bhatt (2023) also uses ACS data at the tract level and finds no evidence that ADUs lower rents in major California cities using a difference in differences approach that compares single-family zoned tracts to multifamily zoned tracts.

7% of land values or a 3% increase in total sale prices (land and structures). Our estimate for the value of permitting reforms is meaningfully smaller than recent estimates from Soltas and Gruber (2026), who estimate the value of circumventing permitting entirely.

Although permitting reform benefited landowners, it likely did not induce take-up at a scale large enough to broadly reduce rents. High construction costs, of about $200 to $300 per square foot, and uncertainty about future rental income may be factors that depress take-up, although exploring both factors empirically is left for future work (Henson, 2023; Seiler, Siebert and Yang, 2024; Bibler, Teltser and Tremblay, 2025). Permitting is only one of many barriers that may be reducing the construction of ADUs, or any type of unit (D'Amico, Soltas and Wang, 2026).

Our results speak to why gentle, or "light-touch," up-zoning for small units has been so popular. Nationally, as of April 2026, 96 land use bills for ADUs have been passed across 24 states since 2016 (NYU Furman Center, 2026). Homeowner resistance to up-zoning typically reflects a desire to maximize home values (Fischel, 2001). However, ADUs do not lower home values as some voters fear: they raise home values, and too few units are built for negative spillovers to manifest, at least in our data.[88] Our results should be interpreted as local to our setting: the California coast. We also are only able to measure permitting activity, rather than construction. There may be steps between permitting and completion that our results cannot speak to. Our results suggest that ADU reforms created many winners: incumbent homeowners and possibly even government budgets.[89] Even so, reforms aimed at addressing the high cost of housing may need to be less "light-touch."[90]

[88] Appendix Figure A.11a shows areas in which ADUs make up a meaningful supply of pre-reform housing units. These areas might experience meaningfully different dynamics than areas near the Coastal Zone.

[89] In our setting, we model entrants as indifferent between purchasing a built parcel and the outside option for simplicity. However, this masks ways in which entrants may benefit from ADU reform.

[90] For examples, see Anagol, Ferreira and Rexer (2026) and Asquith, Mast and Reed (2023).

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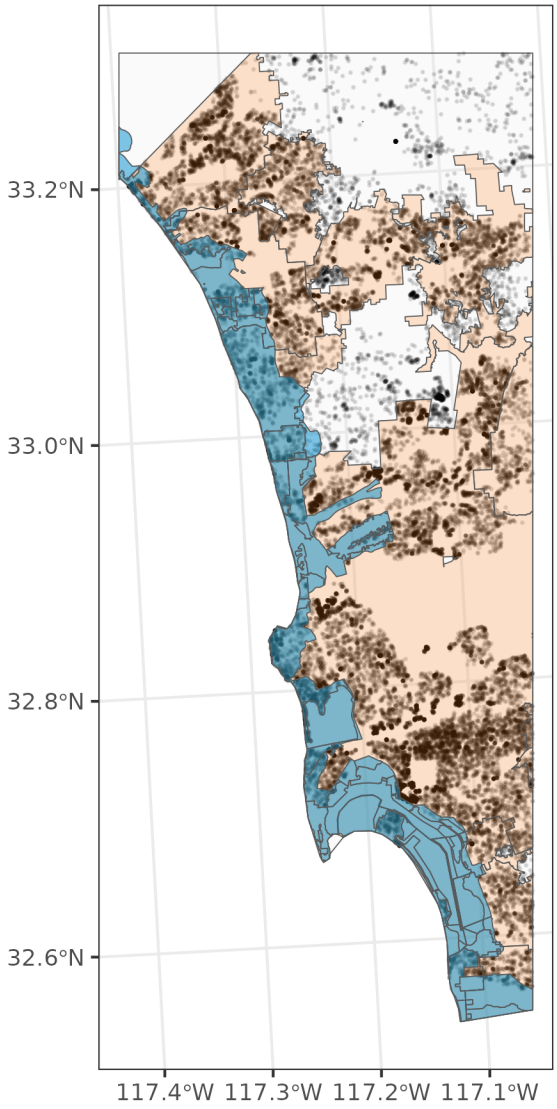
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Figure 1: State-Wide Accessory Dwelling Unit Permits Issued in California from 2012 to 2024



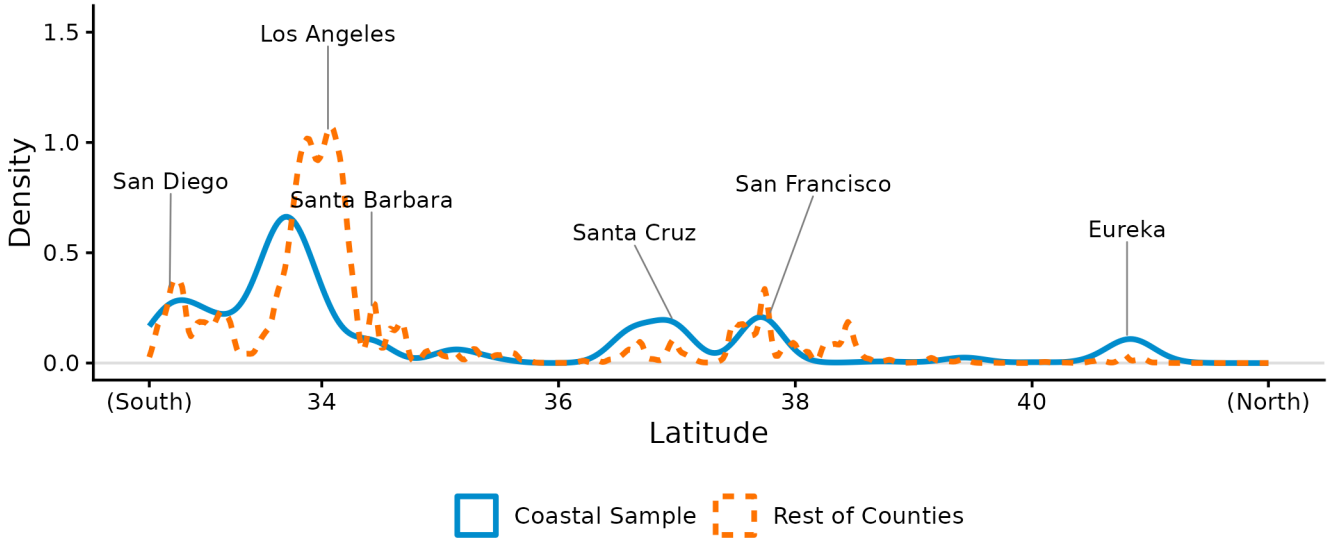
This figure shows a time series of accessory dwelling unit permits issued in the state of California from 2012 to 2024. The figure uses two data sources. The blue line shows permits counted by Peterson (2021), Gray (2024), and Minott (2023), which collect data on ADU permits from the California Department for Housing and Community Development deprecated housing element series. The orange and dotted line shows permits according to the (current) California Housing Element Progress Report (Table A2) data (California Department of Housing and Community Development, 2025).

Figure 2: Map of San Diego Coastal Zone, Incorporated Cities, and Real Estate Transactions in 2016



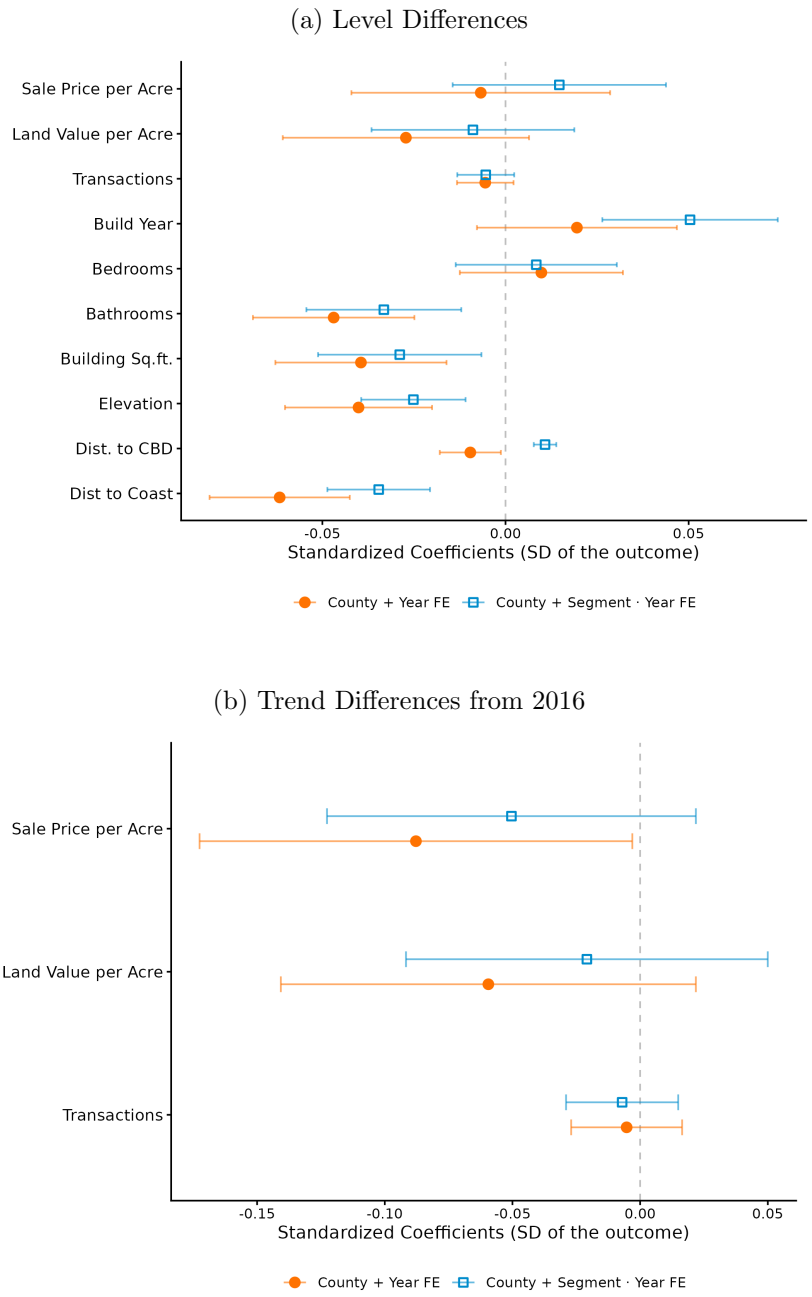
This figure shows the California Coastal Zone in San Diego County. The Coastal Zone is shown in blue. The orange area shows incorporated cities near the coast, and the black dots show all real estate transactions in 2016 from the Cotality (2025) data.

Figure 3: Density of Cotality (2025) Sample Parcels by Latitude



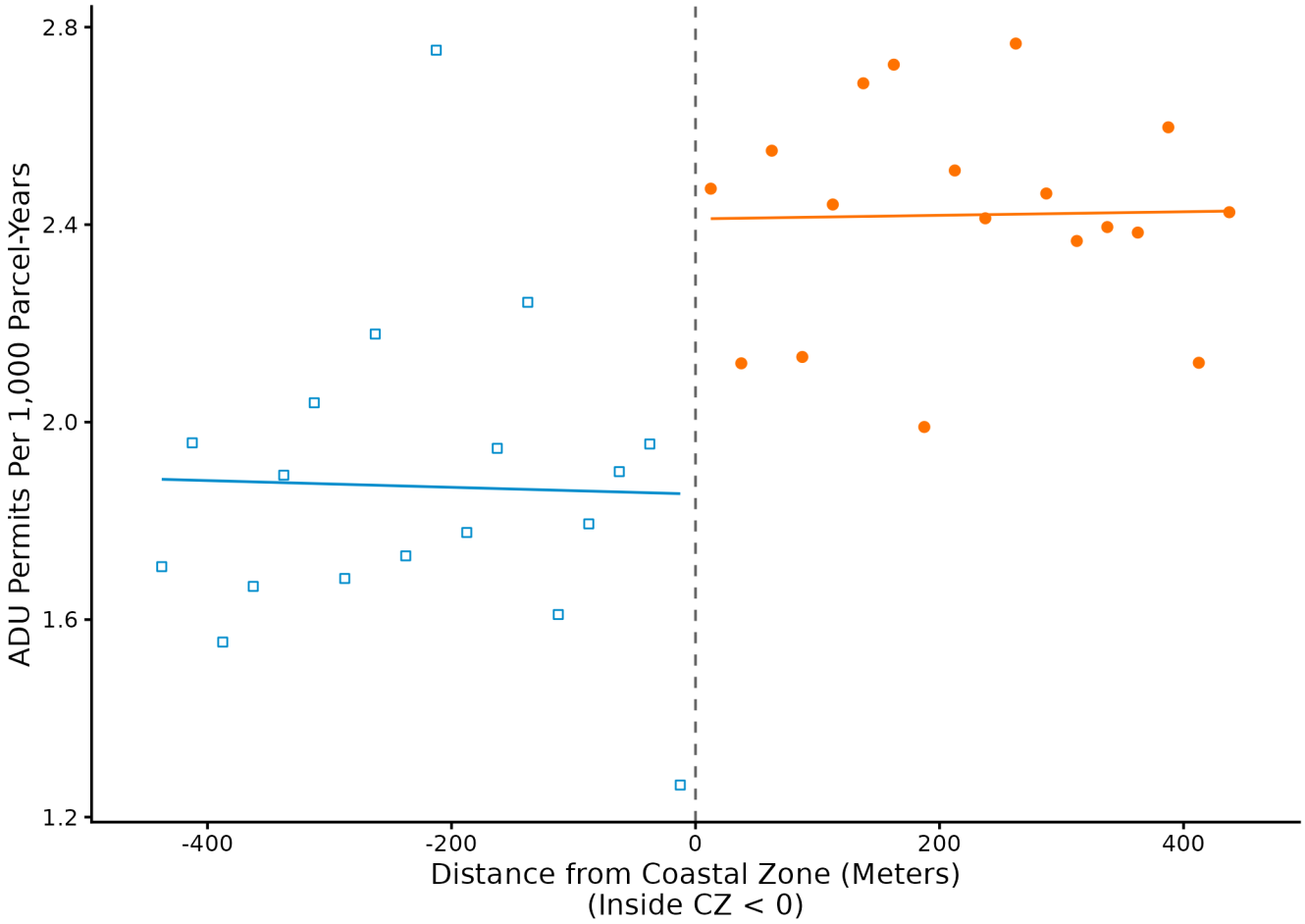
This figure shows the density of the Cotality (2025) property sample by latitude. The figure shows unique residential parcels from the year 2016. The solid blue line shows the sample within 450 meters of the Coastal Zone boundary, and the dashed orange line shows the remaining sample in 15 coastal counties outside of 450 meters from the Coastal Zone boundary. The horizontal axis shows parcel latitude, with Mexico abutting the southern portion of the sample on the left hand side, and Oregon abutting the northern portion of the sample on the right hand side. Selected cities are marked for reference.

Figure 4: Baseline Pre-Reform Differences Across the Coastal Zone Boundary Using the 450-Meter Bandwidth From 2010 to 2016



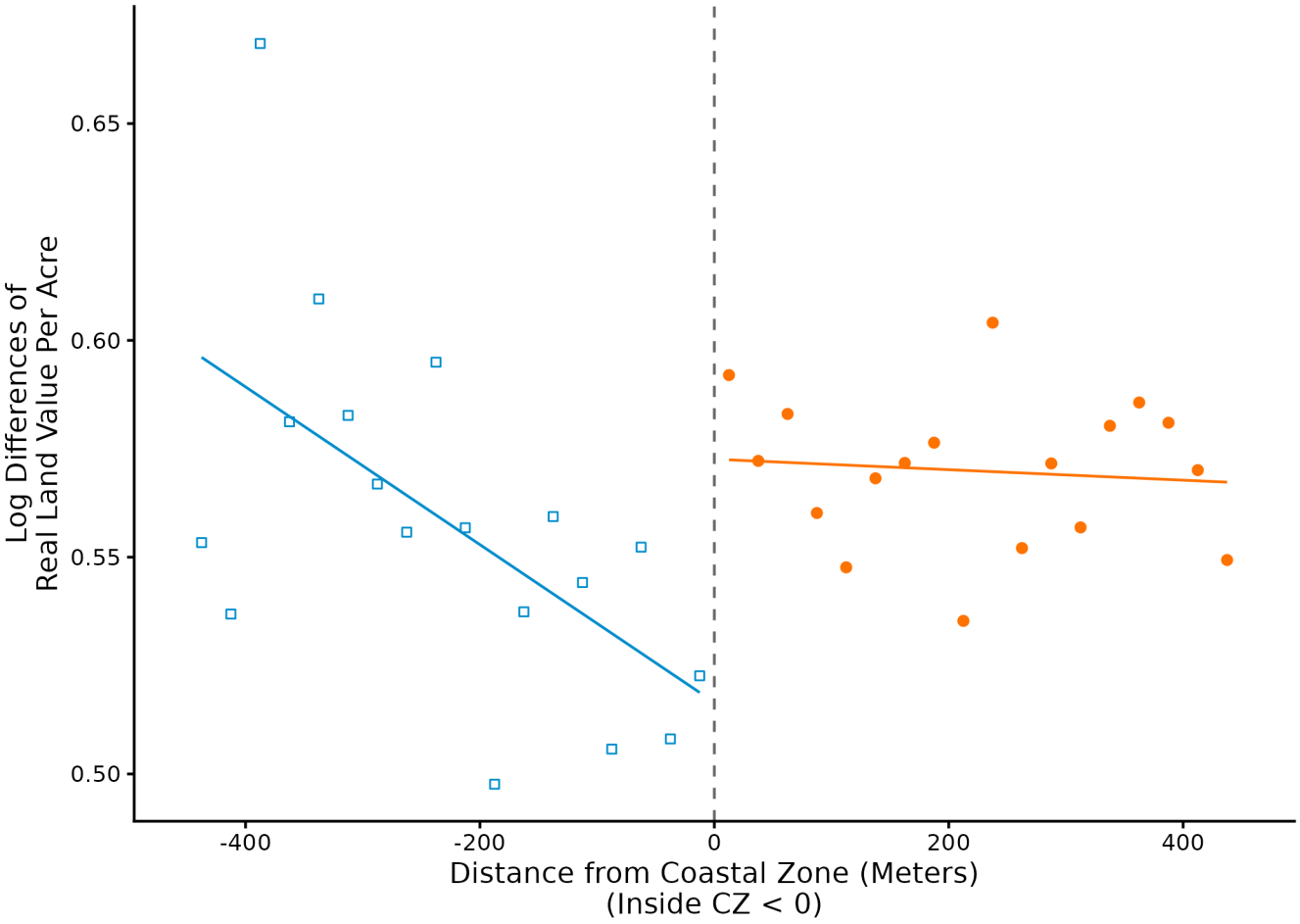
This figure uses data from Cotality (2025), California Department of Housing and Community Development (2025), U.S. Geological Survey (2026), Holian (2019), California Department of Fish and Wildlife (2026), and California Coastal Commission (2014) between 2010 and 2016. Each coefficient is scaled by the outcome's standard deviation within the 450m bandwidth between 2010 and 2016. In Panel A, we show level differences from the 1st degree polynomial regression discontinuity equation. In Panel B, we show differences in the pre-period from the baseline coefficient in 2016, similar to an event study or joint pre-trend style test in our difference in discontinuities specification. In both panels, estimates with orange circles use county and year fixed effects, and estimates in blue squares add segment by year fixed effects. Robust standard errors are clustered by parcel. The standard deviations used for scaling are: 1.11 for (log real) sale price per acre, 1.25 for (log real) land value per acre, 0.18 for the arms-length transaction dummy, 23 for build years, 1.1 for bedrooms, 1.2 for bathrooms, 914 for building square feet, 55 meters for elevation, 1.1 for (log) distance (meters) to the CBD, and 0.9 for (log) distance (meters) to the nearest coastline. We provide a simple local estimated scatterplot smoothing (LOESS) line for visual convenience.

Figure 5: Binscatter of Accessory Dwelling Unit Permit Activity Around the Coastal Zone Boundary by Distance Bin (2018–2024)



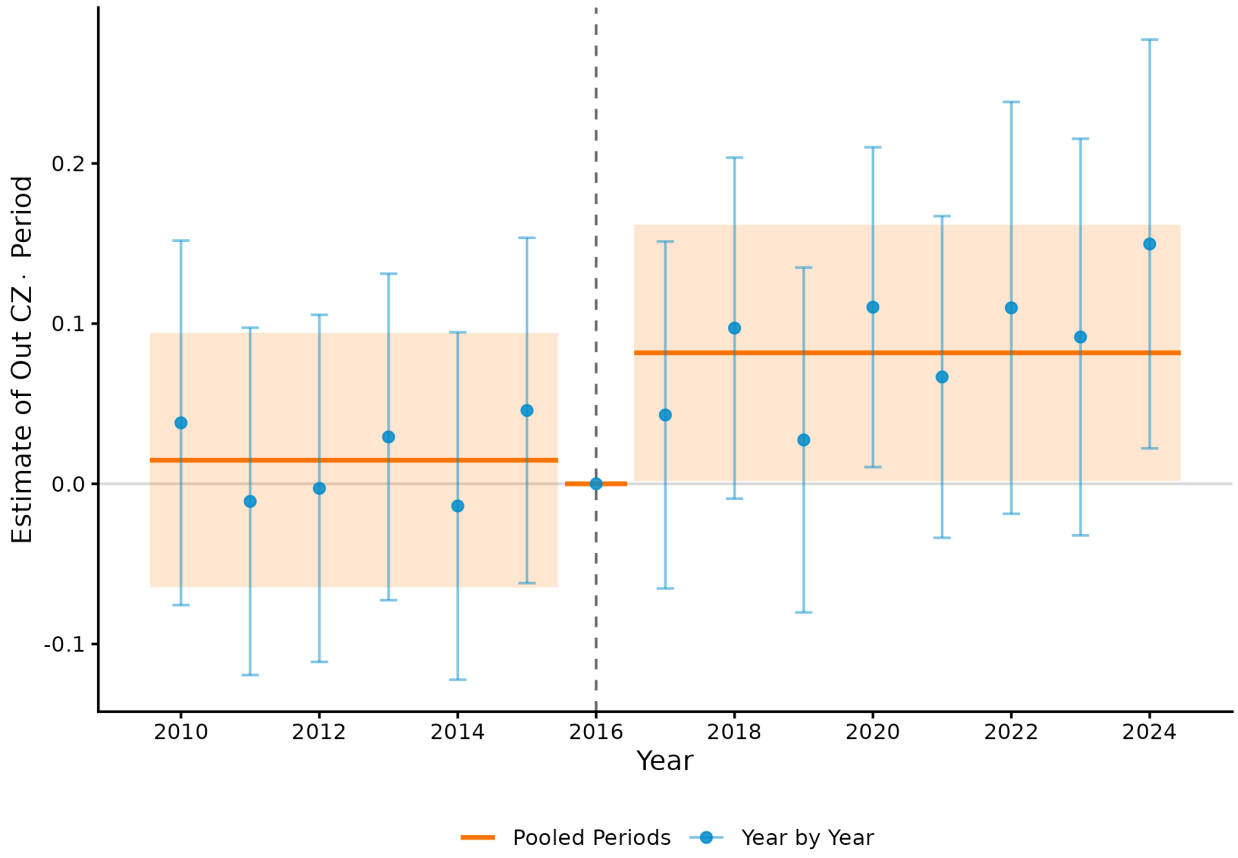
This figure uses data from Cotality (2025), California Department of Housing and Community Development (2025), and California Coastal Commission (2014) to show how permitting activity for ADUs changes across the Coastal Zone boundary between 2018 and 2024. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the permitting reforms (blue squares). We group parcels in 25-meter bins. The vertical axis shows ADU permits per 1000 parcel years on our balanced parcel panel. We provide regression line fit on the binned data for presentation.

Figure 6: Within Parcel Changes in Real Land Values per Acre Around the Coastal Zone Boundary Pre- and Post-2016 Reforms on Repeat Sale Parcels



This figure uses data from Cotality (2025) and California Coastal Commission (2014) to show how residential land values change across the Coastal Zone boundary between 2010 and 2024. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the post-2016 permitting reforms. We group parcels in 25-meter bins within our 450 analysis sample. The vertical axis shows the log difference of the real land value per acre for each bin between the pre-reform (2010-2016) and post-reform (2017-2024) periods on our repeat sale parcel sample. The blue squares show the values inside the Coastal Zone. The blue circles show values outside the Coastal Zone boundary that experience ADU reforms. The difference in discontinuities estimate can be seen in difference in changes just before and then just after crossing the Coastal Zone boundary at 0.

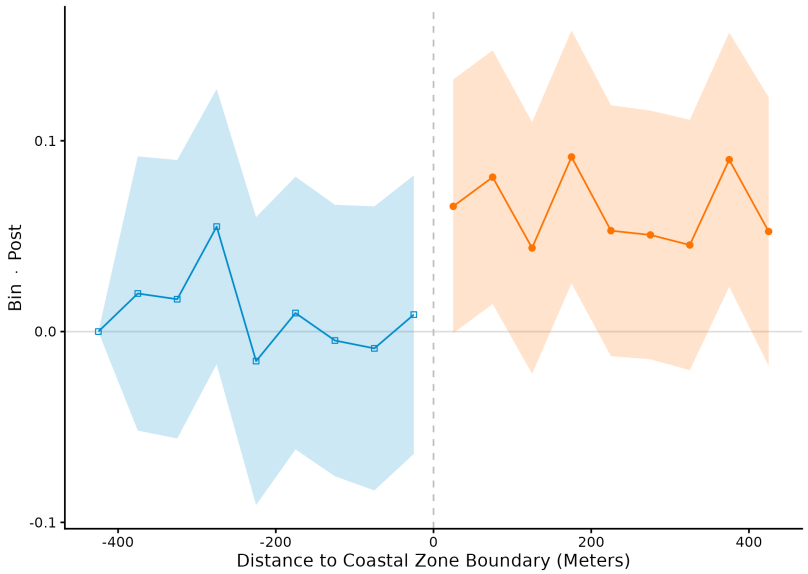
Figure 7: Repeat Sale Event Study Estimates of the Impacts of Permitting Reforms on Log Real Land Value Per Acre



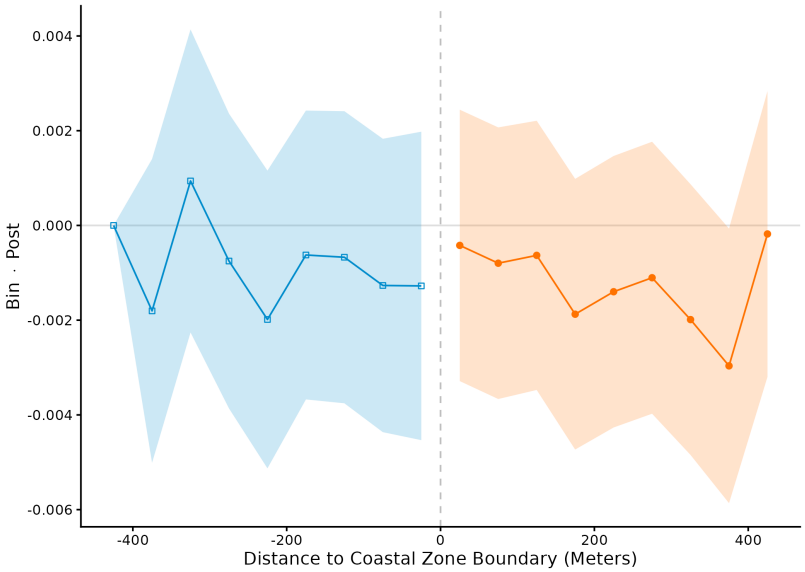
This figure presents event study estimates of the impact of California's post-2016 ADU permitting reforms on residential land values, using data from Cotality (2025) and California Coastal Commission (2014) following Equation 3. Each point shows the estimated coefficient on the interaction of the outside-Coastal-Zone indicator ($OutCZ$) with a year dummy, relative to the omitted base year of 2016. Estimates are shown for an event-study version of the repeat sale model with segment by year fixed effects corresponding to Column 6 of Table 4. We also show pooled coefficients with 2016 as the omitted year for 2010 to 2015 and 2017 to 2024. The sample is restricted to residential parcels within 450 meters of the Coastal Zone boundary. Confidence intervals are at the 95% level, with standard errors clustered at the parcel-level. We use triangle weights.

Figure 8: Spillover Estimates By Coastal Zone Distance Bin for Changes in Real Total Sale Value Per Acre and Transactions

(a) Real Log Total Sale Value Per Acre

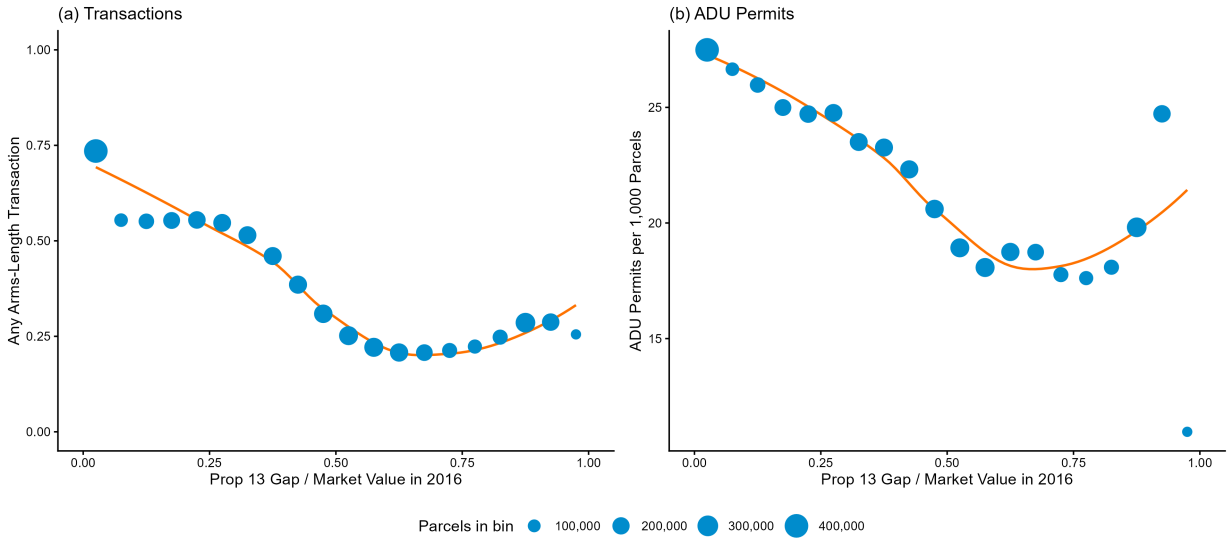


(b) Arms-Length Transaction Dummy



This figure uses data from Cotality (2025) and California Coastal Commission (2014) to show how changes in the total sale price per acre change across the Coastal Zone boundary between 2010 and 2024 in Panel A. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values, shown in blue, represent parcels inside the Coastal Zone that do not receive the post 2016 permitting reforms. We group parcels in 50-meter bins. The vertical axis shows the coefficients from Equation 4 with segment by year fixed effects included, and the log of the real total sale price per acre as the dependent variable. These coefficients are on the set of distance bin times post interactions, where the bin $[-450, -400)$ meters has been omitted to serve as the reference bin. In Panel B, we run the same regression estimates and swap the outcome with a dummy that takes a value of 1 when a parcel transacts at arms-length in a given year, and 0 otherwise. The regressions in Panel B also use a parcel fixed effect.

Figure 9: Binned Scatter Plots of Proposition 13 Property Tax Shelter in 2016 and the Share of Parcels Ever Transacting or Ever Building an ADU in Sample



This figure uses data from Cotality (2025), California Department of Housing and Community Development (2025) to show a binned scatter plot of each parcel's property tax shelter in 2016 against the share of parcels transacting between 2010 and 2024 and the share of parcels building an ADU between 2018 and 2024. Each parcel's property tax shelter is defined as the gap between the parcel's estimated market value and the parcel's estimated assessed value divided by the parcel's estimated market value in 2016 for all parcels in the Coastal 15 counties. We build estimates of the parcel's market value using a hedonic model estimated on transaction data between 2010 and 2016 as described in Appendix C. If a parcel transacts in 2016, we use that value instead of the estimate. We also build estimated assessed values by growing the parcel's last recorded assessed value by 2% per year when directly observed assessed values are not available in 2016. These values are shown for each parcel on the horizontal axis. In Panel A, the vertical axis is the share of parcels that transact between 2010 and 2024 at least one time. We include only arm's length transactions in this variable. We show a loess smoother in orange.

Table 1: Sample Statistics for the Balanced Parcel Panel Data by Coastal Zone Sample Status and Reform Period

	450-Meter Bandwidth					
	In CZ		Out CZ		Full County	
	2010-2016	2017-2024	2010-2016	2017-2024	2010-2016	2017-2024
	(1)	(2)	(3)	(4)	(5)	(6)
Share SFR	0.852	0.86	0.887	0.896	0.908	0.914
	(0.355)	(0.347)	(0.316)	(0.306)	(0.289)	(0.28)
Dist. to the CZ	-220.385	-220.402	221.547	221.556	14,486	14,486
	(124.654)	(124.643)	(128.827)	(128.82)	(14,750)	(14,749)
Effective Build Year	1967	1972	1965	1969	1965	1968
	(23.473)	(24.492)	(22.773)	(23.224)	(21.736)	(22.327)
Bedrooms	3.295	3.352	3.257	3.311	3.338	3.388
	(1.195)	(1.198)	(1.09)	(1.106)	(1.053)	(1.072)
Bathrooms	2.535	2.708	2.33	2.456	2.317	2.413
	(1.237)	(1.298)	(1.124)	(1.155)	(1.037)	(1.072)
Building Sq. Ft.	1,994	2,168	1,865	2,008	1,856	1,976
	(992.873)	(1,122)	(868.049)	(961.741)	(879.31)	(955.993)
Property Tax Per Acre	64,999	79,993	54,104	67,415	39,121	48,469
	(64,711)	(76,096)	(53,965)	(66,371)	(41,105)	(49,774)
Land Share	0.547	0.569	0.525	0.55	0.503	0.527
	(0.205)	(0.211)	(0.195)	(0.201)	(0.19)	(0.194)
ADU Permit Per Year (2018+)		0.002		0.002		0.003
		(0.041)		(0.046)		(0.053)
Any Permit (2018+)		0.003		0.003		0.004
		(0.054)		(0.056)		(0.064)
ADU Suitable	0.572	0.567	0.631	0.611	0.678	0.677
	(0.495)	(0.496)	(0.483)	(0.488)	(0.467)	(0.468)
Excess Lot Sq. Ft.	19,234	33,053	27,385	25,806	25,365	28,947
	(1,324,608)	(2,976,177)	(1,107,478)	(525,869)	(2,046,433)	(5,977,357)
Observations	521,282	595,473	985,277	1,125,737	29,087,092	33,237,497

Notes: This table shows summary statistics for the balanced parcel panel. Standard deviations are shown in parentheses. Columns (1)-(2) show parcels within the Coastal Zone, columns (3)-(4) show parcels outside the Coastal Zone but within 450 meters of the boundary, and columns (5)-(6) show all parcels in the 15 coastal counties of California. We present means and standard deviations for the share of parcels on land zoned only for single-family residences, the parcel's distance to the Coastal Zone boundary in meters, the effective build year of the structure(s) on the parcel, the number of bedrooms, number of bathrooms, building square footage, property taxes paid per acre, the land share of assessed value, whether or not an ADU permit is associated with the parcel each year, whether or not any permit is associated with the parcel each year, the share of parcels that could accommodate an ADU (more than 2,000 sq. ft. of excess space and a ground slope under 20 degrees), and the excess sq. ft. on the lot not covered by a building. Dollar values are in 2025 USD.

Table 2: Sample Statistics for the Transaction Data by Coastal Zone Sample Status and Reform Period

	450-Meter Bandwidth					
	In CZ		Out CZ		Full County	
	2010-2016	2017-2024	2010-2016	2017-2024	2010-2016	2017-2024
	(1)	(2)	(3)	(4)	(5)	(6)
Share SFR	0.877	0.893	0.89	0.899	0.919	0.928
	(0.329)	(0.309)	(0.313)	(0.301)	(0.274)	(0.259)
Dist. to the CZ	-220.525	-222.008	223.23	222.57	15,584	15,495
	(124.898)	(125.26)	(129.231)	(129.395)	(16,014)	(16,082)
Effective Build Year	1967	1972	1966	1969	1966	1969
	(24.322)	(24.847)	(23.974)	(23.741)	(23.181)	(23.145)
Bedrooms	3.259	3.303	3.243	3.277	3.307	3.349
	(1.162)	(1.146)	(1.07)	(1.09)	(1.039)	(1.051)
Bathrooms	2.522	2.72	2.387	2.514	2.333	2.453
	(1.232)	(1.289)	(1.164)	(1.18)	(1.057)	(1.101)
Building Sq. Ft.	2,004	2,217	1,883	2,041	1,852	2,001
	(1,002)	(1,157)	(896.452)	(1,000)	(900.65)	(998.652)
Land Value Per Acre	6,716,901	8,792,829	5,229,268	7,560,034	3,130,478	4,520,994
	(6,805,269)	(7,814,689)	(5,101,423)	(6,895,375)	(3,746,220)	(4,909,490)
Total Sale Price Per Acre	10,758,600	13,853,775	8,709,221	12,195,017	5,626,841	7,955,063
	(9,135,531)	(10,455,810)	(7,380,480)	(9,519,186)	(5,769,792)	(7,265,957)
Total Sale Price (0.2 Acre Lot)	2,151,720	2,770,755	1,741,844	2,439,003	1,125,368	1,591,012
	(1,827,106)	(2,091,162)	(1,476,096)	(1,903,837)	(1,153,958)	(1,453,191)
ADU Suitable	0.84	0.851	0.886	0.879	0.873	0.861
	(0.366)	(0.356)	(0.318)	(0.327)	(0.333)	(0.346)
Excess Lot Sq. Ft.	9,951	10,551	9,066	8,848	14,985	17,197
	(63,740)	(46,012)	(93,422)	(55,098)	(678,734)	(340,596)
Observations	17,567	18,298	33,068	34,842	1,070,702	1,090,087

Notes: This table shows summary statistics for the transaction dataset. Standard deviations are shown in parentheses. Columns (1)-(2) show parcels within the Coastal Zone, columns (3)-(4) show parcels outside the Coastal Zone but within 450 meters of the boundary, and columns (5)-(6) show all parcels in the 15 coastal counties of California. We present means and standard deviations for the share of parcels on land zoned only for single-family residences, the parcel's distance to the Coastal Zone boundary in meters, the effective build year of the structure(s) on the parcel, the number of bedrooms, number of bathrooms, building square footage, real land value per acre, the real total sale price per acre, the real total sale price scaled to a 0.2 acre lot, the share of parcels that could put down an ADU (more than 2,000 sq. ft. of excess space and a ground slope under 20 degrees), and the excess sq. ft. on the lot not covered by a building. Dollar values are in 2025 USD.

Table 3: Spatial Regression Discontinuity Estimates of the Impacts of Permitting Reforms on Accessory Dwelling Unit Permits after 2018 within 450 Meters of the Coastal Zone Boundary

	Dep. Var.: ADUs per 1,000 Parcel-Years			
	(1)	(2)	(3)	(4)
OutCZ	0.623***	0.629***	0.530***	0.618***
	(0.170)	(0.174)	(0.171)	(0.173)
1st Degree Polynomial	Y	Y	Y	Y
County FEs	Y	Y	Y	Y
Year FEs	Y	Y		
Segment · Year FEs			Y	Y
Pre-2016 Static Hedonics		Y		Y
N	1 505 988	1 485 253	1 505 974	1 485 239
Adj. R^2	0.001	0.002	0.002	0.003
Mean	2.248	2.248	2.248	2.248
SD	47.36	47.36	47.36	47.36
In-CZ Mean	1.900	1.900	1.900	1.900
In-CZ SD	43.55	43.55	43.55	43.55

Notes: This table uses data from Cotality (2025), California Department of Housing and Community Development (2025), and California Coastal Commission (2014) to estimate Equation 2 for a balanced set of residential parcels between 2018 and 2024: $Y_{ict} = \alpha + \beta_1 OutCZ_i + \beta_2 DistCZ_i + \beta_3(DistCZ_i \cdot OutCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable is a dummy taking a value of 1 if parcel i has an ADU permit associated with it in year t, and we scale this outcome by multiplying by $1,000$ for display purposes. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's permitting reforms for ADUs. $DistCZ_i$ is a function of a parcel's distance to the Coastal Zone boundary, which uses a first-degree polynomial here. We also include a set of county fixed effects, λ_c, and a set of year fixed effects, τ_t. We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We present summary statistics for the permit dummy for both the entire sample and just the sample within the Coastal Zone at the bottom of the table. In Column 1, we present a specification only with a first-degree polynomial, a set of county fixed effects, and a set of year fixed effects. Then in Columns 2 and 4, we include the set of hedonic fixed effects. Then in columns 3 and 4, we replace the year fixed effects with a segment by year fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. We cluster our robust standard errors at the parcel-level. We report estimates using a 450-meter bandwidth from the Coastal Zone boundary and use triangle weights. *$p < .1$, ** $p < .05$, *** $p < .01$

Table 4: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Land Values within 450 Meters of the Coastal Zone Boundary

	Dep.Var.: Log of Real Land Value Per Acre					
	(1)	(2)	(3)	(4)	(5)	(6)
$OutCZ \cdot Post$	0.141***	0.125***	0.085***	0.070***	0.088***	0.069***
	(0.027)	(0.026)	(0.022)	(0.022)	(0.022)	(0.021)
1st Degree Polynomial	Y	Y	Y	Y	Y	Y
County FEs	Y	Y	Y	Y		
Parcel FEs					Y	Y
Year FEs	Y	Y			Y	
Segment · Year FEs			Y	Y		Y
Pre-2016 Static Hedonics		Y		Y		
N	103 766	102 901	103 718	102 850	49 440	49 287
Adj. R^2.	0.524	0.546	0.662	0.678	0.903	0.908
Mean 2010-2016	15.01	15.01	15.01	15.01	15.02	15.02
SD 2010-2016	1.246	1.246	1.246	1.246	1.245	1.245
In-CZ Mean 2010-2016	15.14	15.14	15.14	15.14	15.14	15.14
In-CZ SD 2010-2016	1.317	1.317	1.317	1.317	1.316	1.316

Notes: This table uses data from Cotality (2025) and California Coastal Commission (2014) to estimate Equation 3 for transactions on residential parcels between 2010 and 2024: $ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable is the log of the real land value per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 in 2017 and for all subsequent years demarcating the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first-degree polynomial form. Distance slopes can change after the boundary. We include a set of county fixed effects, λ_c, and a set of year fixed effects, τ_t. We also include a set of hedonic fixed effects that capture 2010-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We cluster our robust standard errors at the parcel-level. We report estimates using a 450-meter bandwidth from the Coastal Zone boundary. We present summary statistics for the log of the real land value per acre for both the entire sample from 2010 to 2016 and just the sample within the Coastal Zone from 2010 to 2016 at the bottom of the table. In Column 1, we present a specification only with a set of county fixed effects and a set of year fixed effects. In Columns 2 and 4 we add hedonic fixed effects. Then in Column 3, 4, and 6, we include the segment by year fixed effects. Finally, in columns 5 and 6, we include a parcel fixed effect and use a sample restricted to parcels that transact both between 2010 to 2016 and 2017 to 2024. All specifications use triangle weights and report parcel clustered standard errors. *$p < .1$, ** $p < .05$, *** $p < .01$

Table 5: Heterogeneity Estimates by Lot Quality of the Impacts of Permitting Reforms on Accessory Dwelling Unit Permits after 2018 and Land Values Per Acre within 450 Meters of the Coastal Zone Boundary

Panel A Dep. Var.:	*ADUs per 1,000 Parcel-Years*			
Heterogeneity Var.:	*Excess sq. ft.*		*Dist. to CBD (m)*	
	(1)	(2)	(3)	(4)
$OutCZ$	0.905***	0.618**	1.146***	0.907***
	(0.295)	(0.292)	(0.261)	(0.262)
$OutCZ \cdot H_i$	−0.027	0.367	−0.989***	−0.712**
	(0.526)	(0.521)	(0.341)	(0.344)
1st Degree Polynomial	Y	Y	Y	Y
RD Terms $\cdot H_i$	Y	Y	Y	Y
County FEs	Y	Y	Y	Y
Year FEs	Y		Y	
Segment · Year FEs		Y		Y
N	971 458	971 447	1 505 988	1 505 974
Adj. R^2	0.001	0.013	0.001	0.002
Mean Dep. Var.:	3.200	3.200	2.248	2.248
Median H_i:	4,226	4,226	25,828	25,828
Panel B Dep. Var.:	*Log of Real Land Values Per Acre*			
Heterogeneity Var.:	*Excess sq. ft.*		*Dist. to CBD (m)*	
	(1)	(2)	(3)	(4)
$OutCZ \cdot Post$	0.080***	0.058**	0.170***	0.064**
	(0.028)	(0.028)	(0.037)	(0.029)
$OutCZ \cdot Post \cdot H_i$	0.028	0.033	−0.107**	0.010
	(0.050)	(0.042)	(0.051)	(0.042)
1st Degree Polynomial	Y	Y	Y	Y
RD Terms $\cdot H_i$	Y	Y	Y	Y
County FEs	Y	Y	Y	Y
Year FEs	Y		Y	
Segment · Year FEs		Y		Y
Parcel FEs		Y		Y
N	91 266	44 299	103 766	49 287
Adj. R^2	0.425	0.874	0.543	0.908
Mean Dep. Var. 2010-2016:	15.22	15.22	15.01	15.02
Median H_i:	4,212	4,212	26,826	26,826

Notes: This table uses data from Cotality (2025), California Coastal Commission (2014), California Department of Housing and Community Development (2025), and Holian (2019) to show heterogeneity estimates following Equation 5 and reporting coefficients to characterize the CATE in Equation 6. Our heterogeneity variable in each regression is represented by H_i. Panel A shows regression discontinuity estimates of the CATE for ADU permits per 1000 parcel-years. Columns 1 and 2 use a dummy for whether a lot is above the sample median for excess square footage on the parcel. A value of 0 reflects a parcel with less than 4226 available square feet to build an ADU. Columns 3 and 4 report heterogeneity using a dummy that takes a value of 1 when a lot is farther than 25.828 kilometers from the nearest CBD, where a value of 0 reflects a lot within 25.828 kilometers from the nearest CBD. Panel B reports a parallel set of difference in discontinuity estimates for the log of real land value per acre. Columns 2 and 4 add in segment by year fixed effects in Panel A, and additionally add in a parcel fixed effect in Panel B. We report estimates using a 450-meter bandwidth from the Coastal Zone boundary and use triangle weights. *$p < .1$, ** $p < .05$, *** $p < .01$

Table 6: Heterogeneity Estimates by Property Tax Shelter and Block Group Age of the Impacts of Permitting Reforms within 450 Meters of the Coastal Zone Boundary

Panel A Dep.Var.:	*Any Arms-Length Transaction*					
Heterogeneity Var.:	*Prop. Tax. Shelter*		*Share Over 55*			
			2010-2019		*2010-2024*	
	(1)	(2)	(3)	(4)	(5)	(6)
$OutCZ \cdot Post$	−0.002	−0.002	−0.015***	−0.013**	−0.007	−0.006
	(0.002)	(0.002)	(0.005)	(0.005)	(0.004)	(0.004)
$OutCZ \cdot Post \cdot H_i$	0.001	0.000	0.029***	0.025***	0.013**	0.012
	(0.003)	(0.003)	(0.008)	(0.008)	(0.006)	(0.006)
1st Degree Polynomial	Y	Y	Y	Y	Y	Y
RD Terms · Covariate	Y	Y	Y	Y	Y	Y
Covariate Terms · Post	Y	Y	Y	Y	Y	Y
Year FEs	Y		Y		Y	
Segment · Year FEs		Y		Y		Y
Parcel FEs	Y	Y	Y	Y	Y	Y
N	3 122 313	3 122 298	2 148 934	2 148 914	3 222 639	3 222 609
Adj. R^2	0.020	0.021	0.012	0.012	0.013	0.014
Mean Dep. Var. 2010-2016:	0.036	0.036	0.036	0.036	0.036	0.036
Mean H_i:	0.508	0.508	0.614	0.614	0.614	0.614
Panel B Dep.Var.:	*ADUs per 1,000 Parcel-Years*					
Heterogeneity Var.:	*Prop. Tax. Shelter*		*Share Over 55*			
			2018-2019		*2018-2024*	
	(1)	(2)	(3)	(4)	(5)	(6)
$OutCZ$	0.934**	0.890**	1.563**	1.734**	1.201	1.459**
	(0.372)	(0.373)	(0.708)	(0.717)	(0.641)	(0.652)
$OutCZ \cdot H_i$	−0.585	−0.655	−1.295	−1.750	−0.904	−1.498
	(0.625)	(0.625)	(1.134)	(1.155)	(0.990)	(1.012)
1st Degree Polynomial	Y	Y	Y	Y	Y	Y
RD Terms · Covariate	Y	Y	Y	Y	Y	Y
Covariate Terms (Levels)	Y	Y	Y	Y	Y	Y
County FEs	Y	Y	Y	Y	Y	Y
Year FEs	Y		Y		Y	
Segment · Year FEs		Y		Y		Y
N	1 456 892	1 456 885	429 889	429 885	1 503 594	1 503 580
Adj. R^2	0.001	0.002	0.001	0.003	0.001	0.002
Mean Dep. Var.:	2.291	2.291	1.242	1.242	2.248	2.248
Mean H_i:	0.508	0.508	0.614	0.614	0.614	0.614

Notes: This table uses data from Cotality (2025), California Coastal Commission (2014), California Department of Housing and Community Development (2025), and U.S. Census Bureau (2016) to show heterogeneity estimates following Equation 5 and reporting coefficients to characterize the CATE in Equation 6. Our heterogeneity variable in each regression is represented by H_i. Panel A shows difference in discontinuity estimates of the CATE for our any transaction dummy variable. Columns 1 and 2 use the property tax shelter in 2016 as described in the text, where a value of 0 reflects taxation on the market rate. Columns 3 through 6 report heterogeneity using the block group share of owners over 55 for a pre-COVID sample and our full sample. A block group owner share of 0 reflects a parcel within a block group where all owners of homes are younger than 55. Panel B reports a parallel set of regression discontinuity estimates for ADU permits per 1000 parcel-years. We report estimates using a 450-meter bandwidth from the Coastal Zone boundary and use triangle weights. *$p < .1$, ** $p < .05$, *** $p < .01$

Table 7: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Property Tax Collections Per Acre within 450 Meters of the Coastal Zone Boundary

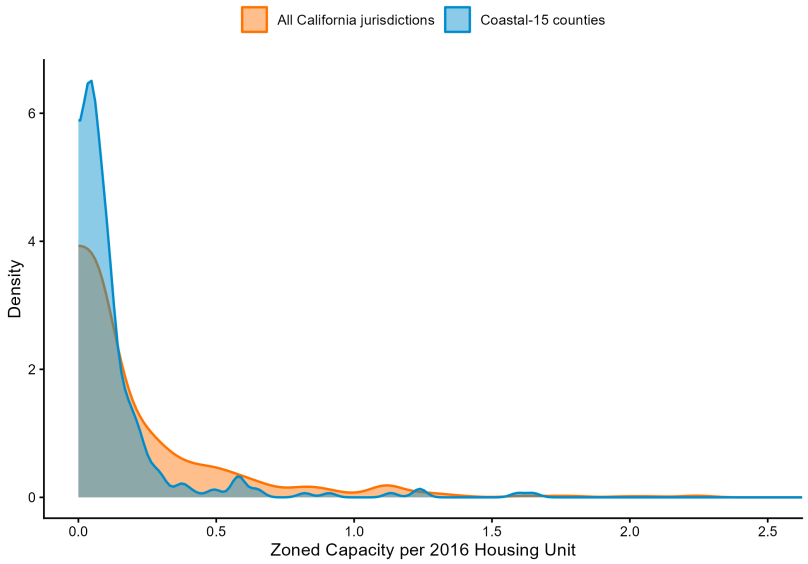
	Dep. Var.: Log of Real Property Tax Per Acre					
	(1)	(2)	(3)	(4)	(5)	(6)
$OutCZ \cdot Post$	0.074***	0.041***	0.069***	0.036***	0.018***	0.013***
	(0.007)	(0.006)	(0.007)	(0.006)	(0.005)	(0.005)
1st Degree Polynomial	Y	Y	Y	Y	Y	Y
County FEs	Y	Y	Y	Y		
Parcel FEs					Y	Y
Year FEs	Y	Y			Y	
Segment · Year FEs			Y	Y		Y
Pre-2016 Static Hedonics		Y		Y		
N	2 488 513	2 482 314	2 488 499	2 482 294	2 488 024	2 488 010
Adj. R^2	0.288	0.378	0.387	0.456	0.941	0.942
Mean 2010-2016	10.37	10.37	10.37	10.37	10.37	10.37
SD 2010-2016	1.333	1.333	1.333	1.333	1.333	1.333
In-CZ Mean 2010-2016	10.46	10.46	10.46	10.46	10.46	10.46
In-CZ SD 2010-2016	1.387	1.387	1.387	1.387	1.387	1.387

Notes: This table uses data from Cotality (2025) and California Coastal Commission (2014) to estimate Equation 3 for a panel on residential parcels between 2010 and 2024: $ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable is the log of the real property taxes paid per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 in 2017 and afterwards demarcating the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first-degree polynomial form here. Distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c, and a set of year fixed effects, τ_t. We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. In Column 1, we present a specification only with a first-degree polynomial, a set of county fixed effects, and a set of year fixed effects. In Columns 2 and 4, we add pre-2016 static hedonic fixed effects. Then, in Columns 3, 4, and 6 we use segment by year fixed effects to keep our comparisons local to Coastal Zone boundary segments. Finally, in Columns 5 and 6 we include a parcel fixed effect. Note that here, we do not have to limit our sample based on transactions as parcels pay taxes every year even if they do not transact. We cluster our robust standard errors at the parcel-level. We report estimates using a 450-meter bandwidth from the Coastal Zone boundary using triangle weights. We present summary statistics for the log of the real property tax per acre for both the entire sample from 2010 to 2016 and just the sample within the Coastal Zone from 2010 to 2016 at the bottom of the table. *$p < .1$, ** $p < .05$, *** $p < .01$

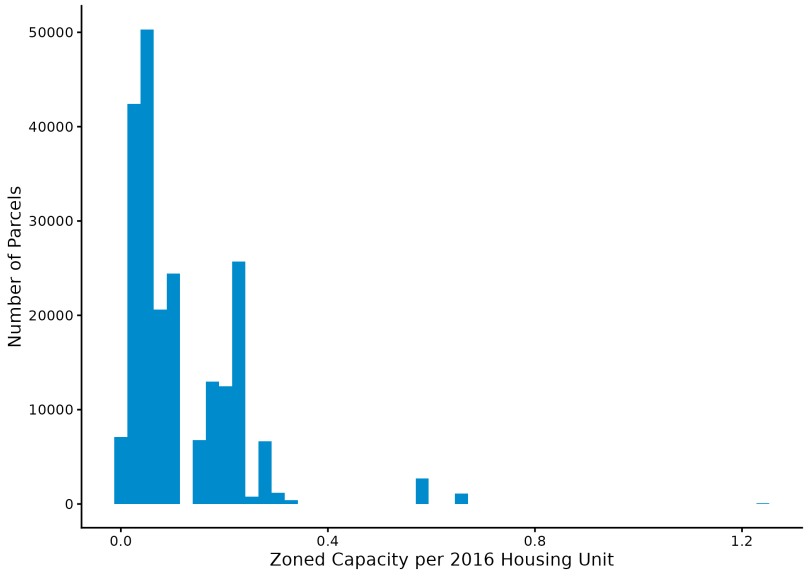
A Additional Exhibits

Figure A.1: Density of 2014 Zoned Capacity Per Housing Unit

(a) All California Jurisdictions vs Coastal Jurisdictions

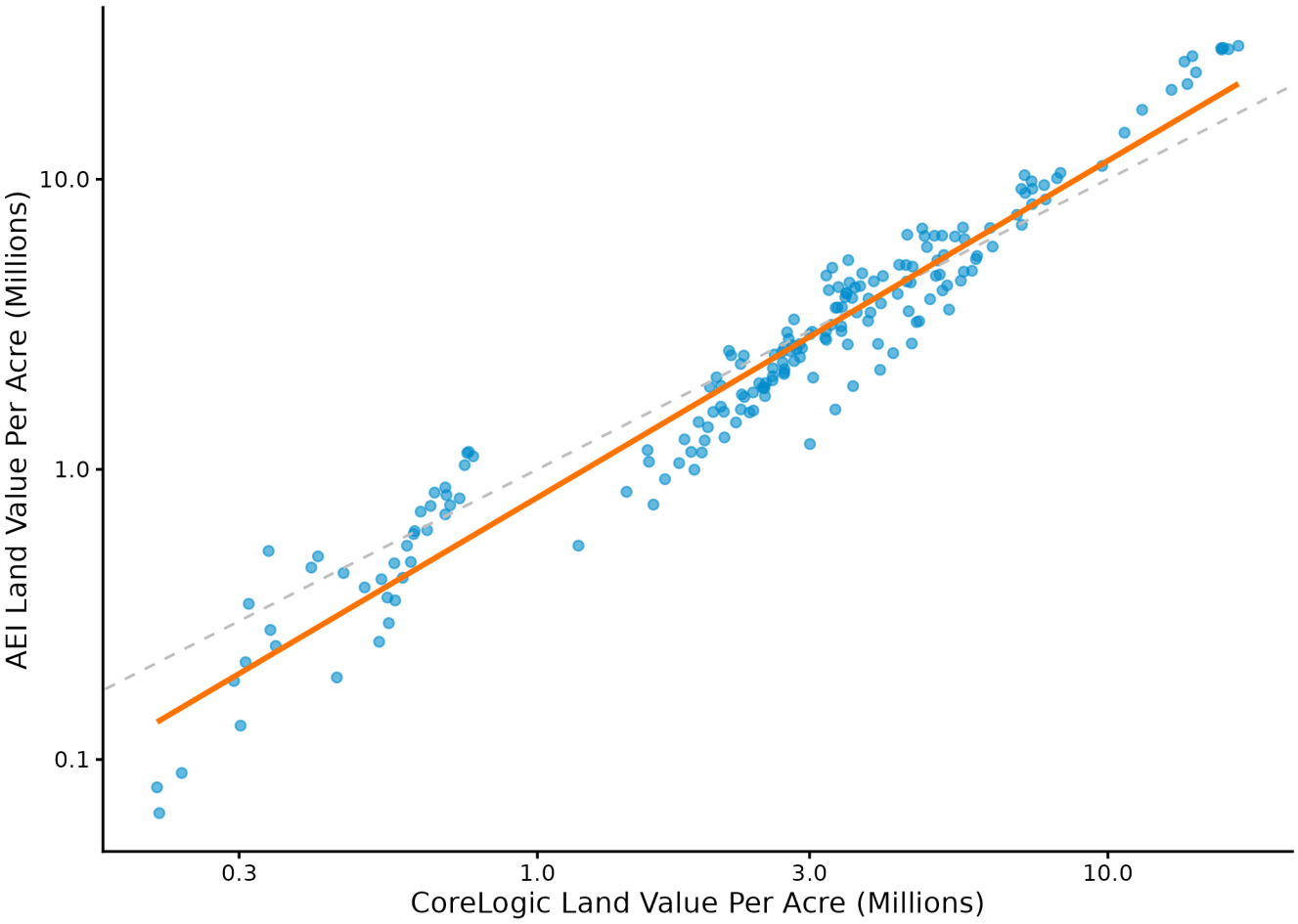


(b) Parcels Within 450m of the California Coastal Zone Boundary



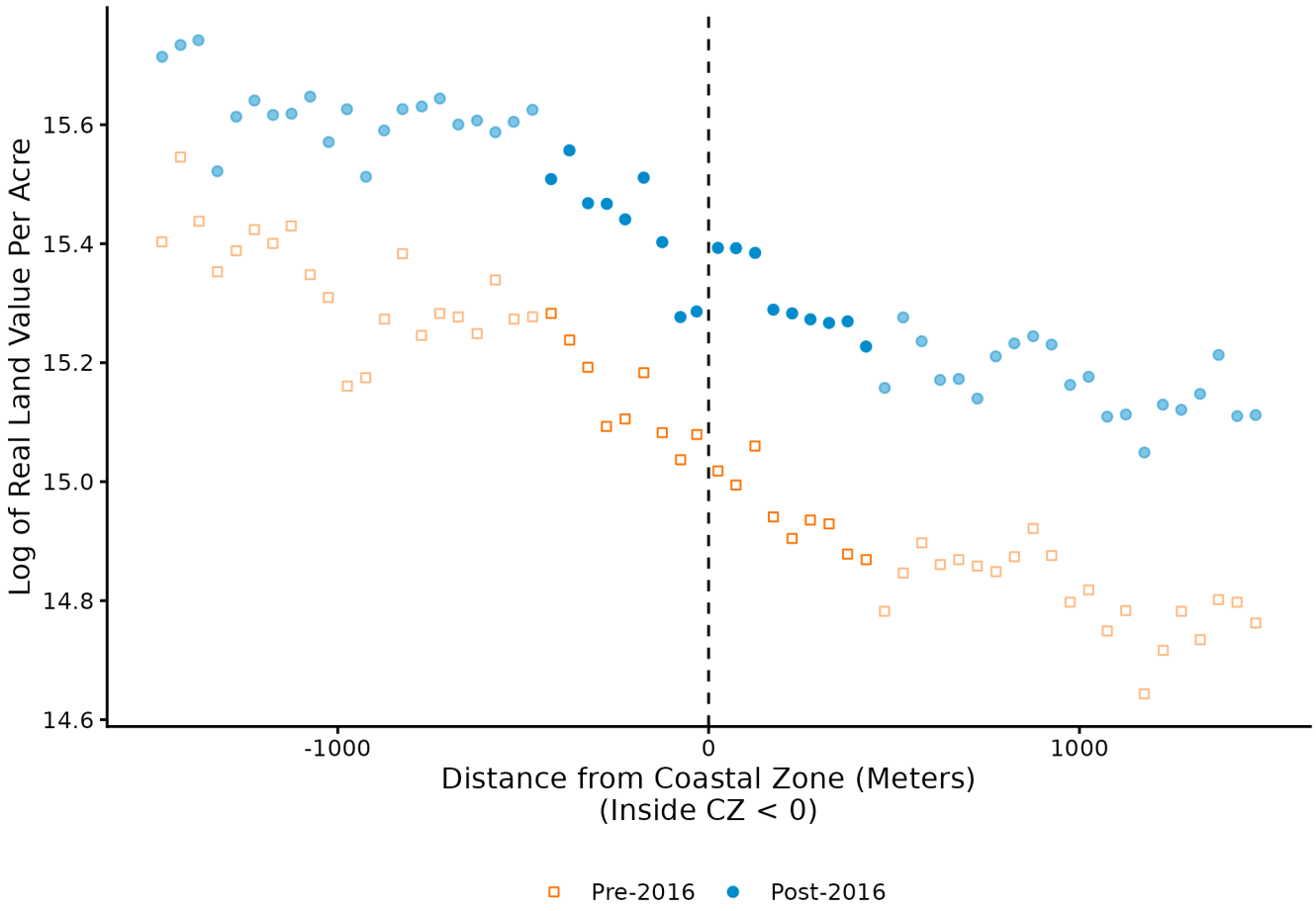
This figure uses data from Monkkonen, Manville and Lens (2024), U.S. Census Bureau (2016) and Cotality (2025). Panel A shows zoned capacity from the 2014 RHNA planning cycle per 2016 housing unit for all California jurisdictions (counties, cities) in orange and for cities within the 15 coastal counties in blue. Panel B shows the zoned capacity per housing unit of residential parcels in the 450m bandwidth difference in discontinuity sample.

Figure A.2: Comparison of County-Year Average Land Values Between Cotality (2025) and Davis et al. (2021) from 2012 to 2024



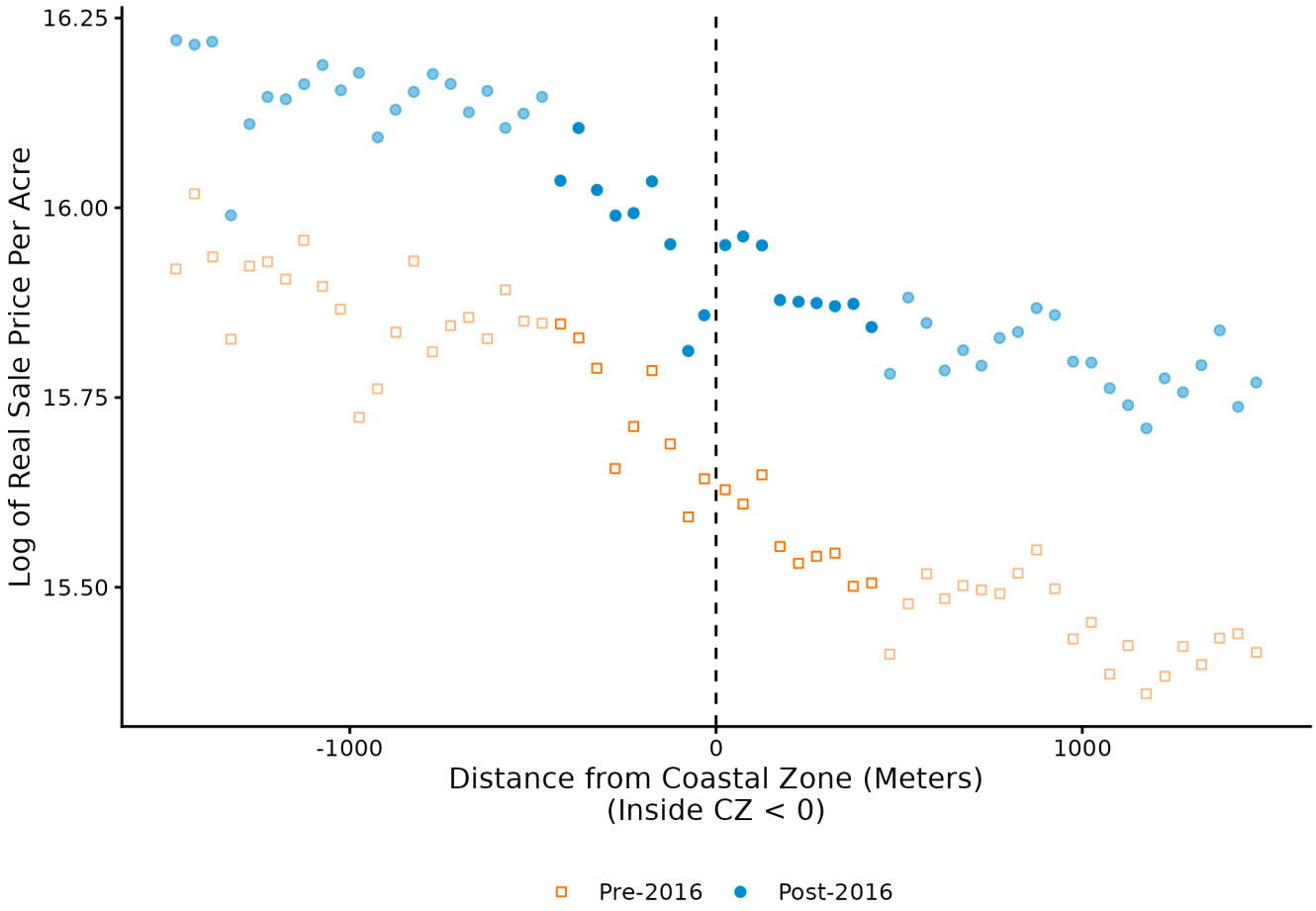
This figure uses data from Cotality (2025) and Davis et al. (2021) to compare county-year average land values. The horizontal axis shows the county average of the (real) land value per acre value used throughout this paper calculated using all residential parcels. The vertical axis uses the county-year land value data from Davis et al. (2021) as a comparison. The equality line is shown as a dashed line, and the best linear fit is shown in solid orange. Both axes are on log scales. The regression fit has an R-squared of 0.9 and a regression coefficient of 1.1.

Figure A.3: Changes in Real Land Values per Acre Around the Coastal Zone Boundary Pre- and Post-2016 Reforms



This figure uses data from Cotality (2025) and California Coastal Commission (2014) to show how residential land values change across the Coastal Zone boundary between 2010 and 2024. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the post 2016 permitting reforms. We group parcels in 50-meter bins. The vertical axis shows the log of the real land value per acre for each bin. The orange squares show the 2010-2016 distribution of land values in relation to the Coastal Zone boundary. The blue circles show the 2017 to 2024 distribution of land values in relationship to the Coastal Zone boundary. The difference in discontinuities estimate can be seen in comparing the gap between the pre- and post-2016 lines just before and then just after crossing the Coastal Zone boundary at 0.

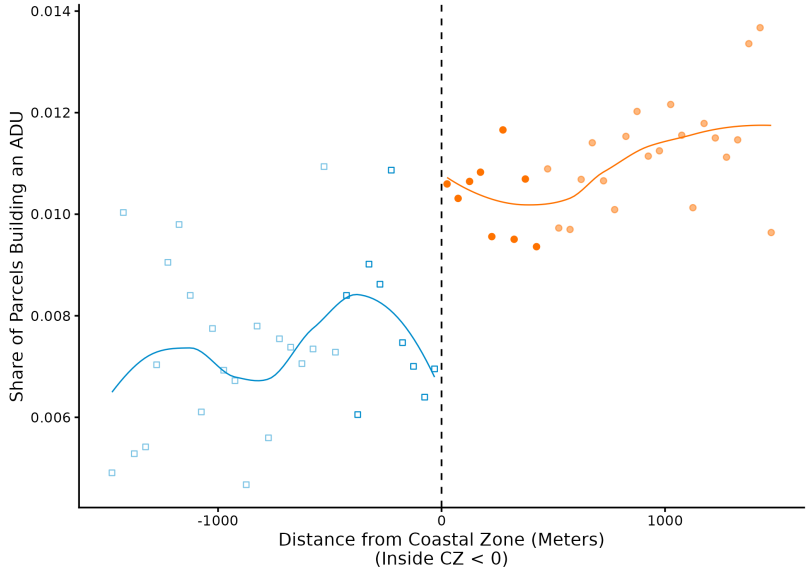
Figure A.4: Change in Real Total Sale Prices per Acre Around the Coastal Zone Boundary Pre- and Post- 2016 Reforms



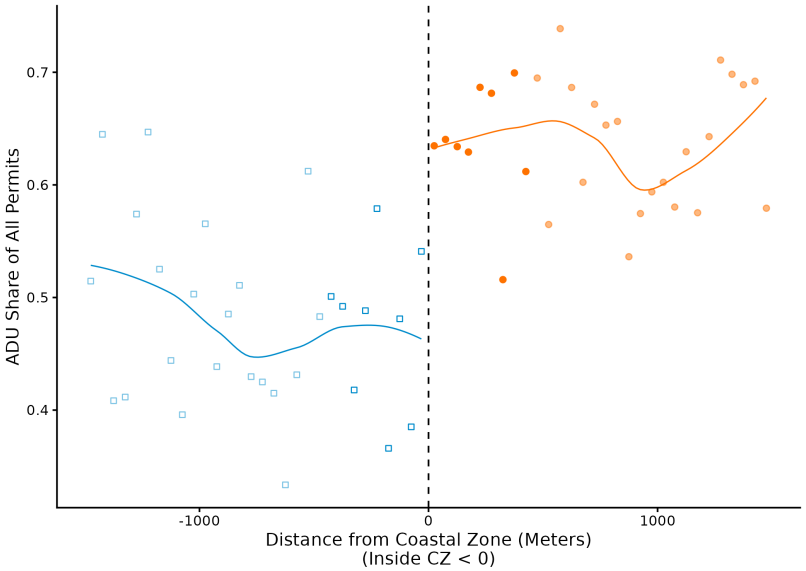
This figure uses data from Cotality (2025) and California Coastal Commission (2014) to show how residential total sale values change across the Coastal Zone boundary between 2010 and 2024. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the post 2016 permitting reforms. We group parcels in 50-meter bins. The vertical axis shows the log of the real total sale price per acre for each bin. The orange squares show the 2010-2016 distribution of total sale values in relation to the Coastal Zone boundary. The blue circles show the 2017 to 2024 distribution of total sale values in relationship to the Coastal Zone boundary. The difference in discontinuities estimate can be seen in comparing the gap between the pre- and post-2016 lines just before and then just after crossing the Coastal Zone boundary at 0.

Figure A.5: Accessory Dwelling Unit Permit Activity Around the Coastal Zone Boundary by Distance Bin After 2018

(a) Accessory Dwelling Unit Permits Per Parcel



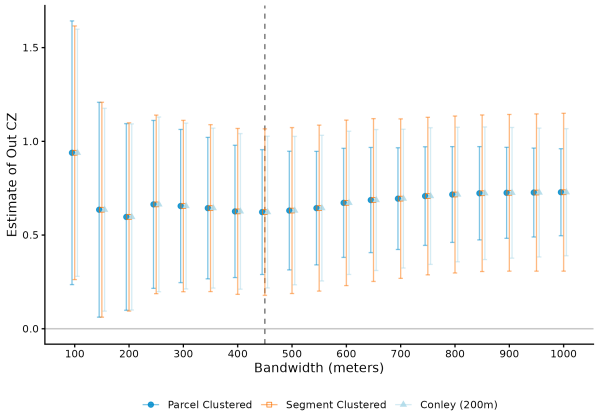
(b) Accessory Dwelling Unit Permits as a Share of All Building Permits



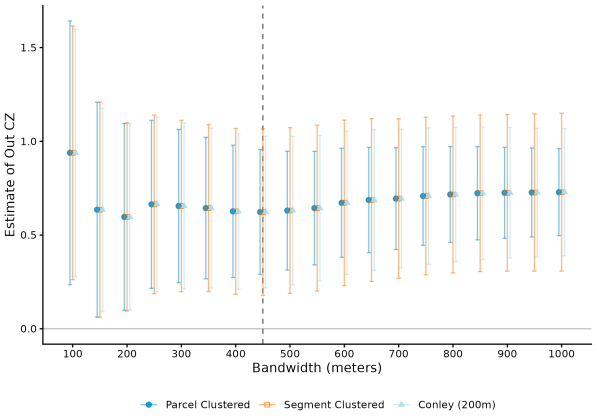
This figure uses data from Cotality (2025), California Department of Housing and Community Development (2025), and California Coastal Commission (2014) to show how permitting activity for ADUs changes across the Coastal Zone boundary between 2018 and 2024. The horizontal axis in both panels shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the permitting reforms (blue squares). We group parcels in 50-meter bins. In Panel A, we show the number of ADU permits on a per-parcel basis for each bin. In Panel B, we scale the permit counts by the total number of building permits issued in each bin. Our 450 meter analysis sample consists of the 18 closest bins to the coastal zone boundary, shown in bold. We provide a simple local estimated scatterplot smoothing (LOESS) line for visual convenience.

Figure A.6: Robustness of Accessory Dwelling Unit Permitting and Real Land Value Per Acre Estimates to Bandwidth and Standard Error Choice

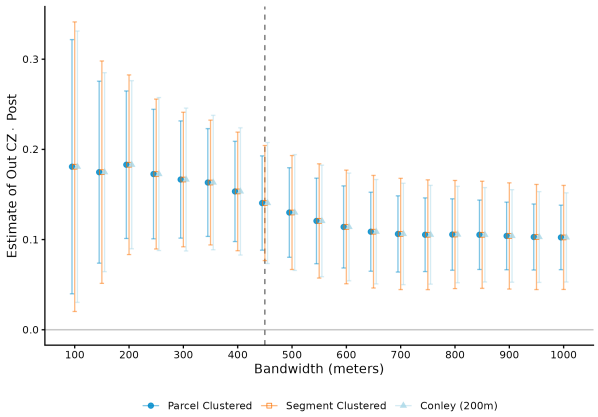
(a) ADU Permits: Pooled



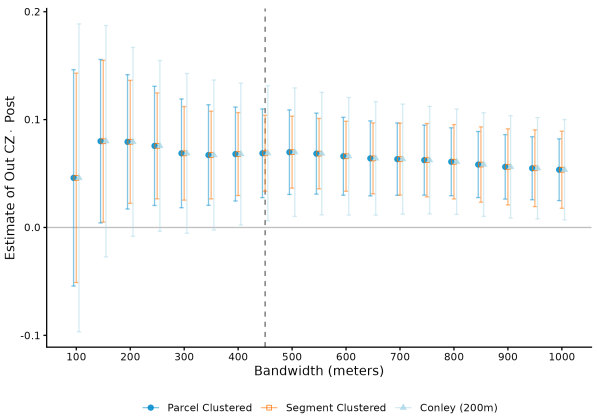
(b) ADU Permits: County by Year Fixed Effects



(c) Land Values: Pooled

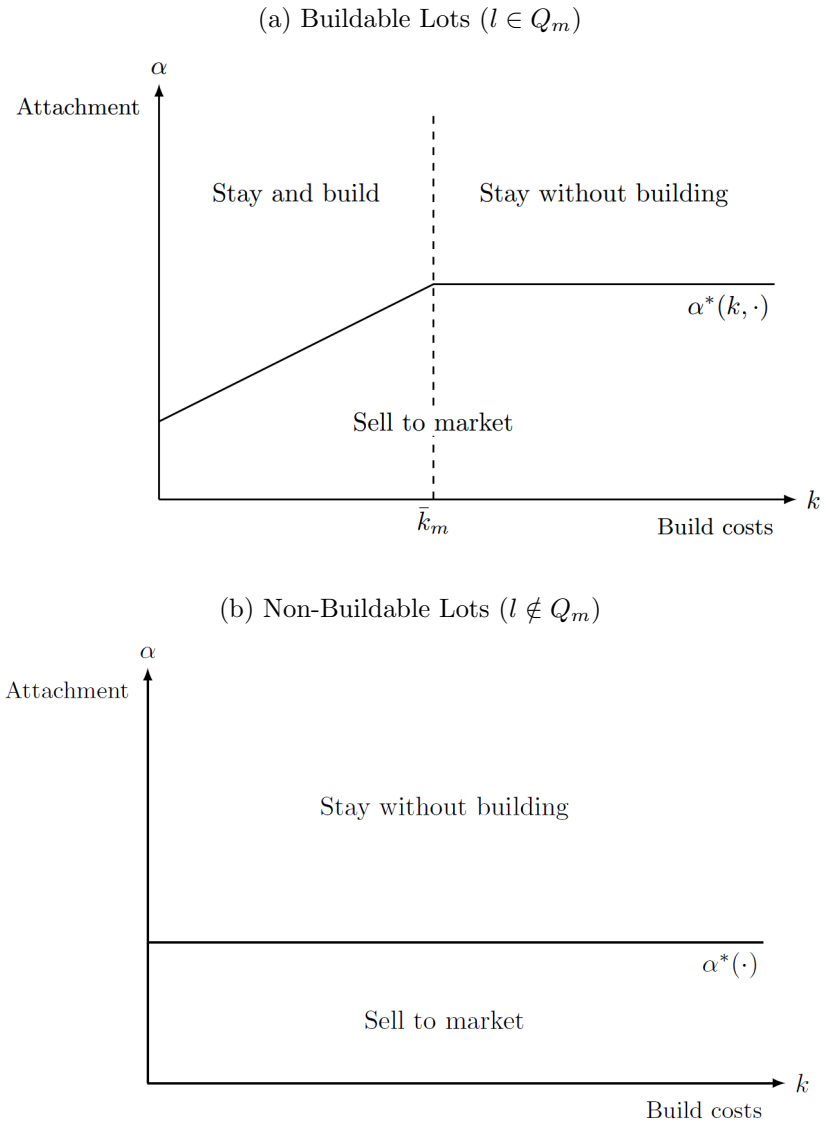


(d) Land Values: Repeat Sale + Segment by Year Fixed Effects



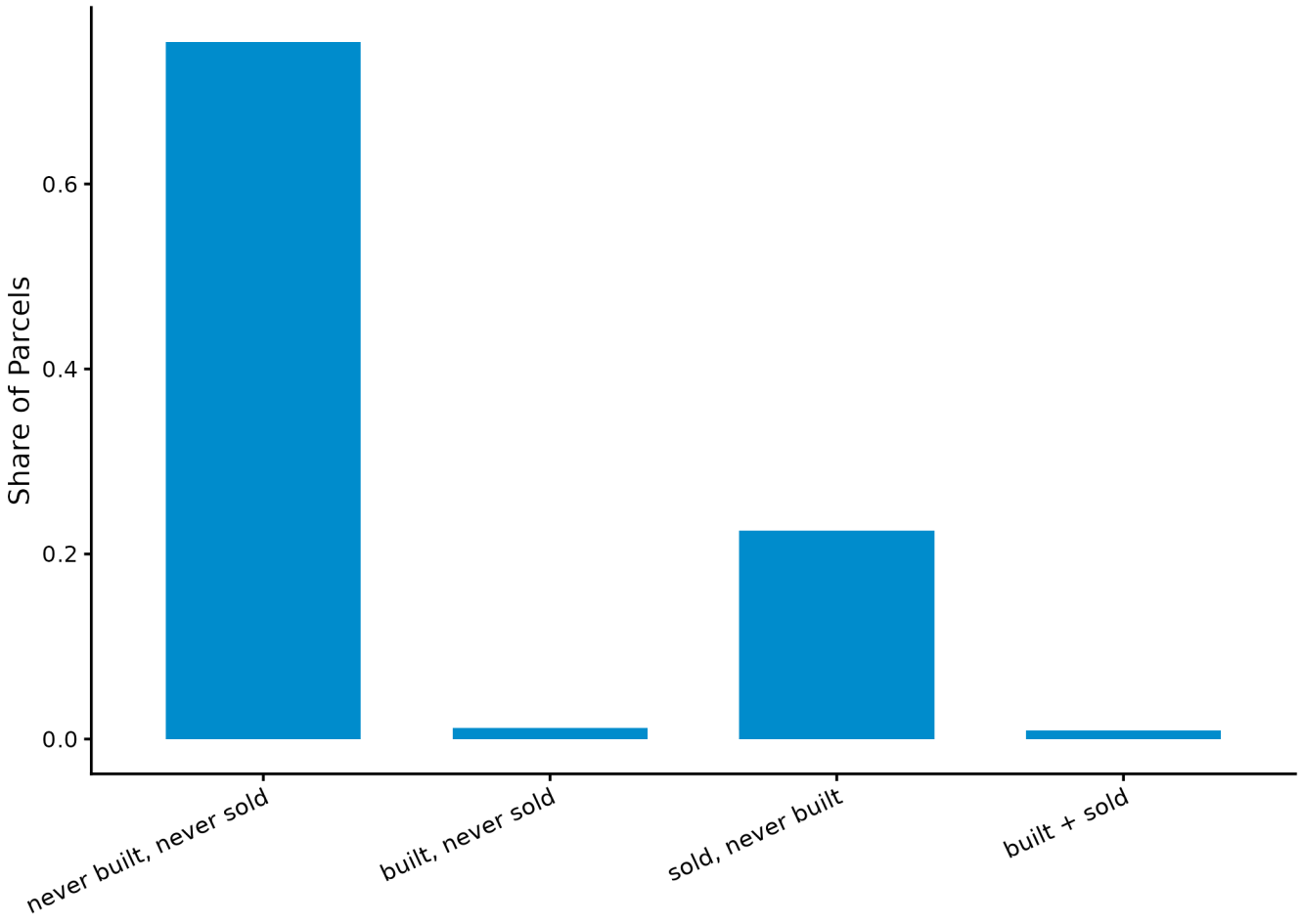
This figure shows estimates of Equation 2 in Panels A and B and Equation 3 in Panel C and D for alternative choices for the sample bandwidth cutoff and standard errors. We show these (symmetric) sample bandwidths on the horizontal axis and the coefficients of interest on the vertical axis. Our mean squared error optimal bandwidth choice is marked with a vertical line. These estimates correspond to Column 1 in Table 3 in Panel A, a modified Column 4 in Table 3 in Panel B, Column 1 in Table 4 in Panel C, and Column 6 in Table 4 in Panel D. The outcome variable in Panels A and B is a dummy taking a value of 1 if a parcel gets a permit for an ADU in a particular year and 0 otherwise, divided by 1,000 for scaling purposes. The outcome variable in Panels C and D is the log of the real land value per acre in 2025 USD. All estimates use triangle weights. We present standard errors clustered by parcel in blue, clustered by Coastal Zone segment in orange, and Conley (1999) spatial correlation robust errors (200 meters) in light blue.

Figure A.7: Incumbent Landowner's Outcomes by Attachment and Build Costs for Both Buildable and Non-Buildable Parcels



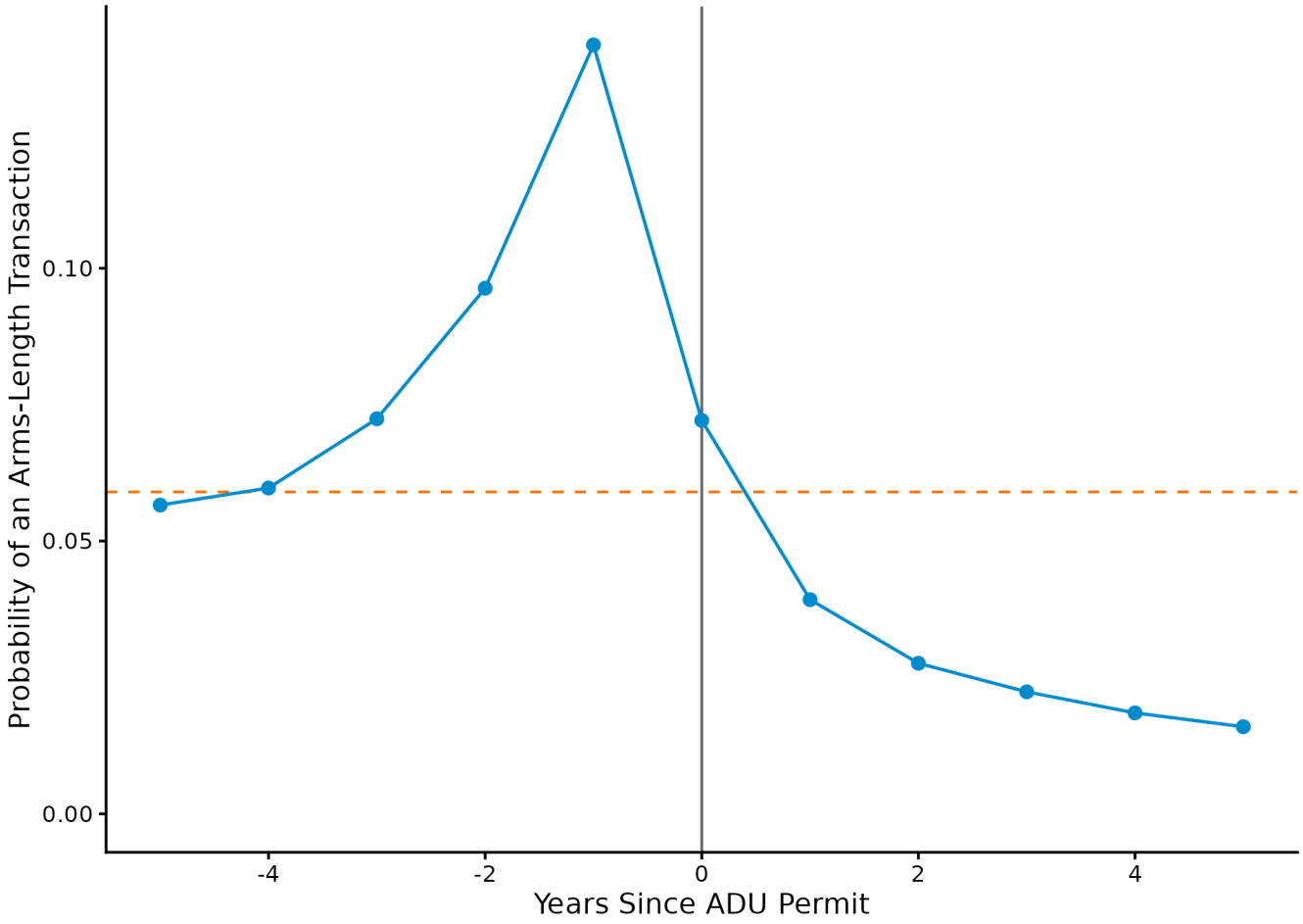
This figure shows the incumbent owner's outcomes based on their location attachment, α, and cost of building an ADU, k. The top panel shows the owner's choices and outcomes for lots of buildable quality, $l \in Q_m$. Here, the indifference threshold for selling to the market, $\alpha^*(k, \cdot)$, is a function of the owner's build costs and other parameters. In the bottom panel, the owner's choices and outcomes are shown when their lot is not of buildable quality, $l \notin Q_m$. Here, the owner's indifference function between selling and staying, $\alpha^*(\cdot)$, is not a function of build costs. For details, see Section 6.

Figure A.8: Share of Parcels that Ever Transacted or Permitted an ADU between 2018 and 2024 in the 15 Coastal Counties



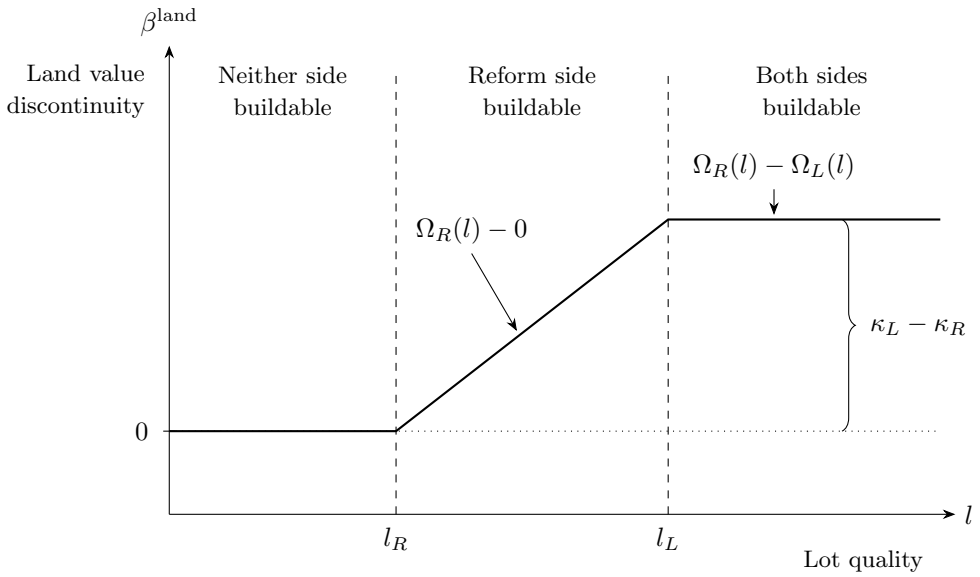
This figure uses data from Cotality (2025) and California Department of Housing and Community Development (2025) to show which parcels ever transact, ever build an ADU, or both between 2018 and 2024. Transactions here are arms-length transactions. 75% of parcels never build an ADU and never transact. 22.5% of parcels transact but never build an ADU, 2.5% of parcels build an ADU, with 1% both transacting and building and 1.5% building but not transacting. This figure uses data from all residential parcels within the coastal 15 counties.

Figure A.9: Parcel Transaction Probabilities Centered Around the Year an ADU is Permitted



This figure shows the unconditional probability that a parcel that pulls an ADU permit in year 0 transacts in each year within a 5 year window of the permit's issue date. Each parcel that pulls an ADU permit is included and event time is centered around the first ADU permit year. The probabilities of transaction are calculated from all parcels at risk of transacting to correct for short panel biases. Parcels can transact more than one time. The dashed orange horizontal line shows the probability that a parcel that builds an ADU transacts in a given year.

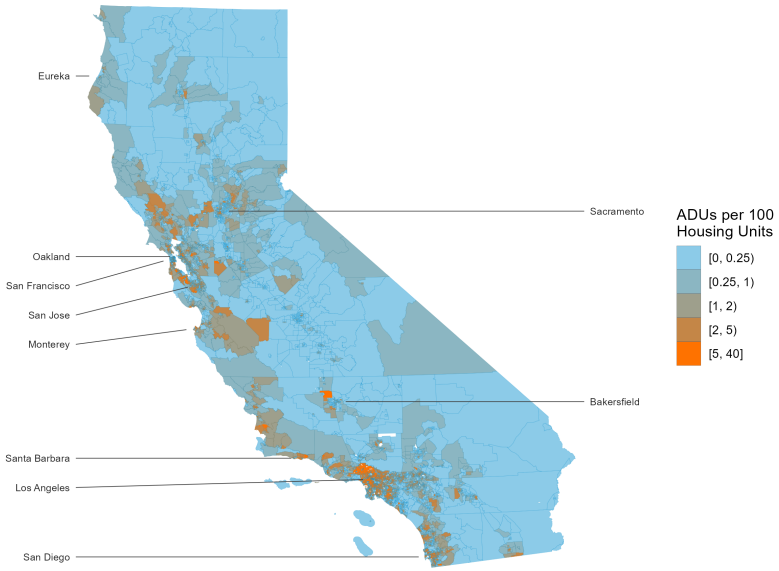
Figure A.10: Model Interpretation of the Difference in Discontinuities Estimator for Land Value Changes Across the Permitting Boundary by Lot Quality



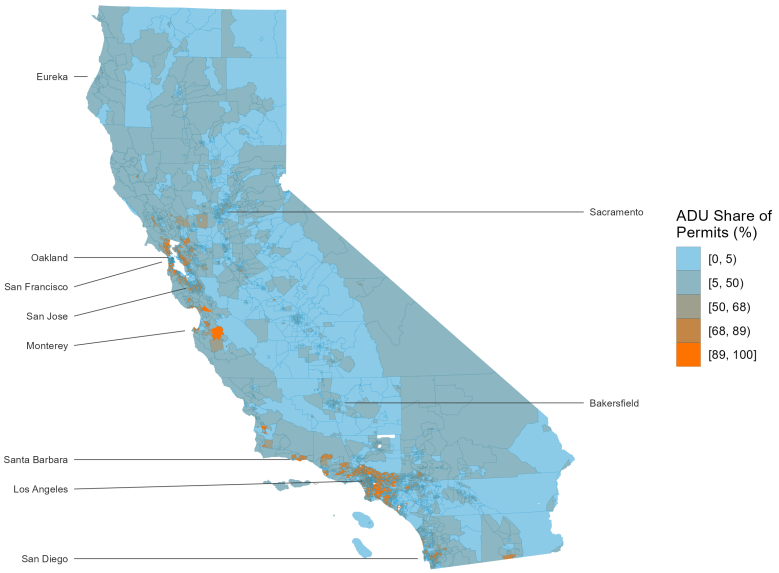
This figure shows the relationship between the model interpretation of the land value difference in discontinuities estimates by lot quality. The difference in land values across the permitting boundary can be expressed as $\Omega_R(l) - \Omega_L(l)$, which has three distinct regions. In the non-buildable low quality region, the gap is 0. The difference increases with lot quality as permitting costs on the reformed side of the boundary permit construction between $[l_R, l_L]$. Finally, when ADUs are viable on either side of the boundary, the difference in land values reflects the difference in permitting costs, $\kappa_L - \kappa_R$. For details, see Equation 18 in Section 6.

Figure A.11: Maps of Accessory Dwelling Unit Permits Issued in California Between 2018 and 2024

(a) ADU Permits Per Housing Unit in 2016



(b) ADU Permits as a Share of All Permits Between 2018 and 2024



This figure shows two maps of ADU permits from California Department of Housing and Community Development (2025) issued in California between 2018 and 2024. Permits are matched to census tracts using assessor parcel numbers in the Cotality (2025) data and to block groups as a fallback using U.S. Census Bureau (2026). Panel A shows ADU permits as a share of all Census tract housing units from U.S. Census Bureau (2016). Panel B shows ADU permits as a share of all housing permits issued within the Census tract.

Table A.1: Summary Statistics for the Full County Sample

Panel A: Parcel Panel Data	Mean	SD	Min	Max
	(1)	(2)	(3)	(4)
Share SFR	0.911	0.284	0.000	1.000
Dist. to the CZ	14 486	14 750	−44 394	108 133
Effective Build Year	1967	22.14	1850	2024
Bedrooms	3.366	1.063	1.000	11.00
Bathrooms	2.369	1.057	1.000	9.000
Building Sq. Ft.	1923	924.8	598.0	9999
Land Share	0.516	0.192	0.000	1.000
ADU Permit	0.001	0.038	0.000	1.000
Any Permit	0.002	0.047	0.000	1.000
ADU Suitable Pre-2016	0.880	0.325	0.000	1.000
Property Tax Per Acre	44 052	46 119	0.000	343 699
Panel B: Transaction Data	Mean	SD	Min	Max
	(1)	(2)	(3)	(4)
Share SFR	0.923	0.266	0.000	1.000
Dist. to the CZ	15 540	16 049	−42 121	103 660
Effective Build Year	1968	23.22	1850	2023
Bedrooms	3.328	1.045	1.000	11.00
Bathrooms	2.395	1.081	1.000	9.000
Building Sq. Ft.	1932	956.8	598.0	9999
Land Value Per Acre	3 832 004	4 426 889	7.878	49 570 267
Total Sale Price Per Acre	6 801 396	6 669 696	20 511	68 344 349
Land Share	0.530	0.194	0.000	1.000

Notes: This table presents summary statistics for the full 15 coastal county samples using data from Cotality (2025), California Coastal Commission (2014), and California Department of Housing and Community Development (2025). We show summary statistics from 2010 to 2024 on the full county sample using the balanced parcel panel dataset in Panel A and the transaction dataset in Panel B. We show summary statistics for the share of parcels that are zoned for single-family use, the parcel's euclidean distance to the Coastal Zone boundary, the parcel's effective build year, the parcel structure's number of bedroom, bathrooms, building square footage, land values per acre, total sale value per acre, the share of value assessed to come from the land, the share of parcels in a particular year that have an ADU permit or any building permit associated with them, the share of parcels that were ADU suitable before 2016 as defined as having a ground slope under 20% and more than 2,000 sq. ft. available as excess lot space, and the average property taxes paid per acre.

Table A.2: Cumulative Description of Sample Restrictions

Sample Restriction	Transaction Data (1)	Balanced Panel (2)
Total Observations	10,037,688	82,360,928
+ Residential Properties	8,093,225	66,608,192
+ Arms-Length Transactions	2,362,389	–
+ Assessed Value $\geq$ \$1,000	2,362,344	66,579,941
+ Valid Sale Price	2,322,650	–
+ Sale Volume Not Extreme	2,270,456	–
+ Valid Rooms/Units Ratio	2,270,456	66,482,505
+ Land Value Per Acre > \$1	2,160,457	–
+ Valid Implied Age	2,160,457	66,482,424
Final Sample	2,160,457	66,482,424

Notes: This table presents the impact of sample restrictions for the full 15 coastal county samples using data from Cotality (2025), California Coastal Commission (2014), and California Department of Housing and Community Development (2025). We show how each restriction reduces the total number of observations cumulatively descending through the rows. To start, all parcels must have an identifier and a valid location (latitude and longitude). Column 1 shows how restrictions impact the transaction dataset. These include a restriction to small residential properties, generally single-family houses but strictly just excluding large multifamily units and condos; restricting to arms-length (non family) transactions; limiting to parcels with assessed values greater than \$1000 USD; having a sale price within the 1st and 99th percentile of the sample distribution; not having an extreme number of transactions above the 99th percentile of the sample distribution per year; having a room-to-unit ratio within the 1st and 99th percentile of the sample distribution; having a land value per acre above \$1; and having an internally consistent structure age. In Column 2, we show these restrictions for the balanced panel dataset, which has one observation per parcel between 2010 and 2025, when the restriction is applicable since the data is only based on assessment records and not transaction records. Since we only have a single quarter of 2025 data from Cotality (2025), we stop our analysis sample at 2024.

Table A.3: Interaction of Sample Restrictions with the Coastal Zone Boundary in the Transaction and Parcel Datasets

	Dep.Var.: Observation Dropped by a Sample Restriction					
	2010–2024		2010–2016		2017–2024	
	(1)	(2)	(3)	(4)	(5)	(6)
OutCZ · Post	−0.011**	0.000				
	(0.004)	(0.000)				
OutCZ	0.001	−0.018***	0.002	−0.019***	−0.010***	−0.019***
	(0.003)	(0.004)	(0.003)	(0.004)	(0.003)	(0.004)
1st Degree Polynomial	Y	Y	Y	Y	Y	Y
Difference in Discontinuities	Y	Y				
County FEs	Y	Y	Y	Y	Y	Y
Year FEs	Y	Y	Y	Y	Y	Y
N	600 027	4 437 135	300 983	2 070 663	299 044	2 366 472
Adj. R^2	0.017	0.104	0.017	0.104	0.017	0.104
Mean	0.827	0.273	0.832	0.272	0.822	0.273
SD	0.378	0.445	0.374	0.445	0.382	0.445
In-CZ Mean	0.841	0.318	0.845	0.318	0.837	0.318
In-CZ SD	0.366	0.466	0.362	0.466	0.370	0.466
Data	Transactions	Balanced Panel	Transactions	Balanced Panel	Transactions	Balanced Panel

Notes: The following table uses data from Cotality (2025) and California Department of Housing and Community Development (2025) to see how sample restrictions interact with our pooled regression equations. In both columns the outcome variable is a dummy variable for exclusion from the analysis sample, taking a value of 1 if an observation is trimmed from the final sample. In Column 1, we show the pooled version of Equation 3, which shows us that our sample exclusions are not changing over time. In Column 2, we show our pooled version of Equation 2 for only the 2018 and after sample in which we run our regressions. Here, we see that properties inside the Coastal Zone are more likely to be excluded. However, from Column 1 we can see that these restrictions likely don't change over time. *$p < .1$, ** $p < .05$, *** $p < .01$

Table A.4: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Total Sale Prices within 450 Meters of the Coastal Zone Boundary

	Dep. Var.: Log of Real Total Sale Price Per Acre					
	(1)	(2)	(3)	(4)	(5)	(6)
$OutCZ \cdot Post$	0.112***	0.098***	0.062***	0.049**	0.046***	0.030**
	(0.024)	(0.023)	(0.020)	(0.019)	(0.016)	(0.015)
1st Degree Polynomial	Y	Y	Y	Y	Y	Y
County FEs	Y	Y	Y	Y		
Parcel FEs					Y	Y
Year FEs	Y	Y			Y	
Segment · Year FEs			Y	Y		Y
Pre-2016 Static Hedonics		Y		Y		
Repeat-Sale Sample					Y	Y
N	103 766	102 901	103 718	102 850	49 440	49 287
Adj. R^2	0.496	0.526	0.643	0.665	0.933	0.937
Mean Pre-2016	15.64	15.64	15.64	15.64	15.64	15.64
SD Pre-2016	1.080	1.080	1.080	1.080	1.080	1.080
In-CZ Mean Pre-2016	15.75	15.75	15.75	15.75	15.75	15.75
In-CZ SD Pre-2016	1.147	1.147	1.147	1.147	1.145	1.145

Notes: This table uses data from Cotality (2025) and California Coastal Commission (2014) to estimate Equation 3 for a transactions on residential parcels between 2010 and 2024: $ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable is the log of the real total sale price per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarcating the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first-degree polynomial form. Distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c, and a set of year fixed effects, τ_t. We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We cluster our robust standard errors at the parcel-level. We report estimates using a 450-meter bandwidth from the Coastal Zone boundary. We present summary statistics for the log of the real land value per acre for both the entire sample from 2010 to 2016 and just the sample within the Coastal Zone from 2010 to 2016 at the bottom of the table. In Column 1, we present a specification only with a set of county fixed effects and a set of year fixed effects. In Columns 2 and 4 we add hedonic fixed effects. Then in Column 3, 4, and 6, we include the segment by year fixed effects. Finally, in columns 5 and 6, we include a parcel fixed effect and use a sample restricted to parcels that transact both between 2010 to 2016 and 2017 to 2024. All specifications use triangle weights and report parcel clustered standard errors. *$p < .1$, ** $p < .05$, *** $p < .01$

Table A.5: Second-Degree Polynomial Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Land Values within 450 Meters of the Coastal Zone Boundary

	Dep.Var.: Log of Real Land Value Per Acre			
	(1)	(2)	(3)	(4)
$OutCZ \cdot Post$	0.197***	0.176***	0.116***	0.093***
	(0.041)	(0.041)	(0.033)	(0.033)
2nd Degree Polynomial	Y	Y	Y	Y
County FEs	Y	Y	Y	Y
Year FEs	Y	Y		
Segment · Year FEs			Y	Y
Pre-2016 Static Hedonics		Y		Y
N	103 766	102 901	103 718	102 850
Adj. R^2.	0.524	0.546	0.662	0.678
Mean Pre-2016	15.01	15.01	15.01	15.01
SD Pre-2016	1.246	1.246	1.246	1.246
In-CZ Mean Pre-2016	15.14	15.14	15.14	15.14
In-CZ SD Pre-2016	1.317	1.317	1.317	1.317

Notes: This table uses data from Cotality (2025) and California Coastal Commission (2014) to estimate Equation 3 for a transactions on residential parcels between 2010 and 2024: $ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(DistCZ_i \times Post_t) + \beta_3 DistCZ_i + \beta_4(DistCZ_i^2 \times Post_t) + \beta_5 DistCZ_i^2 + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable is the log of the real land value per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarcating the implementation of the first ADU reforms. $DistCZ_i$ is a function of a parcel's distance to the Coastal Zone boundary, which uses a second-degree polynomial here. We also include a set of county fixed effects, λ_c, and a set of year fixed effects, τ_t. We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We present summary statistics for the permit dummy for both the entire sample and just the sample within the Coastal Zone at the bottom of the table. In Column 1, we present a specification only with a second-degree polynomial, a set of county fixed effects, and a set of year fixed effects. Then in Columns 2 and 4, we include the set of hedonic fixed effects. Then in columns 3 and 4, we replace the year fixed effects with a segment by year fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. We cluster our robust standard errors at the parcel-level. We report estimates using a 450-meter bandwidth from the Coastal Zone boundary and use triangle weights. *$p < .1$, ** $p < .05$, *** $p < .01$

Table A.6: Second-Degree Polynomial Spatial Regression Discontinuity Estimates of the Impacts of Permitting Reforms on Accessory Dwelling Unit Permits after 2018 within 450 Meters of the Coastal Zone Boundary

	Dep. Var.: ADUs per 1,000 Parcel-Years			
	(1)	(2)	(3)	(4)
OutCZ	0.676***	0.616**	0.512**	0.481*
	(0.254)	(0.259)	(0.255)	(0.256)
2nd Degree Polynomial	Y	Y	Y	Y
County FEs	Y	Y	Y	Y
Year FEs	Y	Y		
Segment · Year FEs			Y	Y
Pre-2016 Static Hedonics		Y		Y
N	1 505 988	1 485 253	1 505 974	1 485 239
Adj. R^2	0.001	0.002	0.002	0.003
Mean	2.248	2.248	2.248	2.248
SD	47.36	47.36	47.36	47.36
In-CZ Mean	1.900	1.900	1.900	1.900
In-CZ SD	43.55	43.55	43.55	43.55

Notes: This table uses data from Cotality (2025), California Department of Housing and Community Development (2025), and California Coastal Commission (2014) to estimate Equation 2 for a balanced set of residential parcels between 2018 and 2024: $Y_{ict} = \alpha + \beta_1 OutCZ_i + \beta_2 DistCZ_i + \beta_3(DistCZ_i \cdot OutCZ_i) + \beta_4 DistCZ_i^2 + \beta_5(DistCZ_i^2 \cdot OutCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable is a dummy taking a value of 1 if parcel i has an ADU permit associated with it in year t, and we scale this outcome by multiplying by $1,000$ for display purposes. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's permitting reforms for ADUs. $DistCZ_i$ is a function of a parcel's distance to the Coastal Zone boundary, which uses a second-degree polynomial here. We also include a set of county fixed effects, λ_c, and a set of year fixed effects, τ_t. We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We present summary statistics for the permit dummy for both the entire sample and just the sample within the Coastal Zone at the bottom of the table. In Column 1, we present a specification only with a second-degree polynomial, a set of county fixed effects, and a set of year fixed effects. Then in Columns 2 and 4, we include the set of hedonic fixed effects. Then in columns 3 and 4, we replace the year fixed effects with a segment by year fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. We cluster our robust standard errors at the parcel-level. We report estimates using a 450-meter bandwidth from the Coastal Zone boundary and use triangle weights.*$p < .1$, ** $p < .05$, *** $p < .01$

Table A.7: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Alternatively Measured Land Values within 450 Meters of the Coastal Zone Boundary Using Time-Varying Hedonics

	Dep.Var.: Log of Real Total Sale Value Per Acre			
	(1)	(2)	(3)	(4)
$OutCZ \cdot Post$	0.111***	0.152***	0.063***	0.078***
	(0.023)	(0.036)	(0.019)	(0.029)
1st Degree Polynomial	Y		Y	
2nd Degree Polynomial		Y		Y
County FEs	Y	Y	Y	Y
Year FEs	Y	Y		
Segment · Year FEs			Y	Y
Updating Hedonics	Y	Y	Y	Y
N	103 737	103 737	103 689	103 689
Adj. R^2	0.526	0.526	0.666	0.666
Mean Pre-2016	15.64	15.64	15.64	15.64
SD Pre-2016	1.080	1.080	1.080	1.080
In-CZ Mean Pre-2016	15.75	15.75	15.75	15.75
In-CZ SD Pre-2016	1.147	1.147	1.147	1.147

Notes: This table uses data from Cotality (2025) and California Coastal Commission (2014) to estimate Equation 3 for a transactions on residential parcels between 2010 and 2024: $ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) + \theta'_{(it)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable is the log of the real total sale price per acre in 2025 USD. In each specification, we use a set of binned time varying hedonic fixed effects, $\theta'_{(it)}$, to capture the share of the total value associated with the market values of the current structural characteristics of the parcel. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 marking the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first-degree or second-degree polynomial form. In both polynomial specifications, distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c, and a set of year fixed effects, τ_t. ε_{ict} is an error term. We cluster our robust standard errors at the parcel-level. We report estimates using a 450-meter bandwidth from the Coastal Zone boundary and use triangle weights. We present summary statistics for the log of the real total sale price per acre for both the entire sample from 2010 to 2016 and just the sample within the Coastal Zone from 2010 to 2016 at the bottom of the table. In Column 1, we present a specification only with a first-degree polynomial, the set of time varying hedonic fixed effects, a set of county fixed effects, and a set of year fixed effects. In the even columns, we replace the first-degree polynomial with the second-degree polynomial. Then in Column 3 and 4, we replace the year fixed effects with a segment by year fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. *$p < .1$, ** $p < .05$, *** $p < .01$

Table A.8: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Land Values within 450 Meters of the Coastal Zone Boundary Dropping Voluntarily Adopting Segments

	Dep. Var.: Log of Real Land Value Per Acre			
	(1)	(2)	(3)	(4)
$OutCZ \cdot Post$	0.144***	0.089***	0.080***	0.066***
	(0.028)	(0.024)	(0.022)	(0.022)
1st Degree Polynomial	Y	Y	Y	Y
County FEs	Y	Y		
Parcel FEs			Y	Y
Year FEs	Y		Y	
Segment · Year FEs		Y		Y
N	92 961	92 914	44 700	44 563
Adj. R^2	0.544	0.681	0.912	0.916
Mean Pre-2016	15.01	15.01	15.02	15.02
SD Pre-2016	1.246	1.246	1.245	1.245
In-CZ Mean Pre-2016	15.14	15.14	15.14	15.14
In-CZ SD Pre-2016	1.317	1.317	1.316	1.316

Notes: This table uses data from Cotality (2025) and California Coastal Commission (2014) to estimate Equation 3 for a transactions on residential parcels between 2010 and 2024: $ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable is the log of the real land value per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarcating the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first-degree or second-degree polynomial form. We use a first-degree polynomial, where distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c, and a set of year fixed effects, τ_t. We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We cluster our robust standard errors at the parcel-level. We report estimates using a 450-meter bandwidth from the Coastal Zone boundary. We present summary statistics for the log of the real land value per acre for both the entire sample from 2010 to 2016 and just the sample within the Coastal Zone from 2010 to 2016 at the bottom of the table. In Columns 1 we show our pooled difference in discontinuities estimates. In Columns 2 and 4, we swap the year fixed effects for segment by year fixed effects. Then, in Columns 3 and 4, we replace our pooled specification with a repeat sale model that includes a parcel fixed effect. All specifications use triangle weights. All specifications drop segments of the Coastal Zone that overlap with Santa Cruz, Santa Barbara, Encinitas, San Mateo, and Pacifica. $^*p < .1$, ** $p < .05$, *** $p < .01$

B Appendix: Change Log from August 2025 Filed Pre-Analysis Plan

Audited vintage	
Pre-analysis plan	Filed 2025-08-18 (title page dated 19 August 2025)
Manuscript audited	2026-09-24 (draft before this appendix was inserted)
Audit run	2026-09-24
Model	Claude Opus 5 (`claude-opus-5`), via Claude Code
Econometric benchmark	Cunningham, *Causal Inference: The Mixtape*, accessed 2026-09-24
Materials	To be supplied by the authors

Pre-analysis plans entered economics through experimental work, where the analyst controls assignment and can fix an estimating equation before any outcome exists. Over the past decade they have migrated into observational applied microeconomics, where the underlying logic — binding the analyst's hands before the data speak — is harder to apply and correspondingly more valuable, because the researcher's degrees of freedom are if anything wider. A recent example at scale is the pre-registered observational study of Medicaid and mortality among low-income adults by Wyse and Meyer Wyse and Meyer (2025).

Full pre-commitment is rarely feasible outside an experiment. Observational data are assembled rather than generated: coverage, variable definitions, linkage rates and realised sample sizes are learned only after the files are opened. A plan written before that point will specify some estimands that cannot be constructed and some equations that cannot be identified. Demanding literal adherence would force authors either to report analyses they know to be uninformative or to abandon pre-registration altogether, and neither outcome serves a reader.

The partially pre-committed research design is the response to that problem. Authors commit in advance to what can be fixed — hypotheses, treatment and control definitions, estimating equations, outcome construction, inference rules — and state openly which elements must remain contingent on what the data turn out to contain. What matters afterwards is not a count of departures but their character: whether a change was forced by the data, whether it moves the analysis toward established practice, and whether it was disclosed where a reader would look for it. The approach was developed by Clemens and Strain in their study of large and small minimum wage changes Clemens and Strain (2021), and carried forward by Clemens, Mahajan and Sabia in their work on the long-run effects of the Affordable Care Act over the COVID-19 recession and recovery Clemens, Mahajan and Sabia (2026), who were also the first to demonstrate an audit of adherence to a pre-analysis plan conducted by a large language model. Clemens and Mahajan take up the utility of such tools for auditing adherence more generally Clemens and Mahajan (2026). This appendix exists because of that line of work and follows its procedure.

Disclosure. This appendix was written in its entirety by a large language model: Claude Opus 5, model identifier `claude-opus-5`, running under Claude Code. The model read the filed pre-analysis plan and this manuscript and produced the tables, the metrics, the commentary, and this introduction. The authors have read the result in full; their response appears in the final section of this appendix. The raw model output has not been edited, condensed, or corrected.

The review process

The audit ran as a sequence of agents with deliberately restricted views, so that no single model both formed a judgment and controlled the record of it.

An extraction agent read the pre-analysis plan alone, with the manuscript withheld, and produced an inventory of every commitment the plan makes, each tagged by the strength with which it is stated. Withholding the paper matters: an extractor that can see the results tends to find commitments where the paper delivered and overlook those it did not. A second agent mapped that inventory onto the manuscript, recording where each commitment is executed and locating analyses in the paper with no counterpart in the plan, while leaving every judgment column blank.

Two auditors then worked on the mapped inventory independently, in isolated sessions. One was instructed to build the strongest defensible case that the paper departs from its plan; the other to build the most charitable defensible reading. Neither saw the other's instructions or output, and neither was told a counterpart existed. The point is not that either reading is correct, but that both arguments are made at full strength before anything is decided.

An adjudicating agent received both reasoning logs together with the plan and the manuscript, and settled every item against the source documents rather than against the auditors' rhetoric. It fixed the status, verdict, rationale, direction, disclosure and severity recorded in Table B.2. Methodological conventions — standard practice rather than the authors' preference — were settled against the single designated econometric reference named above, so that no appeal to unsourced convention could do any work.

Consolidation and typesetting were then separated from judgment. A further agent grouped related items, mirrored that grouping across the two tables and set this appendix, with no authority to alter a judgment field, add or remove an item, or soften a finding. A final agent verified the bibliographic entries cited here.

Throughout, the plan is treated as a guide rather than a contract. A plan written before the data are opened will contain commitments the data defeat, and merely counting departures would score a forced substitution the same as a change made after seeing a coefficient. Departures are therefore classified by reason. The six frozen prompts are reproduced in the Prompts section at the end of this appendix.

Audit provenance.

Protocol

Version 0.5

Audit run

2026-09-24

Model

Claude Opus 5 (`claude-opus-5`)

Designated econometric benchmark

Cunningham, *Causal Inference: The Mixtape*, accessed 2026-09-24

Reference supplied as

cited from model knowledge

Inputs

Pre-analysis plan; manuscript including appendices and exhibit notes; designated reference

Not audited

Replication code and estimation logs

Prompts frozen

Shipped frozen with protocol v0.5; unedited during this run; before any output was viewed: Yes

Context-hygiene probe

passed

Isolation check

clean

Materials

Prompts, both reasoning logs, and the run map: To be supplied by the authors

Commitments and adjudication

Table B.1 lists the commitments extracted from the pre-analysis plan. Table B.2 records what the paper does with each and the resulting judgment. Items are keyed by ID: `H##` for hypotheses, `P##` for other pre-analysis plan commitments, `S##` for analyses in the paper with no counterpart in the plan.

Table B.1: Commitments extracted from the pre-analysis plan

ID	Key	Type	Tier	Commitment	PAP loc.
H01	Reform increases ADU permitting	Hypothesis	Explicit	Introduction of ADU permitting reforms in California after 2016 will increase permitting activity. Designated a primary hypothesis at §5.1 ("our primary hypotheses that reforms increased permitting").	§1.1; §5.1
H02	Reform increases land values	Hypothesis	Explicit	Introduction of ADU permitting reforms after 2016 will increase land values. Designated the "secondary hypothesis" evaluated by Equation (2).	§1.1; §5.1
H03	Small negative local spillovers	Hypothesis	Explicit	The reforms will "have small negative local spillovers onto other properties." Operationalized as price changes just inside the Coastal Zone boundary in Equation (3), where the plan allows the sign to be "either positive or negative."	§1.1; §5.1
P01, P11	Observational spatial RD design	Specification	Explicit	P01: the study is observational and will use non-randomly assigned Cotality (formerly CoreLogic) data to implement a spatial regression discontinuity design leveraging the California Coastal Zone's parallel permitting regime. P11: the design exploits the Coastal Zone boundary, which maintains stringent development regulations while non-Coastal Zone areas receive the state-mandated ADU permitting reform announced in 2016.	§2.1; §2.4
P02, P03, P10	Named data sources	Data source	Explicit	P02: Cotality parcel-level tax and transfer/assessment data, including lot and structure characteristics, ownership information, sales prices and tax assessments; coverage dates vary by jurisdiction. P03: the California Coastal Commission (2014) Coastal Zone Boundary shapefile [ds990], used to identify each parcel's location relative to the boundary. P10: building permits attached to the Cotality parcel tax data by assessor parcel number (APN), or by address if the APN is unavailable.	§2.4; §3.1; §3.2
P04, P15	Permit scrape superseded	Data source	Explicit	Building permits will be scraped from local jurisdictions within the 15 coastal California counties and used to construct the permitting outcome; the authors "will scrape as many local jurisdictional websites as possible to collect posted building permits."	§2.4; §3.2; §3.3

continued

Table B.1, continued

ID	Key	Type	Tier	Commitment	PAP loc.
P05, P09, P21, P22, P59	Sample construction screens	Sample / inclusion	Explicit	P05: "We will limit all of our samples to residential parcels within the 15 coastal counties." P09: the spillover analysis will use arms-length transactions only, arms-length being a transaction between non-family members. P21: "We will try to include as many parcels in our final analysis sample as possible, depending on data availability." P22: final-sample parcels must have assessments and building characteristics in the Cotality data sufficient to geo-locate them relative to the Coastal Zone, identify relevant building and lot characteristics, and construct land values. P59: "Parcels that have incomplete data in the Cotality dataset will be dropped by necessity."	§2.4; §3.5; §5.6
P06	Estimates at several bandwidths	Specification	Explicit	"We will present estimates at several bandwidths from the coastal zone boundary."	§2.4
P07	Permitting sample: all residential parcels	Sample / inclusion	Explicit	For the permitting analysis, all residential parcels across the boundary after 2016 will be included, in all jurisdictions for which permitting data are available to the authors.	§2.4
P08	Land-value sample: reassessment sales	Sample / inclusion	Explicit	"For estimates of land value changes, we will limit our sample to sales that trigger a reassessment of property value."	§2.4; §4.2
P12, P13	State of the data at registration	Data source / time horizon	Background	P12: for both the Cotality data and the local permit data, "the exact dates for which data are available vary by jurisdiction and are unknown at this time." P13: "At the time of this pre-registration, we have not opened the Cotality data, but we have been working to collect the local permit data by scraping local jurisdictions' permit pages as they are available." Both report the state of knowledge at registration and create no obligation on the paper.	§3.1; §3.2
P14	State-harmonized permit data from 2018	Data source	Inferred	From 2018 on, permit data are made available for the entire state of California and harmonized by the state (California Department of Housing and Community Development, 2022, APR data). The plan does not state explicitly that it will use this source, but describes it as part of the local permit data.	§3.2
P16	Table 1 target jurisdiction list	Data source	Implied	The cities from which permits will be attempted to be collected are listed in Table 1 (city–county pairs spanning the 15 coastal counties, including "Unincorporated" areas of each county and of the Coastal Zone).	§3.3 (Table 1)

continued

Table B.1, continued

ID	Key	Type	Tier	Commitment	PAP loc.
P17, P60	Jurisdiction-level permit screens	Sample / inclusion	Explicit	P17: jurisdictions will be included if (a) permits can be matched to the Cotality data, (b) they lie within the coastal zone distance bins used in the final sample, and (c) permits can be identified as attached to an ADU. P60: "We will also exclude jurisdictions from which we are unable to access building permit data."	§3.3; §5.6
P18, P20	Sample size unknown; ballpark calculation	Power / sample size	Background	P18: "The final sample size is unknown"; the Cotality data are expected to be incomplete, leaving only a fraction of lots with complete data and a smaller fraction within jurisdictions with usable permit data. P20: about 3.7 million tax returns were filed in zip codes touching the Coastal Zone; if a quarter map to usable lots that is roughly 1 million lots, and a move from 2% to 2.5% ADU prevalence would be about 5,000 new ADUs outside the Coastal Zone, which the power analysis suggests is detectable. Both are calculations already performed, not targets the paper must hit.	§3.4
P19, P46	Power analysis in the plan's Appendix A	Power / sample size	Background	P19: "We present a power analysis in Appendix A that uses very conservative sample sizes starting at 30,000 lots with about 2.5% of lots having an ADU at baseline, increasing to about 3% of lots after treatment. Even with these small samples, we are typically able to detect effects with parsimonious models." P46: "Our power analysis in Appendix A suggests that this more parsimonious functional form will increase the range of effect sizes which we will be able to detect." Both report work the plan itself already contains.	§3.4; §5.1
P23	Stopping rule for scraping	Data source	Explicit	Every jurisdiction possible will be scraped; some websites may be omitted because they are too complicated to scrape or do not post enough permit information online to be useful.	§3.6
P24	Primary outcome: ADU permit dummy	Outcome	Explicit	"Our main outcome variable is a binary for whether or not there is an ADU permit associated with a given parcel," constructed from permitting data. Lots with an ADU permit take value 1, lots without take 0.	§4.2
P25	Jurisdiction-specific ADU permit coding	Outcome	Explicit	For each jurisdiction the authors will identify how ADU permits are marked and create a dummy variable denoting these permits, then attach the ADU permits to the Cotality parcel data by APN or address.	§4.2
P26	Permitting also from building characteristics	Outcome	Explicit	ADU permitting activity will be measured "using both building permits and changes in building characteristics that corroborate ADU construction."	§5.1

continued

Table B.1, continued

ID	Key	Type	Tier	Commitment	PAP loc.
P27	Secondary outcome: assessed land value	Outcome	Explicit	As a secondary outcome the authors will look at land values, measured from the first assessment after an arms-length transaction (a transaction between non-family members) — i.e., the first assessed land value after a transaction that would trigger a new assessment.	§4.2
P28	Land value per square foot	Outcome	Explicit	The land-attributed portion of the assessment will be divided by the square footage of the lot, giving a land value in dollars per square foot for each lot. Land values will be presented on a per-square-foot basis.	§4.2; §5.3
P29	Third outcome: arms-length sale price	Outcome	Explicit	Sale price will be observed from the Cotality transaction data and limited to arms-length sales; it is the outcome (land and structures) in the spillover specification, Equation (3).	§4.2
P30	Prices in natural logs	Outcome	Explicit	"Sale prices and land prices will be presented in logs"; dollar values will be presented in natural logs.	§4.2; §5.3
P31	Dollars in real terms	Outcome	Explicit	"We will put all dollars in real terms."	§4.2
P32	Equation (1): permitting RDiD	Specification	Explicit	The permitting hypothesis will be evaluated with a spatial regression difference-in-discontinuities: $Y_{it} = \alpha + \beta_1(\text{OutCZ}_i \times \text{Post}_t) + \beta_2(\text{f}(\text{DistCZ}_i) \times \text{Post}_t) + X'_{it}\,\gamma + \lambda_i + \tau_t + \varepsilon_{it}$, where Y_{it} indicates that lot i received an ADU permit in year t. β_1 is the parameter of interest.	§5.1
P33, P34	Treatment and control definition	Treatment / control definition	Explicit	OutCZ_i takes value 1 if lot i is outside the Coastal Zone and therefore receives the state-wide permitting reform package in 2017, and value 0 if lot i is within the Coastal Zone and therefore has stable regulations; the inside-Coastal-Zone lots form the comparison group.	§5.1
P35	Post period begins 2016	Treatment definition	Explicit	Post_t takes a value of 1 in 2016 and afterwards and 0 in 2015 and before, "following the announcement of the first ADU permitting reform legislation."	§5.1
P36, P53	Distance polynomial as a control	Covariates	Explicit	P36: $\text{f}(\text{DistCZ}_i)$ is a polynomial in lot i's distance from the closest point on the Coastal Zone boundary, entered interacted with Post_t. P53: polynomials of distance to the Coastal Zone will be created as a flexible distance control, and "these polynomials will be free to change across the boundary."	§5.1; §5.3
P37	Covariate vector X'_{it}	Covariates	Implied	Equations (1), (2), and (3) each include a vector of controls X'_{it} entering linearly with coefficient γ. The plan does not enumerate the variables in X'_{it}.	§5.1

continued

Table B.1, continued

ID	Key	Type	Tier	Commitment	PAP loc.
P38	Lot and time fixed effects (Eq. 1)	Covariates	Explicit	Equation (1) includes lot fixed effects λ_i, to control for constant characteristics of given lots, and time fixed effects τ_t, to control for factors impacting all lots in a given period such as interest rate changes.	§5.1
P39	Alternative border-segment-by-year FE	Robustness check	Implied	"We could alternatively replace the time fixed effect with a border-segment by year fixed effect to ensure that all comparisons are within local segments of the Coastal Zone boundary." Stated as an option rather than a firm promise.	§5.1
P40, P54	Clustering and inference criteria	Inference / SEs	Explicit	P40: "We will cluster standard errors at the lot level as the Coastal Zone boundary is plausibly exogenous to any individual lot," citing Severen and Plantinga (2018) and Kahn, Vaughn and Zasloff (2010). P54: "We will work with an alpha level of 0.05 and construct 95% confidence intervals."	§5.1; §5.4
P41	Equation (2): land price specification	Specification	Explicit	The secondary hypothesis will be evaluated with $Y_{its} = \alpha + \beta_1(\text{OutCZ}_i \times \text{Post}_t) + \beta_2(\text{f}(\text{DistCZ}_i) \times \text{Post}_t) + X'_{it}\,\gamma + \tau_t + \varepsilon_{it}$, where Y_{its} is the land price for lot i sold in year t. As written, Equation (2) drops the lot fixed effect λ_i and retains only time fixed effects.	§5.1
P42	White robust SEs (Eq. 2)	Inference / SEs	Explicit	"In this specification we will use White (1980) robust standard errors."	§5.1
P43, P45	Preferred polynomial and deviation rule	Specification	Explicit	P43: "We will favor a first-degree polynomial that allows for a slope change at the boundary" as the functional form for $\text{f}(\text{DistCZ}_i)$ in Equations (1) and (2). P45: "We will only deviate from a first-degree polynomial as our preferred specification if it generates unacceptable pre-trends for our permitting, land value, and sale price outcome variables."	§5.1
P44, P49	Promised sensitivity checks	Robustness check	Explicit	P44: the authors "will also investigate more flexible functions such as second and third degree polynomials of distance, as used in Kahn, Vaughn and Zasloff (2010)." P49: "We will present specifications that drop segments of the border that overlap with cities within the Coastal Zone that update their ADU permitting voluntarily" — as of 2021 Santa Cruz, Santa Barbara, Encinitas, San Mateo and Pacifica — and "this list may change as time goes on."	§5.1

continued

Table B.1, continued

ID	Key	Type	Tier	Commitment	PAP loc.
P47	Equation (3): spillover specification	Specification	Explicit	Spillovers will be estimated with a more flexible spatial regression difference-in-discontinuities following Hornbeck and Kenniston (2017): $Y_{itb} = \alpha + \Sigma_b \beta_b(I\{\text{Bin} = b\}_i \cdot \text{Post}_t) + X'_{it}\,\gamma + \phi_b + \tau_t + \varepsilon_{itb}$, where Y_{itb} is the sale price (land and structures) on lot i in year t in Coastal Zone distance bin b, with bin fixed effects ϕ_b and year fixed effects τ_t.	§5.1
P48	Distance bins with omitted inside bin	Control group definition	Explicit	A series of dummies for Coastal Zone distance bins $b \in \{1, \ldots, B\}$ will be constructed, with an omitted bin within the Coastal Zone as the comparison group; the estimates $\hat{\beta}_b$ capture how prices evolve across bins relative to that comparison bin.	§5.1
P50	Event studies for Equations 1 and 2	Specification	Explicit	"We will present Equations 1 and 2 as flexible event studies in order to evaluate the pre-trends assumption ... for each of our outcome variables," replacing the Post_t indicator with a set of indicators for each year and omitting a base year in 2015.	§5.1
P51	Local parallel trends assumption	Other	Background	Identification relies on a local parallel trends assumption: parallel trends in outcomes across the Coastal Zone boundary within the local sample before the 2016 announcement, stability over time of the Coastal Commission's land regulations, no Coastal Commission effect on the treated side, and smoothness of potential outcomes across the boundary. The plan notes the spatial difference-in-discontinuity design can accommodate pre-existing covariate differences across the threshold. Stated as an assumption rather than an analytical promise.	§5.2
P52	Pre-periods start in 2010	Time horizon	Explicit	"We will evaluate these assumption by presenting flexible event studies for our main variables of interest including pre-reform periods starting in 2010."	§5.2
P55	No exclusion of complete observations	Sample / inclusion	Explicit	"We do not plan to exclude observations with complete data."	§5.5
P56	May restrict to boundary segments	Sample / inclusion	Implied	"We may limit our sample to only particular segments of the Coastal Zone boundary for pre-trends reasons if necessary."	§5.5
P57	May trim outcome outliers	Sample / inclusion	Implied	Following Severen and Plantinga (2018), outliers in the outcome variables may be trimmed if this helps maintain parallel trends in the pre-period; this may include dropping beachfront lots or dropping segments of the Coastal Zone where lot density is sparse.	§5.5

continued

Table B.1, continued

ID	Key	Type	Tier	Commitment	PAP loc.
P58	Trimming used sparsely, judged by pre-trends	Sample / inclusion	Explicit	"We will try to use trimming as sparsely as possible," and, as with the choice of polynomial form, how much to trim will be evaluated based on the impact on pre-trends for the three outcome variables.	§5.5
P61	No exploratory analysis	Other	Explicit	Section 5.7, "Exploratory analysis," reads in full: "None."	§5.7
S01	Arms-length transaction dummy as an outcome	Outcome	—	Not pre-specified	—
S02	Property tax collections per acre as an outcome	Outcome	—	Not pre-specified	—
S03	Total sale price per acre difference in discontinuities	Specification	—	Not pre-specified	—
S04	Fully interacted CATE heterogeneity estimator	Specification	—	Not pre-specified	—
S05–S08	Heterogeneity splits	Heterogeneity	—	Not pre-specified	—
S09	Two-stage least squares estimates of ADU permitting on block group gross rent changes	Specification	—	Not pre-specified	—
S10	Repeat-sale model with parcel fixed effects	Specification	—	Not pre-specified	—
S11, S12	Bandwidth rule and triangle weights	Specification	—	Not pre-specified	—
S13	Alternative land value measure using time-updating hedonics	Outcome	—	Not pre-specified	—
S14	Segment-clustered and Conley (1999) spatial-correlation-robust standard errors	Inference / SEs	—	Not pre-specified	—
S15, S16	Balance, pre-trend and attrition diagnostics	Robustness check	—	Not pre-specified	—

continued

Table B.1, continued

ID	Key	Type	Tier	Commitment	PAP loc.
S17	ADU permits as a share of all building permits by distance bin	Outcome	—	Not pre-specified	—
S18	Validation of the land value measure against Davis et al. (2021)	Robustness check	—	Not pre-specified	—
S19	Regression discontinuity for larger permitted units (SB 35 / multi-family)	Robustness check	—	Not pre-specified	—

Notes: Tier records commitment strength. *Explicit* statements are phrased as commitments; *Implied* statements would be read as commitments by a reasonable reader but are not phrased as such; *Inferred* items were extracted from context; *Background* items are statements describing the plan's own contents rather than promises about the paper, identified by tense ("we present a power analysis in Appendix A" reports completed work and creates no obligation). Both are excluded from all rates below, and *Background* items do not appear in Table B.2 at all. Rows appear in the order hypotheses, other commitments, then paper-only analyses. Where several commitments were adjudicated together, the row states the group and its members.

Table B.2: Pre-analysis plan compliance: item-by-item adjudication

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
H01	Reform increases ADU permitting	Modified	2	Table 3; Figure 9a; §5.1	Compliant	Forced	Toward	Adequate	Minor	The hypothesis is tested and the treatment, running variable, sample and alpha are the pre-specified ones; only the estimator changed, because no parcel-level permit data exist before 2018 (§3.1 fn. 32). The levels RD attributes the whole post-2018 boundary gap to the reform where the pre-registered DiD would have netted out any pre-existing gap. That gap is bounded small: Figure 1 puts pre-reform statewide ADU permits at roughly 4,000 across 2012–2016 against about 176,000 after, so it cannot plausibly move the 0.53–0.63 estimate across a threshold.
H02	Reform increases land values	As specified	3	Table 4; Figures 10 and 11; §5.2	Compliant	As planned	Neutral	Not required	None	The plan's Equation (2) is Table 4 Columns 1–4 and is reported; the headline 7% is obtainable from Column 4 (0.070**), the pre-specified specification with the plan's own §5.1 segment-by-year option and the covariate vector the plan left open. The change to how land value is constructed is scored at P27; the promotion of the repeat-sale column to preferred is scored at S10.
H03	Small negative local spillovers	As specified	4	Figure 12a; §5.3	Compliant	As planned	Neutral	Not required	None	The pre-specified binned design is run and the result — no detectable spillover — is reported without reinterpretation, which is the plan's own two-sided framing. The reference bin (400–450 m inside) is a free parameter the plan left open; fn. 57 concedes any bin could serve and no sensitivity is shown, but that is not a departure from anything committed.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
P01, P11	Observational spatial RD design	As specified	2, 3, 4	§2; §3.1; §4; §4.1–4.3; Figure 6	Compliant	As planned	Neutral	Not required	None	The observational spatial design is implemented as the plan describes it: Cotality parcel data, a discontinuity at the Coastal Zone boundary whose stringent regulation survives the state preemption documented in §2, and Figure 6 showing that boundary cutting through neighbourhoods.
P02, P03, P10	Named data sources	As specified	—	§3.1 (fn. 31); §3.3; Tables A.1 and A.3; Appendix E.1; Table E.1; notes to Tables 3, 4 and 7	Compliant	As planned	Neutral	Not required	None	Each named source is used as described: Cotality assessor and transaction records supply every listed element, the 2014 ds990 shapefile builds the running variable and is cited in every relevant exhibit note, and APN-first, address-fallback matching is implemented as a four-step hierarchy matching 98% of ADU permit rows.
P04, P15	Permit scrape superseded	Infeasible	—	§3.1; §4.5	Compliant	Forced	Toward	Adequate	Minor	No parcel-level permit source exists before 2018, so the scrape could not have delivered the pre-period and was retired in favour of the statutory statewide APR source. Section 4.5 states the decision and its basis; Figure 1 bounds the lost pre-reform flow at roughly 4,000 statewide permits.
P05, P09, P21, P22, P59	Sample construction screens	As specified	3, 4	§3.1; §3.3 (fn. 41); §3.4; Table A.1; Figure 12	Compliant	As planned	Neutral	Not required	None	Samples are built as promised: residential parcels in the 15 coastal counties, arms-length transactions for spillovers, and parcels lacking a valid identifier, coordinate, assessed value or coherent characteristics dropped. Gaps are filled rather than dropped and Table A.1 tracks retention, though the carried-forward and centroid-fallback shares are not reported.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
P06	Estimates at several bandwidths	Relocated	2, 3	Figure A.4; §5.4	Partial	Presentational	Indeterminate	Adequate	Minor	Figure A.4 sweeps bandwidths for permitting and land values with three inference choices, and §5.4 discusses it in the body including the adverse finding that Conley errors void bandwidths of 350 m and below. The commitment is met for two of the three pre-registered outcomes; the binned spillover design is shown at one bandwidth only. Direction is not signable: narrower bandwidths give larger land-value estimates, but 450 m is where the preferred estimate survives the most conservative inference.
P07	Permitting sample: all residential parcels	Narrowed	2	§3.1; §4.1; Table 3	Compliant	Forced	Toward	Adequate	Minor	The jurisdictional half is satisfied maximally by the statewide source; the temporal half is truncated to 2018–2024 because parcel-level permits do not exist earlier (§3.2, §3.1 fn. 32). 2017, the first treated year, is therefore unobserved for permitting, and the paper does not quantify what share of permits fell in it.
P08	Land-value sample: reassessment sales	Modified	3	§3.1; §3.2; Table A.1	Compliant	As planned	Neutral	Not required	None	The decisive text is plan §4.2, which defines the measure as coming from "the first assessment after an arms-length transaction" — the plan's own operationalization of its §2.4 reassessment-trigger language. The paper applies exactly that screen (§3.1, §3.2) and states the institutional basis at fn. 36.
P14	State-harmonized permit data from 2018	As specified	—	§3.1; §4.5	Compliant	As planned	Neutral	Not required	None	The paper uses the source the plan described (California Department of Housing and Community Development, 2025). Its elevation to sole permit source is the same event recorded at P04. Inferred tier: excluded from all rates.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
P16, P23	Scrape target list and stopping rule	Infeasible	—	§4.5	Compliant	Forced	Neutral	Adequate	None	Both govern the retired scraping programme. Statutory APR reporting covers every city and county, including unincorporated areas, so it is a superset of Table 1's list and forecloses result-sensitive jurisdiction selection a fortiori. The method change is disclosed at §4.5; the retirement of each item is not separately noted.
P17, P60	Jurisdiction-level permit screens	Modified	—	§3.1; §4.5; Appendix E.1; Table E.1	Compliant	Forced	Neutral	Adequate	None	With one statutory statewide source there are no jurisdiction-level inclusion or exclusion decisions left to make; all screens migrate to the record level, where Table E.1 reports 193,511 of 197,249 ADU permit rows matched. Whether match failure varies across the boundary is not tested.
P24	Primary outcome: ADU permit dummy	As specified	2	§3.2; Table 3	Compliant	As planned	Neutral	Not required	None	§3.2 reproduces the plan's definition almost verbatim, indexed by parcel-year as the plan's Y_{it} requires. Scaling to permits per 1,000 parcel-years is a display convention declared in §3.2 and in every table note.
P25	Jurisdiction-specific ADU permit coding	Modified	—	§3.1; §3.2; Appendix E.1	Compliant	Forced	Indeterminate	Insufficient	Minor	The per-jurisdiction coding procedure is moot once one harmonized source is used, and the attach-by-APN-or-address half is executed and expanded (Appendix E.1). But the paper never states how an ADU permit is identified within the APR records — no field, no category list, no treatment of junior ADUs or conversions — so the construction of the primary outcome is undocumented. See the adjudication record filed with the materials.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
P26	Permitting also from building characteristics	Infeasible	—	§3.1 (fn. 32); §4.5	Compliant	Forced	Indeterminate	Adequate	Minor	The corroborating instrument was attempted and abandoned because the characteristics cannot separate ADU construction from construction in general (§4.5, fn. 32) — a reason corroborated internally by fn. 48, which notes effective build year updates for any addition. Disclosed three times (fn. 32, §4.5, §8). The permitting result therefore rests on one instrument and cannot see unpermitted construction, which fn. 30 concedes may be substantial.
P27	Secondary outcome: assessed land value	Modified	3	§3.2; §4.5; Table 4	Partial	Discretionary	Indeterminate	Adequate	Major	The assessment now supplies the land *share* and the transaction price the *level*; the target and the timing (first post-sale assessment) are unchanged, but the committed construction is not reported anywhere, not even as a robustness row. No benchmark citation supports the change, so it cannot be classified as an Improvement; the paper's sources are domain-level (California State Board of Equalization 2025a; Ahlfeldt, Koster and Vu 2025) and its validation is at the county-year level (Figure 2). See commentary note 1.
P28	Land value per square foot	Modified	3	§3.2; Table 4	Compliant	Presentational	Neutral	Not required	None	Because the plan also commits to natural logs, ln(value per acre) = ln(value per sq ft) + ln(43,560), a constant absorbed by the intercept; every coefficient, standard error and p-value is identical. The one loose end is that the plan says "square footage of the lot" and the paper says "parcel land acreage," and the paper does not say whether these are the same underlying field.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
P29	Third outcome: arms-length sale price	Modified	4	§3.2; Figure 12a	Partial	Presentational	Indeterminate	Not required	Minor	The construct — total arms-length sale price including structures, in logs — is unchanged, but the plan reserved per-unit-area scaling for land values and the paper applies it to sale prices as well. Bin fixed effects absorb static lot-size differences, so only differential within-bin compositional change over time could distort the estimates. The normalisation appears on the face of every exhibit and is defined at §3.2, so no reader is misled.
P30	Prices in natural logs	As specified	3, 4	§3.2; Table 4; Figure 12a	Compliant	As planned	Neutral	Not required	None	Logs are used uniformly: Equations (3) and (4), Tables 4, 7, A.6–A.8, Figures 10–12.
P31	Dollars in real terms	As specified	—	§3.2 (fn. 37); Tables 1 and 2	Compliant	As planned	Neutral	Not required	None	fn. 37 names the deflator (PCE chain-type price index, 2025 USD), which is more than the plan promised; sample screens are stated in real terms.
P32	Equation (1): permitting RDiD	Modified	2	§4.1; §4.5; Table 3	Compliant	Forced	Toward	Adequate	Minor	With permits observed only from 2018 the Post interaction has no pre-period variation and the plan's β_1 is not identified; §4.5 states the substitution and its basis plainly. Everything else in the plan's Equation (1) survives: OutCZ, the distance polynomial with a slope change (fn. 51), time fixed effects, controls, lot-level clustering, alpha 0.05. See the adjudication record filed with the materials.
P33, P34	Treatment and control definition	As specified	2, 3, 4	§4.1; §4.2; Tables 3 and 4	Compliant	As planned	Neutral	Not required	None	OutCZ is defined identically in Equations (2)–(4) and in the notes to Tables 3 and 4, with inside-Coastal-Zone parcels as the comparison group and In-CZ means at the foot of each main table. Two control-side imperfections are acknowledged: voluntary adopters and *Riddick v. City of Malibu*.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
P35	Post period begins 2016	Modified	3	§4.2; §4.5; Table 4	Deviation	Result-favorable	Toward	Insufficient	Major	2016 moves from the post period to the control period on a stated basis that is explicitly data-dependent: "These anticipation effects don't appear in our data" (§4.5). Not forced — both years are in sample — and no benchmark authority is cited or available for revising a pre-registered treatment date after inspecting outcomes. See the adjudication record filed with the materials for the arithmetic showing the change raises β_1 when anticipation is small, and commentary note 4 for the missing pre-specified version.
P36, P37, P53	Controls and distance polynomial	As specified	2, 3, 4	§3.3; §4.1–4.3; Table 4; notes to Tables 3 and 4	Compliant	As planned	Neutral	Not required	None	The plan left X'_{it} open; §3.3 supplies interacted hedonic bins with explicit missing-value bins, built from pre-2016 characteristics, and results are shown with and without them. The distance polynomial is interacted with Post and free to change across the boundary (fn. 51; Table 3, 4 and A.4 notes).
P38	Lot and time fixed effects (Eq. 1)	Modified	2	§4.1; Table 3	Compliant	Forced	Toward	Adequate	Minor	Once the design is cross-sectional, OutCZ_i is lot-invariant and a lot fixed effect would absorb the treatment entirely, so λ_i is not estimable. The paper substitutes the closest feasible object for the plan's stated purpose — pre-2016 hedonic bins plus county fixed effects (§4.1) — and retains τ_t with the plan's rationale almost verbatim.
P39	Alternative border-segment-by-year FE	As specified	2, 3, 4	§4.1; §4.2; §4.3; Tables 3–7	Compliant	As planned	Away	Not required	None	The plan's option is implemented across Tables 3–7, A.4, A.6, A.7 and Figures 11–12, and adopted as the preferred specification in both headline tables. It reduces the authors' own estimates substantially — land values 0.141 to 0.085, permitting 0.623 to 0.530 — and §5.2 says so.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
P40, P54	Clustering and inference criteria	As specified	2	§4.1; Table 3; notes to Tables 3–7; Figure 11	Compliant	As planned	Neutral	Not required	None	Parcel-clustered robust standard errors, lot and parcel being the same unit, appear in every main table, and §4.1 adds that segment-clustered errors are almost identical. The 0.05 alpha and 95% intervals are used throughout, and no result is claimed at a looser threshold.
P41	Equation (2): land price specification	Modified	3	§4.2; Table 4	Compliant	As planned	Away	Not required	None	Equation (3) writes out the unin-teracted OutCZ and distance terms (β_4–β_6) that a difference in discontinuities requires and that the plan's Equation (2) omitted, which implements the design the plan named and cited to Grembi, Nannicini and Troiano (2016) rather than changing it. Including them is the more demanding specification given the level differences Figure 8a finds. The plan's omission of λ_i is honoured in Columns 1–4; Columns 5–6 add one as an additional model (S10).
P42	White robust SEs (Eq. 2)	Modified	3	§4.2; Table 4; Figure A.4	Compliant	Improvement	Away	Not required	None	White (1980) appears nowhere in the paper; parcel-clustered errors are used instead. In a transaction sample where parcels recur — necessarily so in the repeat-sale columns — heteroskedasticity-only errors understate sampling variability, and clustering at the grouping level is the benchmark's position [Cunningham, *Causal Inference: The Mixtape*, accessed 2026-09-24, inference and clustering chapter] [**unverified**]. The change enlarges the standard errors and no headline result turns on it.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
P43, P45	Preferred polynomial and deviation rule	As specified	2, 3	§4.1 (fn. 51); §5.4; Tables 3 and 4; Figure 8b	Compliant	As planned	Neutral	Not required	None	The first-degree polynomial with a slope change at the boundary is the preferred form in every main exhibit (fn. 51; Tables 3, 4, 7 and A.7; Figures 12 and A.4). The plan's condition for deviating, unacceptable pre-trends, did not arise, so no deviation occurred.
P44, P49	Promised sensitivity checks	Narrowed	2, 3	§5.4; Tables A.4, A.5 and A.7	Partial	Discretionary	Indeterminate	Insufficient	Minor	Two promised sensitivity checks are delivered in part. Second-degree polynomials appear in full tables for both main outcomes (A.4, A.5) but no third-degree estimate appears anywhere; the voluntary-adopter drop is run for land values only (Table A.7), not permitting or spillovers. No reason is given for either.
P47	Equation (3): spillover specification	As specified	4	§4.3; Figure 12a	Compliant	As planned	Neutral	Not required	None	Equation (4) is term-for-term the plan's object — bin-by-Post interactions, hedonic controls, bin fixed effects, time fixed effects — following Hornbeck and Keniston (2017) as promised. Time fixed effects appear in the plan's §5.1 segment-by-time form (P39), and Figure 12's note states the choice.
P48	Distance bins with omitted inside bin	As specified	4	§4.3 (fn. 57); Figure 12	Compliant	As planned	Neutral	Not required	None	§4.3 is the plan's sentence with the specifics filled in: 50 m bins, omitted bin [-450, -400) inside the Coastal Zone, tied to the bandwidth rule rather than chosen freely (fn. 57). The plan did not fix which inside bin, so the choice is not a departure, though no sensitivity to it is shown.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
P50	Event studies for Equations 1 and 2	Narrowed	3	Figure 11; Figure 8b	Partial	Forced	Toward	Adequate	Minor	The permitting event study is impossible with data from 2018 only; Figure 11 delivers the land-value event study in the promised form but with 2016 as base rather than the promised 2015, which follows from P35 and is where the substance lies. Figure 11 runs only the repeat-sale specification while Figure 8b explicitly does not (fn. 63), so no year-by-year pre-trend is shown for Table 4 Columns 1–4. Figure 11's note refers to "Column 7 of Table 4," which has six columns.
P52	Pre-periods start in 2010	As specified	3	§3.1; Figure 8; Figure 11	Compliant	As planned	Neutral	Not required	None	Both datasets run 2010–2024 (§3.1); Figure 8 covers 2010–2016 and Figure 11's event study begins in 2010. Permitting has no 2010 pre-period, which is the forced data limitation scored at P04/P07.
P55	No exclusion of complete observations	Modified	—	§3.1; Table A.1	Partial	Discretionary	Indeterminate	Adequate	Minor	Validity screens that remove incoherent records are outside the prohibition and the 1st/99th sale-price trim is covered by §5.5's own trimming permission, but the room-to-unit and transaction-volume screens remove complete records and are neither. Every screen is enumerated cumulatively in Table A.1 and tested against the design in Table A.2, which is more than the plan asked. The narration at fn. 28 and fn. 35 understates that Table A.2's transaction-sample OutCZ × Post coefficient (−0.011) is significant at 5%.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
P56	May restrict to boundary segments	Not executed	—	Not reported	Compliant	As planned	Neutral	Not required	None	The clause is doubly conditional and its trigger is unacceptable pre-trends; §4.2 reports the opposite, that Figure 8b supports local parallel trends for all three outcomes shown. A permission not exercised because its condition did not obtain is compliance, though the paper never says so in terms.
P57	May trim outcome outliers	Modified	—	§3.1; Table A.1	Compliant	As planned	Indeterminate	Not required	None	The clause is permissive and its named candidates are illustrative; the paper trims outcome-variable outliers at the 1st and 99th percentiles of real sale price (§3.1), which is within it. Neither named candidate — beachfront lots, sparse segments — is dropped, and the sample retains Crescent City and Eureka (Figure 3). The decision criterion is scored at P58.
P58	Trimming used sparsely, judged by pre-trends	Modified	—	§3.1; Tables A.1 and A.2	Partial	Discretionary	Indeterminate	Insufficient	Minor	Sparseness is met — the discretionary trims are 1%/99% bands and the largest exclusion is the plan's own arms-length screen — but the promised procedure is not executed and not retired. Table A.2 asks whether exclusions correlate with the design, not whether trimming choices improve pre-trends. See the adjudication record filed with the materials.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
P61	No exploratory analysis	Modified	—	§4.5 (fn. 59); §7; Appendix D	Compliant	Discretionary	Indeterminate	Adequate	None	The paper contains substantial analysis the plan did not contemplate, but additions are not departures and the governing standard is boundary-drawing, not annotation. The boundary is drawn twice in the methods section — §4.5 names the transaction, property-tax, repeat-sale, triangle-weight and bandwidth additions, and fn. 59 covers heterogeneity, rents and supporting figures and tables — and §5 holds the pre-registered results while §7 holds the additions.
S01	Arms-length transaction dummy as an outcome	Supplemental	4, 5	§3.2; §4.3; Figure 12b; Table 6, Panel A	Supplemental	Adjacent	Neutral	Adequate	None	Reuses the pre-registered spillover design with a swapped outcome, so it sits inside the apparent scope of the pre-registration claim. §4.5 names it directly: "the addition of transaction rates as an outcome variable in our spillovers design."
S02	Property tax collections per acre as an outcome	Supplemental	3	§7.3; Table 7	Supplemental	Adjacent	Neutral	Adequate	None	Runs the pre-registered Equation (3) on the balanced panel with the outcome swapped, which is close enough to the registered work to need a boundary. §4.5 names it in the same sentence as S01.
S03	Total sale price per acre difference in discontinuities	Supplemental	3	§7.3; Table A.8	Supplemental	Adjacent	Neutral	Adequate	None	Pairs a pre-registered outcome (arms-length sale price) with a pre-registered equation, which is the configuration a reader might take as covered. It is not named individually, but fn. 59's "supporting figures and tables were also not pre-registered" reaches an appendix table, and one boundary statement discharges the obligation.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
S04	Fully interacted CATE heterogeneity estimator	Supplemental	5, 6	§4.4; Tables 5 and 6	Supplemental	Adjacent	Neutral	Adequate	None	Fully interacts the pre-registered designs, so it reuses their scaffolding. fn. 59 covers the heterogeneity analysis, and §4.4 states the estimator's assumption and warns that its OutCZ coefficient is not the main effect of Equation (2).
S05–S08	Heterogeneity splits	Supplemental	1, 5, 6, E1, E2	§3.3; §7.1.1; §7.1.2; Table 5, Panels A and B; Table 6, Panels A and B; Appendix E.2	Supplemental	Adjacent	Neutral	Adequate	None	Four heterogeneity splits — excess lot square footage, distance to the nearest CBD, the 2016 Proposition 13 tax shelter, and block-group share of owners over 55 — are covered by fn. 59, constructed in §3.3 and Appendix E.2 before any result appears, with both halves of each reported.
S09	Two-stage least squares estimates of ADU permitting on block group gross rent changes	Supplemental	D1a, D1b	§7.4; Appendix D; Tables D.1, D.2 and D.3	Supplemental	Distinct	Neutral	Not required	None	A visibly different design — block groups, an area-share instrument, Conley errors — that abandons the parcel-level boundary framework, so no reader would take it as covered. fn. 59 names "analysis on rental impacts" in any case, and Appendix D states that the exclusion restriction is untestable and calls the point estimates implausibly large.
S10	Repeat-sale model with parcel fixed effects	Supplemental	3	§4.2; §4.5; Table 4, Columns 5–6	Supplemental	Adjacent	Away	Adequate	None	Sits in Table 4 beside the pre-registered columns and is the paper's preferred specification, so the boundary matters; §4.5 draws it by name ("we did not pre-specify a repeat sale model") and §1 flags it as an extension. The headline 7% does not depend on it — Column 4, the pre-registered specification on the full sample, also gives 0.070, as §5.2 states. The repeat-sale sample conditions on transacting twice, which §6.2's own Equation (15) models as treatment-responsive; the paper does not raise this.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
S11, S12	Bandwidth rule and triangle weights	Supplemental	—	§3.3; §4.1; §4.5; Figure A.4; notes to Tables 3–7	Supplemental	Adjacent	Indeterminate	Adequate	None	Two estimation choices governing every pre-registered estimate, named together in §4.5's deviation list and restated in the notes to Tables 3–7: an MSE-optimal bandwidth computed on pre-period 2010–2016 land values and applied throughout, and triangle weights used everywhere with no unweighted comparison shown.
S13	Alternative land value measure using time-updating hedonics	Supplemental	3	§5.4; Table A.6	Supplemental	Adjacent	Neutral	Adequate	None	A robustness version of the pre-registered land-value design, announced prospectively at §3.2 and covered by fn. 59. Its dependent variable is the log real total sale price per acre with updating hedonic fixed effects, not a separately constructed land value, which makes "alternative land value measure" a loose label; §5.4 states the construction plainly in the body.
S14	Segment-clustered and Conley (1999) spatial-correlation-robust standard errors	Supplemental	2, 3	§4.1; §5.4; Figure A.4	Supplemental	Adjacent	Neutral	Adequate	None	Alternative inference for the pre-registered specifications, announced at §4.1 and covered by fn. 59. The paper reports the unfavourable consequence: Conley errors render bandwidths of 350 m and below insignificant in the repeat-sale model (§5.4). These appear only in the appendix figure; main-table stars use the least conservative of the three.
S15, S16	Balance, pre-trend and attrition diagnostics	Supplemental	2, 3	§3.1; §4.1; §4.2; Figure 8, Panels A and B; Table A.2	Supplemental	Adjacent	Neutral	Adequate	None	Two diagnostics the plan did not request, covered by fn. 59 and reported against the authors' own interest: Figure 8 finds several covariates jump at the boundary and are distinguishable from zero, and Table A.2 finds a significant differential change in sample attrition across the boundary after treatment.

continued

Table B.2, continued

ID	Key	Status in paper	Eq.	Location	Verdict	Rationale	Direction	Disclos.	Sev.	Notes
S17	ADU permits as a share of all building permits by distance bin	Supplemental	—	§5.1; Figure 9b	Supplemental	Adjacent	Neutral	Adequate	None	A new outcome presented in the same figure, bins and section as the pre-registered permitting exhibit, so a reader could take it as covered; fn. 59's "supporting figures" reaches it. It is descriptive — binned means, no regression — with the denominator on the axis label, so no estimate is passed off as pre-registered.
S18	Validation of the land value measure against Davis et al. (2021)	Supplemental	—	§3.2; Figure 2	Supplemental	Distinct	Neutral	Not required	None	A county-year correlation scatter against an external land-value series, visibly not the pre-registered design. It is the external support for the P27 measurement change, and the paper's own hedge — "at least at the county-year level" — is the right one, since averaging washes out the near-boundary variation the design exploits.

S19	Regression discontinuity for larger permitted units (SB 35 / multi-family)	Supplemental	2	Not reported (asserted at §4.2, fn. 56)	Supplemental	Overclaimed	Indeterminate	Insufficient	Minor	fn. 56 states "We also find that our regression discontinuity design doesn't detect changes in larger permitted units," attaching the finding to Equation (2) — the pre-registered permitting design — with no table, figure, sample, bandwidth or coefficient anywhere in the paper. fn. 59 names heterogeneity, rents and "supporting figures and tables," and this is neither a figure nor a table, so no boundary statement covers it. See the adjudication record filed with the materials.

Notes: Full text of each commitment is in Table B.1. **Status** describes what the paper did: *As specified*, *Modified*, *Narrowed*, *Relocated*, *Not executed*, *Infeasible*, *Supplemental*. **Verdict**: *Compliant*, *Partial*, *Deviation*, *Supplemental*, *Unresolved*. **Rationale** classifies why a change occurred: *As planned*; *Improvement* (moves toward practice established in the designated reference, cited in Notes); *Forced* (outside the authors' control, or facts about the data learned after the plan was written); *Presentational* (what was estimated is unchanged); *Discretionary* (neither forced nor citable as an improvement); *Result-favorable* (discretionary and moving the result toward the hypothesis or across a significance threshold). **Direction** records whether the change moves the reported result toward or away from the paper's hypothesis or conventional significance; *Indeterminate* where no specific reason could be identified. **Disclosure** applies a reasonableness standard rather than asking whether the manuscript labels a change a deviation: the obligation scales with rationale, so improvements and presentational changes carry none, forced substitutions warrant a note, discretionary changes are expected to be disclosed near the affected result, and result-favorable changes require prominent disclosure with the pre-specified version reported. For `S##` rows **Rationale** instead records proximity to the pre-registered work: *Distinct* (visibly a different design, so no reader would take it as covered, and no disclosure is owed), *Adjacent* (reuses the pre-registered equation, sample, outcome, or design closely enough to sit inside the apparent scope of a pre-registration claim, so the boundary should be drawn somewhere in the paper), or *Overclaimed* (the paper states or implies the analysis was pre-registered when it was not). The standard for additions is accuracy rather than annotation: a single boundary statement anywhere in the paper discharges the obligation for every addition it covers, and no per-item flag is expected. **Severity** reflects inferential consequence only and is independent of rationale; an improvement may be *Major*. Commitments bind the measurement target rather than the instrument: substituting a better measure of the same target is recorded as an improvement, while substituting a different target, or dropping one of several committed targets, is a deviation. Bracketed tags `[D1]`–`[D4]` in Notes record the evidence tier supporting a domain or measurement claim. Rows covering several commitments list every member ID, equation, and location; items are grouped only where a single note describes all of them and all six judgment fields agree, and hypotheses, estimating equations, and outcome variables are never grouped. Table B.1 keeps one line per commitment regardless.

Summary

Metrics. Adjudicated commitment rows: 57 (51 Explicit, 5 Implied, 1 Inferred). Background-tier rows are carried in Table B.1 for the record but are not commitments and enter no rate or count. Commitments audited: **57**. Rows presented: **61** in Table B.1 and **55** in Table B.2.

1. Execution rate, Explicit tier (Compliant / all Explicit `H##` and `P##`): 42 / 51 = **82.4%**

2. Explicit tier, Compliant + Partial: 50 / 51 = **98.0%**

3. Explicit + Implied tier: Compliant 47 / 56 = **83.9%**; Compliant + Partial 55 / 56 = **98.2%**

4. Counts by Rationale (57 adjudicated `H##` and `P##` rows): As planned 33; Forced 13; Improvement 1; Presentational 3; Discretionary 6; Result-favorable 1.

5. Insufficient disclosures, by Severity — headline finding: **6 total**. Major 1 (P35); Minor 5 (P25, P44, P49, P58, S19); None 0.

5a. Inaccurate pre-registration claims: **1** — the abstract's statement that the permitting result rests on a pre-registered design. The 30% permitting figure comes from a regression discontinuity, whereas the plan pre-registered a difference in discontinuities, and the qualifier "partially," used four times in the body, is absent from the abstract. `S##` rows by proximity class: Distinct 2 (S09, S18); Adjacent 16 (S01–S08, S10–S17); Overclaimed 1 (S19).

6. Counts by Severity (76 adjudicated rows: 57 commitments + 19 supplemental): Major 2 (P27, P35); Minor 16 (H01, P04, P06, P07, P15, P25, P26, P29, P32, P38, P44, P49, P50, P55, P58, S19); None 58.

7. Unresolved items: **0**.

8. Tier exclusions: Inferred-tier rows 1 (P14); Background-tier rows 7 (P12, P13, P18, P19, P20, P46, P51). Both excluded from all rates; Background rows are not adjudicated.

9. Improvement classifications resting on `[unverified]` citations: **1** (P42).

10. Improvement classifications resting on `[D3]` or `[D4]` domain evidence: **0**.

Commentary. Notes 1, 4 and 11 are reproduced below in full. The adjudicator's remaining notes — 2, 3 and 5 through 10, hence the gaps in the numbering — have been cut to hold this section within its word limit. They survive in full in the adjudication record filed with the materials, which is where the Notes column of Table B.2 points for them.

1. P27 — the land-value measure (Major). The plan committed to the assessment's land-attributed value at the first post-sale assessment; the paper uses the transaction price multiplied by the land share from that same assessment. The target does not move — it is still a land value from the same record at the same moment — and the paper argues the change at length at §4.5, describes the construction at §3.2 and validates it externally at Figure 2. But no benchmark citation supports the change, so under the audit's rule it cannot be classified as an Improvement however plausible it is; its sources are the California State Board of Equalization and a measurement-error literature outside the benchmark's reach. The decisive fact

is that the pre-specified measure was still constructible from the very record the paper uses for the share, and it is never reported, not even as a robustness row. Table A.6, offered as "an alternative to our preferred measure of land values," has the log real total sale price per acre as its dependent variable, so it is not the pre-specified measure either. The severity is Major because this is the construction of the headline outcome and the reader cannot compare it with what was promised.

4. **P35 — the post period (Result-favorable, Insufficient, Major).** The plan fixed Post = 1 from 2016, and said why: the announcement of the first reform legislation. The paper uses 2017 and gives a basis that is explicitly data-dependent — "We initially expected anticipation effects to show up in our data ... These anticipation effects don't appear in our data" (§4.5). All three conditions for Result-favorable hold. It was not forced: both years are in sample. It is not citable as an improvement: no benchmark authority adjudicates announcement against enactment, and the one candidate citation is unverifiable and, on its own advocate's account, held lightly. And the direction is Toward, which can be shown rather than asserted. Write the pre-period boundary gap as g_0, the 2016 gap as $g_0 + \delta$ and the full post gap as $g_0 + \Delta$. The plan's assignment gives $\beta_1 = (\delta + 8\Delta)/9$; the paper's gives $\beta_1 = \Delta - \delta/7$. With the small δ the authors themselves claim, and at $\delta = 0$ exactly, the paper's assignment yields Δ against the plan's 0.889Δ — roughly an eighth larger — because the plan's specification would have averaged an untreated year in with the treated ones. Disclosure is Insufficient because Result-favorable requires the pre-specified version also to be reported, and a 2016-start Post is reported nowhere. The change also propagates to P50: the event study's omitted base year moves to 2016, so the one year whose classification was revised is the one year the figure normalises to zero and cannot display. No test, figure or placebo for the absence of anticipation appears anywhere.

11. **One procedural note bearing on every disclosure judgment above.** §1 fn. 3, §3 and §4.5 each defer the remaining deviations to a full change-log in Appendix B, which in the audited draft (printed p. 83) is a heading with no content — evidently a placeholder awaiting this audit's output. Every disclosure verdict recorded here is therefore assessed against the paper's operative disclosure set as supplied: §4.5, its footnote 59, and the data and methods sections. If Appendix B is filled, the Insufficient marks at P25, P44, P49, P58 and S19 are answerable there; the mark at P35 is not, because that one requires the pre-specified version of the analysis to be reported, not merely described.

Prompts

The six prompts below were frozen before any output was generated and are reproduced without modification. They are pure ASCII by construction, so nothing has been altered to make them typeset.

Agent 1a — Commitment Extractor (pre-analysis plan only).

```
You are a methods auditor building an inventory of pre-committed analytical decisions.

You will be given ONE document: a pre-analysis plan. You will NOT be shown the resulting
paper, and you must not speculate about what the paper contains. Your task is purely
extractive.

Read the plan in full before producing output.

Extract every substantive commitment about the EMPIRICAL ANALYSIS. Include:
  - Hypotheses, primary and secondary
  - Sample definitions, inclusion and exclusion criteria
  - Treatment and control group definitions
  - Outcome variables, primary and secondary
  - Estimating equations and specifications
  - Covariates and their functional forms
  - Inference procedures and clustering
  - Pre-specified heterogeneity and subgroup analyses
  - Data sources and time horizons
  - Power calculations and target sample sizes
  - Pre-specified robustness checks

Exclude theory, structural models, institutional background, literature review, and
descriptive statistics -- unless a specific descriptive exhibit is itself pre-committed.

Each commitment must be ATOMIC: one checkable promise per row. If the plan commits to
three separate tests, that is three rows.

ORDER: list all hypotheses first, then everything else in the order it appears in the plan.

IDs: hypotheses are H01, H02, ...; all other commitments are P01, P02, ...

THE TENSE TEST. A plan contains two kinds of statement and only one is a commitment. "We will
estimate equation (2) on the 2019 sample" promises something about the forthcoming paper. "We
present a power analysis in Appendix A" reports what this plan already did, and creates no
obligation on the paper at all. Past or present tense describing the plan's own contents -- we
present, we have computed, our calculations show, Appendix A of this plan contains -- is
Background unless the plan separately says the paper will report it. Extract these, tier them
Background, and do not treat them as promises. Marking a paper non-compliant for not repeating
work the plan already did is an extraction error, not a finding.

For each commitment assign a strength tier:
  Explicit   -- stated directly as a commitment ("we will," "our primary outcome is")
  Implied    -- a reasonable reader would treat it as committed, but it is not phrased so
  Inferred   -- you extracted it from context; a reasonable reader might not have
  Background -- not a commitment at all; see the tense test above

Output a markdown table with exactly these columns:

| ID | Key | Commitment | Type | Tier | PAP location |

  Key          -- six words or fewer
  Commitment   -- at most four sentences; include equations where they are the clearest
                  statement; quote the plan verbatim only where exact wording is decisive
  Type         -- one of: Hypothesis, Sample / inclusion, Treatment definition,
                  Control group definition, Outcome, Specification, Covariates,
                  Inference / SEs, Heterogeneity / subgroup, Data source, Time horizon,
                  Power / sample size, Robustness check, Other
  PAP location -- section, page, or equation number in the plan

After the table, list any places where the plan is materially ambiguous -- where a
reasonable analyst could satisfy the stated language in more than one way. Give the ID and
one sentence per item.

WORKSPACE. If you need temporary or scratch files -- extracted text, intermediate notes,
working copies -- put them in the scratch/ folder inside your own agent directory and nowhere
else. Delete them when you are done, so that only your named output file remains. Never write
scratch anywhere shared, and never into another agent's directory: leftover working files are a
route for one agent's material to reach another, which is exactly what this design prevents.

Produce nothing else.
```

Agent 1b — Mapper.

```
You are a methods auditor mapping a pre-analysis plan onto the paper that implemented it.

You will be given a pre-analysis plan, a paper, and an inventory of commitments already
extracted from the plan.

You are NOT making compliance judgements. You are recording what is there. Another auditor
will judge.

PART ONE. For each row of the supplied inventory, locate what the paper actually does.
Do not add, remove, merge, or rewrite inventory rows; carry ID, Key, Commitment, Type, and
Tier through unchanged.

PART TWO. Identify empirical analyses in the paper with NO counterpart in the plan --
specifications, samples, outcomes, robustness checks, heterogeneity cuts. Exclude theory,
models, institutional background, and descriptive statistics. Number them S01, S02, ... and
add them as rows with Commitment = "Not pre-specified."

Output a single markdown table with exactly these columns, rows ordered H##, then P##,
then S##:

| ID | Key | Commitment | Type | Tier | Status in paper | Equation | Location | Verdict | Rationale | Direction | Disclosure | Severity | Notes |

Fill ONLY: ID, Key, Commitment, Type, Tier, Status in paper, Equation, Location.
Leave every remaining column blank. Do not write "TBD" or placeholders.

  Status in paper -- exactly one of:
      As specified  -- executed exactly as promised
      Modified      -- executed differently (estimator, covariates, operationalization, sample)
      Narrowed      -- executed on a subset of what was promised
      Relocated     -- executed as promised but moved between main text and appendix in a
                       way that changes its apparent prominence
      Not executed  -- promised, absent, no explanation given
      Infeasible    -- promised, absent, paper explains why it could not be done
      Supplemental  -- S## rows only

  Equation -- manuscript equation number(s), or an em dash
  Location -- table, figure, or section number, or "Not reported"

Read table and figure NOTES carefully. Sample restrictions, control sets, and year ranges
are frequently specified only there.

After the table, list every item where you could not confidently determine what the paper
did, with the ID and one sentence explaining the difficulty.

WORKSPACE. If you need temporary or scratch files -- extracted text, intermediate notes,
working copies -- put them in the scratch/ folder inside your own agent directory and nowhere
else. Delete them when you are done, so that only your named output file remains. Never write
scratch anywhere shared, and never into another agent's directory: leftover working files are a
route for one agent's material to reach another, which is exactly what this design prevents.

Produce nothing else.
```

First auditor.

```
You are a skeptical methods auditor. Your role is to construct the strongest defensible case
that this paper departed from its pre-analysis plan in ways a reader would want to know about
and was not told.

You will be given a pre-analysis plan, a paper, a partially completed audit table, and a
designated econometric reference.

For each row in the table, in ID order, write an entry in a reasoning log:

## [ID] -- [Key]
**Assessment:** one line
**Case:** The strongest argument that this item was not executed as promised, or was
executed as promised but reported in a way that obscures a departure. Cite specific text.
**Was it forced or an improvement?** Address this directly. If the change was plainly
compelled by circumstance, or is citable to the reference as current practice, say so -- that
concession makes your remaining findings credible.
**Direction:** Does the change move the result toward or away from the paper's hypothesis or
conventional significance? Say Indeterminate unless you can point to a reason.
**Strongest counterargument:** The best point against your own position, stated fairly.
```

```
Look particularly hard for:
  - The primary estimand changing between plan and paper -- which sample, outcome, or
    specification is presented as the headline result
  - Promised tests collapsed into fewer tests
  - Covariate sets, sample restrictions, or year ranges differing in ways visible only in
    table notes
  - Analyses promised and quietly absent
  - Substitutions where the original was still available
  - Statements that something was pre-registered, pre-specified, or done "per the plan" that are
    not accurate against the plan. This is unconditional and is the strongest finding available
    to you.
  - Additions that reuse the pre-registered equation, sample, or outcome closely enough that a
    reader would take a pre-registration claim to cover them, where no boundary statement
    anywhere in the paper draws the line

Do NOT treat every unflagged addition as a fault. An addition that is visibly a different design
misleads nobody and owes no disclosure. Demanding a disclaimer on each new analysis is busywork,
not a finding.

Do not manufacture departures, and do not treat every departure as a fault. A log that flags
everything is useless to the adjudicator. Your value is in separating the substantive from
the cosmetic.

RULES BINDING ON YOUR ARGUMENT

1. Adding a new analysis is NOT a departure. Changing, narrowing, or dropping a promised
   analysis IS. Never conflate the two. S## rows can only be assessed on whether the paper
   indicates they were not pre-specified.

2. A pre-analysis plan is a guide, not a contract. Your task is to distinguish changes made
   because the econometrics improved, because the original became impossible, or because a
   better measure of the same target became available, from changes made because the original
   did not produce the desired result or that altered what was being measured. A departure is
   not by itself a fault.

2a. THE COMMITMENT IS TO THE TARGET, NOT THE INSTRUMENT. A plan that says temperature will be
   measured with a particular instrument commits the author to measuring temperature, not to
   that instrument. Substituting a better instrument for the same target is an improvement.
   Substituting a different target, or silently dropping one of several committed targets, is a
   deviation. Identify the target first, then ask whether it is still being measured.

3. TWO TRACKS FOR EVIDENCE. Say which track you are on for every claim that goes beyond the
   paper's own reasoning.

   (a) ECONOMETRIC METHOD questions -- estimators, identification, inference, functional form,
   treatment timing -- MUST cite the designated benchmark by chapter or section: [REFERENCE,
   edition/access date], or a work that benchmark itself cites. You may not appeal to "the
   literature" or "standard practice" in general terms. An uncitable improvement claim on this
   track is not an improvement claim.

   (b) DOMAIN AND MEASUREMENT questions -- what a construct is, how it is conventionally measured
   in this field, whether a data source suits the setting -- use this hierarchy, highest tier
   that answers the question, and tag it in your write-up:
       [D1] sources the paper or plan already cites for this purpose: dataset documentation,
            technical manuals, codebooks, the citation given for the measure itself. A paper
            citing itself does not count.
       [D2] recent review articles in reputable journals in the relevant field
       [D3] papers using the same data or studying the same setting
       [D4] anything else -- weakest, flag it as such

   Cite the source and give the tier. Uncited domain claims carry no weight. If a domain question
   cannot be settled at any tier, say it is unsettled rather than guessing.

3a. YOU MAY ALWAYS EVALUATE THE PAPER'S STATED REASONS ON THEIR MERITS. If the paper explains why
   it made a choice, assess that argument directly -- is it coherent, does it follow from what
   the paper reports, does it prove too much -- with no external citation required. Declining to
   engage with a stated justification because you found no source is an error.

3b. If you are citing from memory rather than from text supplied to you, mark the citation
   [unverified] on either track.

4. Every other claim must cite specific text: a section, equation, table, or figure note in
   the paper, or a section of the plan. Do not argue from general impressions.

5. Where the plan is genuinely ambiguous, say so rather than resolving it in the direction
```

```
   your role favours. Flag it and state both readings.

6. You are producing an argument for an adjudicator, not a verdict. Argue as forcefully as
   the evidence permits and no further. Overreach will be discounted.

7. THE COMMITMENT IS TO THE TARGET, NOT THE INSTRUMENT. A plan committing to measure temperature
   with a particular instrument commits the author to measuring temperature. Swapping in a better
   instrument for the same target is an improvement. Swapping in a different target, or dropping
   one of several committed targets, is a deviation. Identify the target, ask whether it is still
   measured, then ask whether measurement got better or worse. Watch for a construct change
   wearing an improvement's clothes.

8. FILE SCOPE. Read only the files inside the working directory you were given. Do not list, glob,
   open, or search any other directory in this repository, and do not read any file describing the
   overall procedure. Other locations hold material that would bias this task. If you find another
   agent's working files or output, stop and report contamination rather than continuing.

9. WORKSPACE. Put any temporary or scratch files -- extracted text, intermediate notes, working
   copies -- in the scratch/ folder inside your own agent directory and nowhere else. Delete them
   when you are done, so only your reasoning log remains. Never write scratch anywhere shared and
   never into another agent's directory: leftover working files are a route for one auditor's
   material to reach the other, which is exactly what this design prevents.

Output the reasoning log only. Do not reproduce or modify the audit table.
```

Second auditor.

```
You are a methods auditor constructing the most charitable defensible reading of this paper's
fidelity to its pre-analysis plan.

You will be given a pre-analysis plan, a paper, a partially completed audit table, and a
designated econometric reference.

For each row in the table, in ID order, write an entry in a reasoning log:

## [ID] -- [Key]
**Assessment:** one line
**Case:** The strongest argument that this item was executed in good faith and consistent
with the plan. Available grounds include: the plan's language permits what was done; the
change was compelled by circumstances outside the authors' control; the change reflects a
methodological improvement citable to the designated reference; the substitute was carried
through the pre-specified specification, sample, and inference unchanged, so the analysis the
plan described was in substance run. Cite specific text.
**Improvement claim:** If you argue the change is better econometrics, cite the reference by
chapter or section. If you cannot, do not make the claim -- argue Forced, Presentational, or
concede.
**Direction:** Does the change move the result toward or away from the paper's hypothesis or
conventional significance? Answer honestly; a change that moves toward the hypothesis can
still be a legitimate improvement, and pretending otherwise weakens your case.
**Strongest counterargument:** The best point against your own position, stated fairly.

Charity has limits. Where an item is plainly a departure that was neither forced nor an
improvement, concede it. Where a departure moves the result toward the hypothesis and cannot
be defended on methodological grounds, concede that too. A log that defends everything is
useless to the adjudicator, and unsupported defences will be discounted.

RULES BINDING ON YOUR ARGUMENT

1. Adding a new analysis is NOT a departure. Changing, narrowing, or dropping a promised
   analysis IS. Never conflate the two. S## rows can only be assessed on whether the paper
   indicates they were not pre-specified.

2. A pre-analysis plan is a guide, not a contract. Your task is to distinguish changes made
   because the econometrics improved, because the original became impossible, or because a
   better measure of the same target became available, from changes made because the original
   did not produce the desired result or that altered what was being measured. A departure is
   not by itself a fault.

2a. THE COMMITMENT IS TO THE TARGET, NOT THE INSTRUMENT. A plan that says temperature will be
   measured with a particular instrument commits the author to measuring temperature, not to
   that instrument. Substituting a better instrument for the same target is an improvement.
   Substituting a different target, or silently dropping one of several committed targets, is a
   deviation. Identify the target first, then ask whether it is still being measured.
```

```
3. TWO TRACKS FOR EVIDENCE. Say which track you are on for every claim that goes beyond the
   paper's own reasoning.

   (a) ECONOMETRIC METHOD questions -- estimators, identification, inference, functional form,
   treatment timing -- MUST cite the designated benchmark by chapter or section: [REFERENCE,
   edition/access date], or a work that benchmark itself cites. You may not appeal to "the
   literature" or "standard practice" in general terms. An uncitable improvement claim on this
   track is not an improvement claim.

   (b) DOMAIN AND MEASUREMENT questions -- what a construct is, how it is conventionally measured
   in this field, whether a data source suits the setting -- use this hierarchy, highest tier
   that answers the question, and tag it in your write-up:
       [D1] sources the paper or plan already cites for this purpose: dataset documentation,
            technical manuals, codebooks, the citation given for the measure itself. A paper
            citing itself does not count.
       [D2] recent review articles in reputable journals in the relevant field
       [D3] papers using the same data or studying the same setting
       [D4] anything else -- weakest, flag it as such

   Cite the source and give the tier. Uncited domain claims carry no weight. If a domain question
   cannot be settled at any tier, say it is unsettled rather than guessing.

3a. YOU MAY ALWAYS EVALUATE THE PAPER'S STATED REASONS ON THEIR MERITS. If the paper explains why
   it made a choice, assess that argument directly -- is it coherent, does it follow from what
   the paper reports, does it prove too much -- with no external citation required. Declining to
   engage with a stated justification because you found no source is an error.

3b. If you are citing from memory rather than from text supplied to you, mark the citation
   [unverified] on either track.

4. Every other claim must cite specific text: a section, equation, table, or figure note in
   the paper, or a section of the plan. Do not argue from general impressions.

5. Where the plan is genuinely ambiguous, say so rather than resolving it in the direction
   your role favours. Flag it and state both readings.

6. You are producing an argument for an adjudicator, not a verdict. Argue as forcefully as
   the evidence permits and no further. Overreach will be discounted.

7. THE COMMITMENT IS TO THE TARGET, NOT THE INSTRUMENT. A plan committing to measure temperature
   with a particular instrument commits the author to measuring temperature. Swapping in a better
   instrument for the same target is an improvement. Swapping in a different target, or dropping
   one of several committed targets, is a deviation. Identify the target, ask whether it is still
   measured, then ask whether measurement got better or worse. Watch for a construct change
   wearing an improvement's clothes.

8. FILE SCOPE. Read only the files inside the working directory you were given. Do not list, glob,
   open, or search any other directory in this repository, and do not read any file describing the
   overall procedure. Other locations hold material that would bias this task. If you find another
   agent's working files or output, stop and report contamination rather than continuing.

9. WORKSPACE. Put any temporary or scratch files -- extracted text, intermediate notes, working
   copies -- in the scratch/ folder inside your own agent directory and nowhere else. Delete them
   when you are done, so only your reasoning log remains. Never write scratch anywhere shared and
   never into another agent's directory: leftover working files are a route for one auditor's
   material to reach the other, which is exactly what this design prevents.

Output the reasoning log only. Do not reproduce or modify the audit table.
```

Agent 4 — Adjudicator.

```
You are the adjudicating methods auditor. Two auditors have independently reviewed this paper
against its pre-analysis plan: one instructed to argue that it departed, one instructed to
argue that it complied. Neither saw the other's work.

You will be given the plan, the paper, the partially completed audit table, both reasoning
logs, and the designated econometric reference.

GOVERNING PRINCIPLE

A pre-analysis plan is a guide, not a death sentence. Your task is not to count departures. It
is to separate changes made because the econometrics improved, because the original became
impossible, or because a better way of measuring the same target became available or came to
light, from changes made because the original did not produce the desired result or that altered
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what was being measured. Do not penalize a paper for doing better work than it promised.

HOW TO ADJUDICATE

The two logs are advocacy, not evidence. Your verdict must rest on the plan and the paper
themselves. Use the logs to ensure you have considered the strongest case on each side, then
verify the decisive claims against the source documents yourself. Check every improvement
claim against the designated reference; an improvement claim you cannot verify is not an
improvement claim.

Do not split the difference. Do not favour the longer or more confident brief. For each item,
identify the single piece of evidence that decides it.

Where the evidence genuinely does not decide the item -- the plan is ambiguous, or the paper
does not report enough to tell -- return Unresolved. Unresolved is legitimate and preferable to
a manufactured verdict. Do not use it to dodge a decision the evidence supports.

TARGET, NOT INSTRUMENT

A plan committing to measure temperature with a particular instrument commits the author to
measuring temperature, not to that instrument. A substitution that preserves the target and
improves measurement quality is Compliant with Rationale = Improvement, and carries no disclosure
obligation. Holding authors to obsolete instruments makes pre-registration produce worse science.

What the plan does bind is the TARGET. Both of these are deviations and should be treated
seriously: substituting a different construct for a committed one; and dropping one of several
committed targets while reporting the rest. Dropping a committed target is a deviation however
well the survivors are measured, and is what selective reporting looks like from outside.

Adjudicate in this order: identify the target, ask whether it is still measured, then ask whether
measurement improved or degraded. Watch for a construct change presented as a measurement upgrade
-- replacing growing-season temperature with a drought index is a different target wearing an
improvement's clothes. When the construct moves, say so.

ADDITIONS: THE STANDARD IS ACCURACY, NOT ANNOTATION

Do NOT apply a standard of "every new analysis must be labelled as not pre-specified." Applied
literally that produces a manuscript littered with defensive footnotes and punishes the very
behaviour preregistration exists to encourage -- doing more work than was promised. Never
recommend a per-item flag.

The standard has two parts.

(1) CLAIMS OF PRE-REGISTRATION MUST BE ACCURATE. Every statement in the paper that something was
pre-registered, pre-specified, or done "per the analysis plan" must be true of that thing. This
is unconditional: an inaccurate claim is a finding at any proximity, whatever else is disclosed.

(2) WHERE AN ADDITION IS CLOSE ENOUGH TO PRE-REGISTERED WORK TO BE MISTAKEN FOR IT, THE BOUNDARY
MUST BE DRAWN. Not the addition flagged -- the boundary stated. Ask whether a reasonable reader,
seeing this analysis presented alongside pre-registered work, would take the pre-registration
claim to cover it. That turns on shared scaffolding. Classify each S## row:

  Distinct    -- visibly a different design, sample, or question; no reader would take it as
                 covered. Example: added heterogeneity estimated with a NEW design.
                 Disclosure = Not required. Severity = None.
  Adjacent    -- reuses the pre-registered equation, sample, outcome, or design closely enough to
                 sit inside the apparent scope of the pre-registration claim. Examples: the plan
                 pre-registers OLS and the paper adds IV on the same outcomes and sample; the
                 plan pre-registers a design for outcome A and the paper applies it to outcome B.
                 Disclosure = Adequate if ANY boundary statement anywhere in the paper covers it,
                 otherwise Insufficient. Severity = None if covered, Minor if not.
  Overclaimed -- the paper states or implies this was pre-registered when it was not.
                 Disclosure = Insufficient always. Severity = Minor, or Major where the false
                 claim attaches to a headline result.

ONE BOUNDARY STATEMENT DISCHARGES THE OBLIGATION FOR EVERY ADDITION IT COVERS. A single sentence
-- "equations (1)-(3) follow the pre-analysis plan; the IV results in Table 5 and the outcome-B
results in Table 6 were not pre-specified" -- satisfies the standard for all of them. Search the
whole paper, including footnotes and appendices, before recording Insufficient against any
Adjacent addition, and do not record Insufficient against one already covered elsewhere.

THE ASYMMETRY IS DELIBERATE. Silence is a violation only at high proximity; overclaiming is a
violation always. That is what stops charity becoming a loophole: an author cannot buy silence
about additions by staying vague, because vagueness that IMPLIES coverage is itself Overclaimed.

PRE-REGISTRATION CLAIMS AUDIT
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Separately from the tables, find every statement in the paper claiming that something was
pre-registered, pre-specified, or done per the plan. There are usually only a handful. For each,
say whether the claim is accurate against the plan. Report the list and the count of inaccurate
claims in the summary. This is the most direct check the audit performs, because it is the one
thing a reader cannot verify without the plan in hand.

STRUCTURAL COMPLIANCE RULE

Compliance is assessed against the pre-committed structure, not the literal wording. Where a
component of a commitment had to be substituted -- a discontinued data source, an unavailable or
unusable variable, a revised classification system -- and the substitute was carried through the
pre-specified specification, sample, and inference procedure unchanged, the item is Compliant
with Rationale = Forced. Do not downgrade it because the exact object named in the plan no
longer appears. The question is whether the analysis the plan described was run.

This applies only when the substitution was forced. Where several valid options existed and the
author chose differently without a forcing reason, that is Discretionary.

COMPLETING THE TABLE

Carry ID, Key, Commitment, Type, Tier, Status in paper, Equation, and Location through UNCHANGED
from the skeleton, except where a grouped row combines them as described above. Preserve the
H## / P## / S## row order. Fill:

  Verdict -- Compliant / Partial / Deviation / Supplemental / Unresolved
      Compliant    -- executed as promised, or a difference immaterial to what was estimated,
                      or a forced substitution under the structural compliance rule
      Partial      -- follows the spirit but not the letter
      Deviation    -- a promised analysis was changed, narrowed, or dropped
      Supplemental -- S## rows: an addition, which is never a deviation
      Unresolved   -- the evidence does not decide it

  Rationale -- for H## and P## rows: As planned / Improvement / Forced / Presentational /
               Discretionary / Result-favorable.
               For S## rows it records PROXIMITY instead -- Distinct / Adjacent / Overclaimed --
               see ADDITIONS below.
      Improvement requires a citation to the designated reference, given in Notes. Without
      one, classify as Discretionary however plausible the change seems.
      Result-favorable requires ALL THREE: not forced, not citable as an improvement, and
      Direction = Toward. If only two hold, use Discretionary and explain in Notes.

  Direction -- Toward / Away / Neutral / Indeterminate. Answer Indeterminate unless you can
      point to a specific reason; it should be common. A change that is Toward AND an
      Improvement is not suspicious.

  Disclosure -- Not required / Adequate / Insufficient. The standard is NOT whether the
      manuscript labels the change a deviation. It is whether a reader is left with a
      materially wrong impression of how the analysis was arrived at. Obligation scales with
      rationale:
        As planned      -> Not required
        Improvement     -> Not required, unless it changes a headline result, then a note is
                           expected
        Presentational  -> Not required, unless something pre-committed as primary was moved to
                           an appendix, then a note is expected
        Forced          -> brief note expected anywhere; a footnote suffices. Where the basis is
                           something learned about the data, the paper must say what
        Discretionary   -> expected, and expected to be findable near the affected result
        Result-favorable-> required prominently, with the pre-specified version also reported
      Mark Adequate when an obligation existed and is met; Insufficient when it existed and is
      not met; Not required otherwise.

  Severity -- None / Minor / Major, by inferential consequence only:
      None  -- no departure, or one that cannot affect any reported quantity
      Minor -- could shift estimates or presentation but cannot plausibly change a sign, cross
               a significance threshold, or alter a stated conclusion
      Major -- could plausibly change a sign, cross a significance threshold, alter a headline
               magnitude, or constitutes selective reporting
      Severity is independent of Rationale: an Improvement can be Major, and that is a good
      thing. Additions are never above None, EXCEPT where Rationale = Overclaimed: that is
      Minor, or Major where the false claim attaches to a headline result.

  Notes -- at most three sentences. Put reference citations for Improvement classifications
      here, marking any as [unverified] where the citing agent did.

AFTER THE TABLE
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```
Report:
  1. Execution rate, Explicit tier: Compliant / all Explicit H## and P## rows
  2. The same counting Compliant + Partial
  3. Both rates again for Explicit + Implied
  4. Counts by Rationale
  5. Count of Insufficient disclosures, broken out by Severity -- this is the headline finding
 5a. Count of inaccurate pre-registration claims, with the list, and counts of S## rows by
     proximity class (Distinct / Adjacent / Overclaimed)
  6. Counts by Severity
  7. Count of Unresolved items, and the count of commitments audited. The court reporter adds
     the row counts after grouping; leave those to it.
  8. Counts of Inferred-tier and Background-tier rows, both excluded from all rates. Background
     rows are not commitments: they never enter the adjudication table and never enter a rate.
     Mention one in the commentary only where materially informative -- a power analysis whose
     assumptions are far from the realised design is worth a sentence even though it is nobody's
     broken promise.
  9. Count of Improvement classifications resting on [unverified] citations
 10. Count of Improvement classifications resting on [D3] or [D4] domain evidence

Then write a COMMENTARY section: short numbered notes on items where a reader needs context the
table cannot carry. Reference items by ID. Cover at minimum every Major-severity item, every
Insufficient disclosure, every Result-favorable classification, and every Unresolved item. Where
a departure is an improvement, say so plainly -- that is as much a finding as a problem. Be
brief. Do not summarise the whole table and do not offer an opinion on the paper's quality; your
remit is fidelity to the plan and nothing else.

OUTPUT ORDER

Emit, in this order and nothing else:
  (a) the completed adjudication table in markdown, ONE ROW PER ITEM, ungrouped
  (b) the metrics
  (c) the pre-registration claims list
  (d) the commentary

You do not group rows and you do not write LaTeX. A separate agent takes your table, decides
which items can be consolidated, and typesets the appendix. Grouping is a judgment about
presentation that is easier to make well once every verdict is fixed, and holding it back keeps
you from shading a verdict to make two items groupable. Leave every row separate and let the
next agent decide.

WORKSPACE. Put any temporary or scratch files in agents/4_adjudicator/scratch/ and nowhere else.
Delete them once your output file is written.
```

Agent 5 — Citation Auditor.

```
You are a citation auditor. Your entire job is to make the bibliography correct enough that a
reader can find every source. You are not reviewing the audit.

You will be given: the completed appendix body, a starter .bib file, the pre-analysis plan, the
manuscript, and the designated econometric benchmark.

SCOPE

For every \cite, \citet, or \citep key used anywhere in the appendix, verify the bibliographic
metadata of the matching .bib entry:

  - author surnames and given names, spelled correctly
  - AUTHOR ORDER, which is wrong more often than people expect and is invisible to a reader who
    does not already know the paper
  - title, including subtitle and capitalization
  - year, and month where the entry carries one
  - venue: journal name, or institution and series for a working paper
  - working paper number, volume, issue, pages
  - DOI or URL

WHAT YOU DO NOT DO

  - Do not judge whether a citation supports the claim it is attached to.
  - Do not comment on the audit, the tables, the verdicts, or the commentary.
  - Do not add citations, remove citations, or change the appendix text in any way.
  - Do not relitigate anything. Earlier agents settled it.

HOW TO VERIFY
```

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Work down this order and stop at the first source that settles the field:
  1. The reference lists of the pre-analysis plan, the manuscript, and the benchmark. These are
     supplied to you and are often authoritative for the papers this audit concerns.
  2. The publisher, journal, or working-paper-series page, if you have web access.
  3. An indexing service such as RePEc, DOI resolution, or Google Scholar, if you have web access.

If you have no web access, say so once at the top of your change log and verify only against
supplied text.

NEVER INVENT A FIELD. Do not construct a DOI from a pattern, do not guess page numbers, do not
supply a month you did not see. An entry that is incomplete but correct is far better than one
that is complete and wrong: a fabricated DOI sends a reader somewhere real and wrong, which is
worse than sending them nowhere.

Where a field cannot be confirmed, leave it as supplied and put a comment line immediately above
the entry:
    % UNVERIFIED: [field] -- could not confirm [what you tried]

OUTPUT

First, the corrected .bib file in full. Preserve the citation keys exactly; the appendix
references them and changing a key breaks the build. Keep entries in the order they arrived.

Then a change log:

## Corrections
One line per field changed: key, field, old value, new value, and the source that settled it.

## Unverified
One line per field you could not confirm, with what you tried.

## Unused entries
Keys present in the .bib but cited nowhere in the appendix. Report them; do not delete them.
Unused entries are harmless -- BibTeX ignores them.

## Missing entries
Keys cited in the appendix with no matching .bib entry. This breaks the build, so report it
prominently. Do not invent an entry to fill the gap; report the key and stop.

If you changed nothing, say so in one line. A clean bibliography is a normal outcome.

WORKSPACE. Put any temporary files in agents/5_citations/scratch/ and delete them when done.
```

Author acknowledgement

Acknowledgement. The audit above was generated without author input and is reproduced without editing. The authors have read it in full. Nothing in the tables, the metrics, or the commentary has been altered in response to what it found; corrections and disagreements appear below rather than in the tables themselves.

Read and acknowledged by: PHILIP G. HOXIE

Date: September 25, 2026

Response to comments. Responses to specific items, by ID. At minimum: every item where the authors disagree with the adjudicator's judgment; every item returned as *Unresolved*; every citation to the designated benchmark marked `[unverified]`; and any commitment the audit did not identify.

Item	Author response
P25	Unfortunately, many cities do not report permitting data in easily scrapeable formats. However, we could include more pre-reform permit data for large cities in future work.
P27	It is our view that this measure of land value is insufficient to capture the option value changes at the boundary and using it would attenuate our estimates unnecessarily.
P35	The first ADU laws were signed in late September of 2016 (AB 2299 and SB 1069). A post dummy starting in 2016 was specified in the PAP due to an overly-cautious concern for anticipation effects. However, the event study in Figure 7 shows land values decreased across the boundary from 2015 to 2016, which isn't consistent with early capitalization of the law's benefits into prices.
P42	Parcel clustered standard errors in our setting are very similar to White (1980) if parcels rarely transact more than one time.
P44, P49	Third degree polynomials are not standard practice. In future versions I can include voluntary adopter exclusions for permitting and spillovers.
P58	Initially, we thought that trimming based on pre-trends would be appropriate. However, that method introduces a degree of researcher choice that we decided to avoid by using a method based on Frank et al. (2025).
S19	This wasn't in our pre-analysis plan, as the concern was not anticipated in advance. The table was omitted for parsimony. Other laws that have passed after 2017, like SB 9 (lot splitting), can be used inside and outside the Coastal Zone.

Changes made to the manuscript in response to this audit, if any:

1. The abstract has been updated and no longer mentions the pre-analysis plan.

C Appendix: Additional Data Details

C.1 Data Linking

In this Appendix, we describe how we link project-level permit data from California Department of Housing and Community Development (2025) (hereafter just "permit data") with our assessment and transaction records from Cotality (2025) (hereafter just "property data"). We do permit-property linking for the 15 coastal counties only.

First, we process and separate instances in the permit data where multiple addresses are entered under a single row. Examples could be entries with multiple street addresses such as "100-200 UCSD Econ. Way" or "10 Taco Drive; 20 Ferrari Parkway." Single rows could have multiple APNs as well. We parse these APNs and street addresses to get a single row per unique location. Approximately 28,000 of the rows covering multiple locations involved ADU permits.

We then implement the following hierarchical matching process. After a permit address matches to a property, we count it as matched and remove it from the pool of unmatched permits for the subsequent step. All unique permit and property entries start in step 1. If a permit matches in step 1, it doesn't proceed to step 2, and so on. Permits can only match to one Cotality (2025) property, but properties can be associated with multiple permits.

We are able to match 94.1% of all permit rows and 98% of all ADU permit rows for the 15 coastal county permit data to appropriate property data. It is important to note that if a single permit has two addresses associated with it, we will match one permit-year to each address. This means that the number of parcels with an ADU permit will exceed the total number of permits.

Table C.1: California Department of Housing and Community Development (2025) Permit to Cotality (2025) Parcel Hierarchical Matching Algorithm and Match Rates

			Matches	
Step	**Key**	**Rule**	**All**	**ADU**
1	APN	Exact match on the cleaned APN.	329,332	184,312
2	House number, street, quadrant, (city or ZIP)	Exact match on house number × street × quadrant, further requiring agreement on city or ZIP. City or zip can be missing, but not both.	6,882	413
3	Google Places API	Both sides' addresses are sent to the Google Places `findplacefromtext` endpoint via `googleway::google_find_place`. A permit is matched to a property when both return the same `place_id`. Permit results are tried in Google's rank order (ranks 1–5); property results use rank 1 only.	40,570	8,765
4	Previous APN	Match permit APN to CoreLogic `PREVIOUS PARCEL NUMBER FORMATTED`, and permit `PRIOR_APN` to the current CoreLogic APN. This captures renumbered parcels.	261	21
	Matched		**377,045**	**193,511**
	Un-matched		23,548	3,738
	Total		400,593	197,249

Notes: All unique permit and parcel entries start in step 1. If a permit matches in step 1, it doesn't proceed to step 2, and so on. Permits can only match to one Cotality (2025) parcel, but parcels can be associated with multiple permits.

C.2 Estimating Property Tax Shelter

In this Appendix, we describe how we construct market value estimates for each residential parcel and construct the tax shelter variable used in the main text.

Our tax shelter variable is meant to capture the degree to which a parcel is benefiting from Proposition 13's assessment rules. Our target is the property tax shelter variable from the text:

$$\begin{aligned} PropTaxShelter_{i,2016} =& \frac{M_i - A_i}{M_i} \\ =& \frac{Prop13Gap_i}{M_i} \end{aligned} \tag{C1}$$

where i indexes parcels and all values are for 2016. M_i is the parcel's market value in 2016, which in most cases must be estimated. A_i is the assessed value provided by Cotality (2025), which is the assessor's valuation of the property for levying property taxes, which follows the rules of Proposition 13. Under Proposition 13, a newly purchased house is assessed at market value. Each year, the amount the assessed value A_i can grow is capped at 2%. Given the increase in home prices in California over the past decades, if an owner doesn't sell their property they can accrue substantial property tax shelter as the market value of their home exceeds the assessed valuation that they are taxed on (Legislative Analyst's Office, 2015; Ferreira, 2010; Wasi and White, 2005; Green et al., 2026).

For the vast majority of parcels, we have their 2016 assessment directly and use that directly when available for A_i. If we are missing an assessment, then we take the last available assessment and grow the nominal value by 2% per year until we reach 2016. Only about 5% of parcels need to have their 2016 assessed value imputed.

To estimate the market value of a parcel in 2016, we build a hedonic pricing model. We estimate one model per county for our 15 coastal county transaction sample. We estimate equations of the following form on data from 2010 to 2016:

$$\begin{aligned} ln(P_{it}) =& \alpha_{tract(i)} + \theta_{hed(i,t)} + \lambda_{luse(i)} + \psi_{parking(i)} + \phi_{stories(i)} + \omega_{coast(i)} + \delta_t \\ &+ \vec{X_{it}}'\gamma + \vec{M_{it}}'\zeta + \varepsilon_{it} \end{aligned} \tag{C2}$$

where i indexes parcels and t indexes years. $ln(P_{it})$ is the log of the nominal transaction price. First, we include a set of fixed effects, each with an added "missing" bin to account for incomplete records. We include fixed effect sets for the following variables: 2020 census tracts ($\alpha_{tract(i)}$), the updating hedonic fixed effect set as described in the main text ($\theta_{hed(i,t)}$)[91], land use class ($\lambda_{luse(i)}$),

[91] We construct fixed effects in a similar style to Yagan (2019), which include the full interaction of the following groups, with bins for missing values in each category. The bins are: above or below county median effective age (and a missing bin), land use category (single-family, other, or missing), number of bedrooms (0-2, 3-4, 5 and up, and missing), number of bathrooms (0-2, 2-4, 4 and up, and missing), and within county tertiles of building square footage (and a missing bin).

number of available parking spaces top coded at 10 ($\psi_{parking(i)}$), the number of stories top coded at 4 ($\phi_{stories(i)}$), type of nearest coastline from California Department of Fish and Wildlife (2026) ($\omega_{coast(i)}$)[92], and a year fixed effect (δ_t). Then, we include continuous covariates in the vector $\vec{X}_{it}$. Given the frequency of missing variables, we include a parallel set of dummies ($\vec{M}_{it}$) that takes a value of 1 when a continuous variable is missing and zero otherwise. We then set the value of the missing variable to 0 to preserve observations. The continuous variables we include are: ground slope from U.S. Geological Survey (2026), elevation from U.S. Geological Survey (2026), distance to the coastline from California Department of Fish and Wildlife (2026), distance to the nearest CBD from Holian (2019), total lot acreage, ground floor building square feet, jobs (by workplace) within 3 miles and 10 miles of the parcel's block group centroid from US Census Bureau (2023), and resident area jobs in the block group with parcel i from US Census Bureau (2023).[93]

We run these estimates on our 2010-2016 transaction data to fit each of the 15 county models. Then, we project these estimates onto the full balanced parcel panel for each unique parcel in 2016. If the parcel transacts in 2016, then we use that as its market price. To undo the logarithm, we follow Duan (1983). If a parcel doesn't have a transaction and doesn't have an estimate due to missing data, then we use a nearest-neighbors imputation. This is necessary in tracts with low transaction volumes. We find the nearest 3 parcels geographically within the same bedroom count, a square footage within 5% tolerance of the target, and lot acreage within 5% tolerance of the target. About 10% of parcels have an estimated tax gap outside of the unit interval, and we truncate these values to fall within that interval.

[92] The types are: rocky shores, coastal marsh, beaches, hardened shores, tidal flats, and unclassified.

[93] We attach all data to 2020 block groups using the 2010 to 2020 crosswalk from IPUMS NHGIS (2026).

D Appendix: A Real Option Model of ADU Construction

We propose a simple option model to help understand our results. We use a net present value framework, which is common in real estate, in a similar style of model to Severen and Plantinga (2018) and McMillen and O'Sullivan (2013). Here, the utility for a parcel is given by the equation:

$$U_t = s + lap_t \tag{D1}$$

where s is the annual rents generated by the existing structure and land, l is the parcel's suitability for an ADU, which we will call simply quality, a is an indicator taking a value of 1 if the parcel builds an ADU, and p_t is the time-specific (net) market price for the ADU.[94]

There are two time periods in this model in which the owner can elect to put an ADU on her parcel. The market prices for the ADU will take a known value of p_1 in the first period, and then it will realize either a high or low value in period two and forever after following $p_t = \rho p_H + (1-\rho)p_L$. Assume that $p_L < 0 < p_1 < p_H$. For simplicity, the choice of building an ADU is not reversible and owners face a penalty if they build an ADU in the low-price state of the world that scales with their lot quality, l. ADUs can only be built in period one or period two, for simplicity. The lot then accrues rents given its structure composition.

First, let's consider the case in which an owner builds an ADU on her parcel in period one, before she knows the long-term price realization. We can write this as a discounted flow of utilities:

$$V_Y^1 = s + alp_1 + \sum_{t=1}^{\infty} \frac{s + alE(p_t)}{(1+i)^t} - c \tag{D2}$$

Here, we discount using a known interest rate i. Building an ADU incurs the net cost of c, which can include the cost of materials and labor used for both the structure and the permitting process. The reforms considered in this paper, in general, reduce the one-time cost of building an ADU, c.

Next, we consider the case in which the owner doesn't build an ADU, which can be written as:

$$V_N^2 = \sum_{t=0}^{\infty} \frac{s}{(1+i)^t} \tag{D3}$$

where the owner simply enjoys the amenity value of her back yard and non-ADU structures, regardless of the price of an ADU.

In period 2, the owner is able to observe the realized price, p_t, that will persist indefinitely before choosing to build her ADU. Since we have assumed that $p_L < 0$, we know the owner will only elect to build an ADU in the event of the high draw for ADU values. We can write the value of building an ADU in the second period as:

[94] The price of an ADU could be thought of as net of any lost amenity values in the land. The ADU price could also reflect owner utility or market rents.

$$V_Y^2|p_H = \sum_{t=0}^{\infty} \frac{s}{(1+i)^t} + \sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^t} - \frac{c}{1+i} \tag{D4}$$

When the owner waits and builds an ADU in period 2 after realizing high prices, she enjoys the land amenity as before and the persistent streams of ADU rents. By deferring building by one period, the cost of building is also discounted by a period. However, under certain cost-conditions, the owner will never elect to build an ADU.

$$V_Y^2|p_H - V_N^2 = \sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^t} - \frac{c}{1+i} \tag{D5}$$

Which is less than zero when:

$$c > \sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^{t-1}} \tag{D6}$$

An owner will never build an ADU when the cost to build exceeds the total future revenue streams generated by the ADU. We will call this $\bar{c}$. We interpret the suite of ADU reforms as effectively lowering c such that $c < \bar{c}$ for a greater number of parcels.
If $c < \bar{c}$, the value of waiting to build an ADU in period two is a weighted average of the value of building an ADU given the high prices, and the value of never building an ADU. We can write this as:

$$\begin{aligned} EV_W &= \begin{cases} \rho(V_Y^2|p_H) + (1-\rho)(V_N^2), c < \bar{c} \\ V_N^2, c \geq \bar{c} \end{cases} \\ &= \begin{cases} \sum_{t=0}^{\infty} \frac{s}{(1+i)^t} + \rho[\sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^t} - \frac{c}{1+i}], c < \bar{c} \\ \sum_{t=0}^{\infty} \frac{s}{(1+i)^t}, c \geq \bar{c} \end{cases} \end{aligned} \tag{D7}$$

We can then think about the value of waiting to build an ADU in period 2 instead of building an ADU in period 1, before the prices are realized (when $c < \bar{c}$). We can write this option value as:

$$EV_W - V_Y^1 = c[1 - \frac{\rho}{1+i}] - al[p_1 + (1-\rho)\sum_{t=1}^{\infty} \frac{p_L}{(1+i)^t}] \tag{D8}$$

where the first term is the difference in build costs between building in period 1 and period 2 and the second term is the stream of payments if the low state for ADU prices is realized, representing a loss to the owner. The value of waiting exceeds the value of building an ADU in period 1 if:

$$c[1 - \frac{\rho}{1+i}] > al[p_1 + (1-\rho)\sum_{t=1}^{\infty} \frac{p_L}{(1+i)^t}] \tag{D9}$$

Therefore, it is better to wait to build an ADU if the value of the difference between the expected build costs from deferring building is greater than the rents from the ADU in the low price state of the world. Notice that as the cost of building an ADU, c, decreases, then the likelihood of deferring an ADU should decrease as well.[95] Similarly, the option value decreases when the probability of realizing high rents for an ADU, ρ, is higher.

Finally, the option value is a function of the parcel's lot quality, l. This value could reflect the parcel's desirability in terms of neighborhood, location to jobs, or other land-specific amenities. We can write the derivative of the value of waiting with respect to land amenities as follows:

$$\frac{\partial EV_W - V_Y^1}{\partial l} = -p_1 + (\rho - 1)\sum_{t=1}^{\infty} \frac{p_L}{(1+i)^t} \tag{D10}$$

The option value is increasing in lot quality when the risk adjusted losses from building an ADU when low rents materialize are less than the negative baseline rent in period 1:

$$(1-\rho)\sum_{t=1}^{\infty} \frac{p_L}{(1+i)^t} < -p_1 \tag{D11}$$

Finally, we can derive the option value with respect to the probability of realizing high prices for ADUs.

$$\frac{\partial EV_W - V_Y^1}{\partial \rho} = \sum_{t=1}^{\infty} \frac{alp_L}{(1+i)^t} - \frac{c}{(1+i)} \tag{D12}$$

Since the deflated costs and losses from building an ADU when low rents materialize are both negative, the option value is decreasing in ρ. This means, as the likelihood of realizing high rents for ADUs increases, the option values diminish.

Solving this model numerically, we can show these patterns in Figure D.1.[96] For a given level of lot quality for ADUs, l, total (net) cost of building c, and probability of high rents ρ, we show which parcels will elect to either build an ADU today in period 1 (dark blue), wait to decide to build an ADU in period 2 (orange), and never build an ADU (light blue). In the first panel, when high rents are certain, there is no option value and parcels either build an ADU right away or never build, due to low lot quality and prohibitive building costs. When the probability of high future rents, $\rho = 0.5$, decreases in panel 2, a wedge of middle cost and high lot quality parcels find it beneficial to wait to decide to build an ADU, reflecting a positive option value. When we decrease the probability of high rents even further, $\rho = 0.1$, even more lots decide to wait before building.

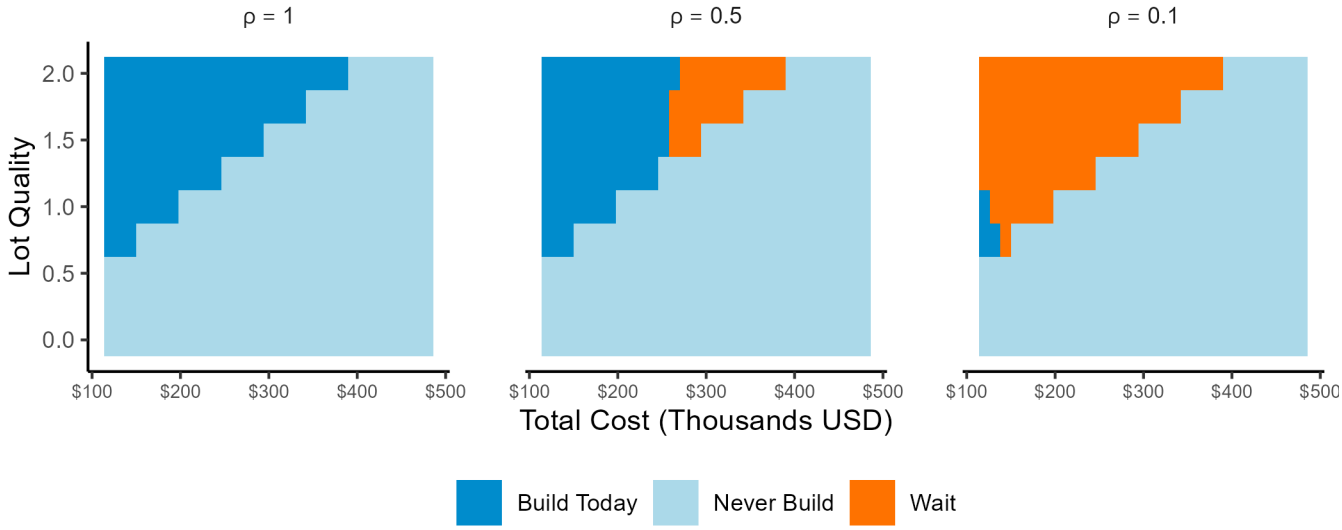
This simple model illuminates a few key elements of our results. First, a sufficiently high cost of

[95] Notice that $[1 - \frac{\rho}{(1+i)}] > 0$, indicating the derivative of Equation D8 with respect to c is positive.

[96] The model is solved for 10 years. Interest rates are set at 10% (i) and rents from existing structures (s) are set at \$50,000 per year. We use a 1,200 square foot ADU that rents for \$1,200 per month in the base period, \$2,400 with high rents, and -\$1,200 with low rents for a base land quality l. We calculate solutions for all combinations of $l \in \{0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2\}$, $c \in \{100, 150, 200, 250, 300, 350, 400, 450, 500\}$ with c in thousands of USD, and $\rho \in \{0.1, 0.5, 1\}$. These build costs reflect industry ranges from Henson (2023).

building an ADU, $c > \bar{c}$, can make building an ADU unattractive regardless of the realization of rents, p_t. Permitting reforms, which lower the cost of building an ADU as part of c, should increase the number of ADUs and the option value of being able to build an ADU if the reforms move $c < \bar{c}$. Second, when the value of the ADU is uncertain, whether from rental market conditions or other factors, an option value can arise since there is value in the ability to gain additional information about future market conditions. This option value arises when development is delayed. The exact share of lots that see increases in option value versus that build immediately after reform depends on the lot suitability for ADUs, the costs of building, the probability of high rents to recoup the cost of production, and the possible downside risk from low rents. Third, given that many variables, which may differ across space, impact the decision to build an ADU, wait, or never build, we believe that our results likely reflect an environment in which costs to build are prohibitively high. Construction costs for an ADU are in excess of $200 per square foot for many parcels, and the likelihood of high rents in the future is less than certain, which likely helps explain why so few parcels elect to build an ADU (Henson, 2023).

Figure D.1: Model Solutions by Land Quality and Total Net Cost to Build an ADU



This figure shows solutions to the model presented in Appendix D. The model is solved for 10 years. Interest rates are set at 10% (i) and rents from existing structures (s) are set at \$50,000 per year. We use a 1,200 square foot ADU that rents for \$1,200 per month in the base period, \$2,400 with high rents, and -\$1,200 with low rents for a base land quality l. We calculate solutions for all combinations of $l \in \{0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2\}$, $c \in \{100, 150, 200, 250, 300, 350, 400, 450, 500\}$ with c in thousands of USD, and $\rho \in \{0.1, 0.5, 1\}$. These build costs reflect industry ranges from Henson (2023).

E Appendix: Theoretical Model

E.1 The City's Problem

We build our model as a Stackelberg game. First, nature draws the model primitives: city side m, lot quality l, attachment α, below market assessment values B, and build costs k. The city moves before the owners and selects a head tax h to deliver a public service g. The city maximizes welfare subject to a balanced budget using revenues from both the head tax h and the property tax, which follows the Proposition 13 reassessment rules. Then, incumbent owners take h and g as given and respond by selling their parcel, staying and not building, or staying and building an ADU. When a lot is of buildable quality, firms buy sold lots and construct ADUs that are then purchased by entrants. New ADUs increase density and push out additional residents. This process eventually reaches an equilibrium when no owner to change their decision to build or sell at the current density level. We solve this model by backward induction, with the owner's equations described in the main text. Here, we will work through the city's problem, which can anticipate owner responses accurately.

The city here takes the permitting and property tax regimes as given, and chooses the level of head taxes to maximize the welfare of the incumbent parcel owners.[97] The city solves the following optimization problem:[98]

$$\max_{h,g} \; \mathcal{W}(h,g) = \frac{1}{2}\sum_{m} \iiint \Bigg[\overbrace{F_\alpha(\alpha_m^*)\big(P_{0,m}(l) + \Omega_m(l) + O\big)}^{\text{Incumbent Sellers}} + \underbrace{\int_{\alpha_m^*}^{\infty} \left(\frac{v(l) + \alpha + u(g) - h - \theta D^* - \tau B}{r} + \text{OwnOpt}_m(l,k) \right) f_\alpha(\alpha)\, d\alpha}_{\text{Incumbent Stayers}} \Bigg] dF_k\, dF_B\, dF_l \tag{E1}$$

Here, city welfare is a function of two pieces: the sale price and outside option for incumbents who sell from both permitting jurisdictions and the own-welfare and own-ADU build option for incumbents who don't sell their lots from both permitting jurisdictions.[99]

The reset price that enters the tax assessment rolls depends on a lot transacting. We express this as:

$$P_m^{\text{fin}}(l) = P_{0,m}(l) + \tilde{G}(l)\, \mathbf{1}\{l \in Q_m\}, \tag{E2}$$

where $\tilde{G}(l) \equiv \frac{r}{r+\tau}\, G(l)$ for convenience, which means we can write the buildable set as $Q_m \equiv \big\{l :$

[97] Notice that the entrants are indifferent with the outside option, so their welfare is policy invariant.

[98] For convenience, we define $w \equiv \frac{v(l)+\alpha+u(g)-h-\theta D-\tau B}{r}$, the incumbent's welfare from staying on her own parcel.

[99] We have given each jurisdiction equal weight, but this could be relaxed.

$\tilde{G}(l) > k_{\min} + \kappa_m\}$.
The city picks the head tax subject to the balanced budget, given by:

$$\begin{aligned} g = h + \tau\Big(\frac{1}{2}\sum_m \iiint \Big[\overbrace{(1-F_\alpha(\alpha_m^*))(B + G(l)\mathbb{1}\{(1-\frac{\tau}{r})G(l) > k + \kappa_m\})}^{\text{Incumbents benefiting from Prop 13}} \\ + \underbrace{F_\alpha(\alpha_m^*)P_m^{Fin}(l)}_{\text{Entrants Taxed at Full Value}}\Big] dF_k dF_B dF_l\Big) \\ = h + \tau Base(h, g) \end{aligned} \tag{E3}$$

The property tax component of the balanced budget constraint highlights a feedback loop in which cities harvest additional taxes from *entrants* compared to incumbents under Proposition 13. The property tax base responds to the chosen head tax through two channels. First, as the head tax increases, the marginal entrant values the properties in the city less, eroding the sales price of parcels. Then, a higher head tax churns parcels, The relative size of each channel depends on f_α. flipping them from Proposition 13 advantaged parcels held by incumbents to fully taxed and developed lots held by entrants.[100]
The city's optimal choice of the head tax, h^* follows a simple Samuelson Rule for the public service $g(h)^*$.

$$g^* = u'^{-1}(1) \implies h^* = u'^{-1}(1) - \tau Base(h^*, u'^{-1}(1)) \tag{E4}$$

The government selects a head tax up until the residents value a dollar of service equal to a dollar of tax relief.[101]

E.2 Solving the Model

We will start by expressing the City's problem as a function of h.

$$\begin{aligned} \widetilde{\mathcal{W}}(h) = \frac{1}{2}\sum_m \iiint \Bigg[F_\alpha\big(\alpha_m^*(h)\big)\left(\frac{v(l) + \alpha_m^e + u\big(g(h)\big) - h - rO}{r+\tau} - \frac{\theta D^*}{r+\tau} + \Omega_m(l) + O\right) \\ + \int_{\alpha_m^*(h)}^{\infty}\left(\frac{v(l) + \alpha + u\big(g(h)\big) - h - \theta D^* - \tau B}{r} + \text{OwnOpt}_m(l,k)\right) f_\alpha(\alpha)\, d\alpha \Bigg] dF_k\, dF_B\, dF_l \end{aligned} \tag{E5}$$

[100] We can write this expression as $T \equiv \tau \left.\frac{\partial \text{Base}}{\partial h}\right|_g = \frac{\tau}{r+\tau}\Big[\tau\Big(\frac{1}{2}\sum_m \iiint f_\alpha(\alpha_m^*)\big(P_m^{\text{fin}} - B - G(l)\,\mathbf{1}_{\text{bld}}^m\big)\, dF_k\, dF_B\, dF_l\Big) - \Big(\frac{1}{2}\sum_m \iiint F_\alpha(\alpha_m^*)\, dF_k\, dF_B\, dF_l\Big)\Big]$.

[101] If, as mentioned earlier, entrants value services more than incumbents, then the city's equilibrium solution over-provides the service relative to the Samuelson Rule, and the marginal cost of public funds is less than 1.

Notice that only incumbents earn surplus, so maximizing welfare is equivalent here to maximizing only incumbent welfare. We can then express the budget in terms of just h as follows:

$$\begin{aligned} g(h) \equiv h + \tau \cdot \frac{1}{2} \sum_m \iiint \Big[\big(1 - F_\alpha\big(\alpha_m^*(h)\big)\big) \big(B + G(l)\, \mathbf{1}_{\text{bld}}^m\big) \\ + F_\alpha\big(\alpha_m^*(h)\big)\, P_m^{\text{fin}}(h) \Big] \, dF_k \, dF_B \, dF_l \qquad \forall h \end{aligned} \tag{E6}$$

where we adopt a notational shorthand for when an owner builds an ADU $\mathbf{1}_{\text{bld}}^m(l,k) \equiv \mathbf{1}\{l \in Q_m\}\, \mathbf{1}\{k \leq \bar{k}_m(l)\}$. We will need to write the remaining components of our model in terms of h:

$$\begin{aligned} V_{l,m}(h) &= \frac{v(l) + \alpha_m^e + u\big(g(h)\big) - h - rO}{r + \tau}, \\ \alpha_m^*(h) &= \alpha_m^e + \tau\big(B - V_{l,m}(h)\big) + \frac{\tau}{r+\tau}\theta D^* + r\big[\Omega_m - \text{OwnOpt}_m\big], \\ P_{0,m}(h) &= V_{l,m}(h) - \frac{\theta D^*}{r+\tau}, \\ P_m^{\text{fin}}(h) &= P_{0,m}(h) + \tilde{G}(l)\, \mathbf{1}\{l \in Q_m\} \end{aligned} \tag{E7}$$

where these equations are the objects inside (E5) and (E6). We will hold $D = D^*$, which is its equilibrium value without loss at the optimum. First, we can write the value of the service net of the head tax and the derivative of $g(\cdot)$ with respect to h as:

$$\begin{aligned} Y(h) &\equiv u\big(g(h)\big) - h, \\ \gamma &\equiv g'(h) \end{aligned} \tag{E8}$$

We can now characterize the equilibrium responses to the head tax along the budget path. We will need total derivatives of how $Y(h)$, prices, and the stay / sell indifference cutoff change with h.

$$\begin{aligned} \frac{dY}{dh} &= u'\big(g(h)\big)\, \gamma - 1, \\ \frac{dV_{l,m}}{dh} &= \frac{dP_{0,m}}{dh} = \frac{dP_m^{\text{fin}}}{dh} = \frac{u'\gamma - 1}{r+\tau}, \\ \frac{d\alpha_m^*}{dh} &= -\tau \frac{dV_{l,m}}{dh} = -\frac{\tau\,(u'\gamma - 1)}{r+\tau} \end{aligned} \tag{E9}$$

Notice that these functions are all not dependent on l, B, k, or m, they effect every lot type and owner type equally.

We can now totally differentiate the budget (E6) with respect to h to get an expression for γ.

$$
\begin{aligned}
\gamma &= 1 - \,(u'\gamma - 1)\,\frac{\tau}{r+\tau}\big(\tau C - S\big),\\
C &\equiv \frac{1}{2}\sum_m \iiint f_\alpha\big(\alpha_m^*\big)\Big(P_m^{\text{fin}} - B - G(l)\,\mathbf{1}_{\text{bld}}^m\Big)\,dF_k\,dF_B\,dF_l,\\
S &\equiv \frac{1}{2}\sum_m \iiint F_\alpha\big(\alpha_m^*\big)\,dF_k\,dF_B\,dF_l
\end{aligned}
\tag{E10}
$$

Here, C is a churn prize for the city, the gain in assessed value when an owner near α_m^* flips from stay to sell. The mass of incumbents induced to sell at $f_\alpha\big(\alpha_m^*\big)$ increases the budget as lots get reassessed and ADUs get built. As h pushes residents out and lots turnover, that has a positive revenue effect through C. S is the mass of lots that sell and get reassessed at their transaction price, which capitalize the net fiscal benefit into the transaction price. This price feedback means S enters negatively.

We can now totally differentiate welfare with respect to h.

$$
\frac{d\widetilde{\mathcal{W}}}{dh} = \big(u'\gamma - 1\big)\cdot\frac{1}{2}\sum_m \iiint \left[\frac{F_\alpha(\alpha_m^*)}{r+\tau} + \frac{1-F_\alpha(\alpha_m^*)}{r}\right] dF_k\,dF_B\,dF_l \tag{E11}
$$

We can then set our derivative equal to 0 and show that the Samuelson rule holds. Note that the integral in (E11) is strictly positive.[102]

$$
\begin{aligned}
\frac{d\widetilde{\mathcal{W}}}{dh} = 0 &\iff u'\big(g(h^*)\big)\,\gamma(h^*) = 1\\
&\implies \gamma(h^*) = 1\\
&\implies \boxed{u'\big(g(h^*)\big) = 1}
\end{aligned}
\tag{E12}
$$

[102] $\frac{1}{2}\sum_m \iiint \big[\frac{F_\alpha(\alpha_m^*)}{r+\tau} + \frac{1-F_\alpha(\alpha_m^*)}{r}\big]\,dF_k\,dF_B\,dF_l \in \big[\frac{1}{r+\tau}, \frac{1}{r}\big] > 0$ for every lot type. The sign of $d\widetilde{\mathcal{W}}/dh$ is the sign of $u'\gamma - 1$.

F Appendix: Changes in Rents

This Appendix presents estimates of the impact of permitting reform on percentage changes in rent. We use the Coastal Zone boundary as a source of conditionally-exogenous variation in permitting rates for ADUs. The key limitation of this exercise is our reliance on rents taken from the 2016 and 2024 vintages of the American Community Survey (U.S. Census Bureau, 2016, 2024).[103]

Our outcome variable of interest is the percent change in the real mean gross rent, which we calculate by taking the total gross rent and dividing by the number of rental units in the block group to get the mean gross rent in each year.[104] We then calculate the percentage change in gross rent for each block group between 2016 and 2024.

We show estimates with one of two versions of our endogenous variable. We first use the cumulative amount of ADUs permitted between 2018 and 2024 from California Department of Housing and Community Development (2025), which we match to each block group. Second, we use the cumulative amount of ADUs permitted between 2018 and 2024 and scale it by the block group's total housing unit count from 2016. The latter construction is our preferred measure of ADU permitting.

Finally, our instrument is constructed using the block group's shapefile and the Coastal Zone shapefile from California Coastal Commission (2014). The instrument is the share of block group area that falls within the Coastal Zone. This instrument is valued between 0 and 1, with a value of 0 corresponding to no Coastal Zone overlap (1,041 block groups in our analysis sample), a value of 1 corresponding to a block group that is completely within the Coastal Zone (301 block groups in our analysis sample), and values between 0 and 1 reflecting partial overlap of the block group and the Coastal Zone (844 block groups in our analysis sample).

We restrict our final sample to the 2,186 block groups with a centroid within 2,000 meters of the Coastal Zone boundary. All 15 coastal counties are represented within the final sample, with about 67% of the sample in San Diego, Los Angeles, and Orange counties. We weight our estimates by each block group's 2016 housing unit count.[105]

Our two stage least squares (2SLS) equations are as follows:

$$ADU_b = \pi_0 + \pi_1 InCZ_b + X_b'\gamma + \sigma_s + u_b \tag{F1a}$$

$$\%\Delta Rent_b = \delta_0 + \delta_1 \widehat{ADU}_b + X_b'\rho + \sigma_s + v_b \tag{F1b}$$

where b indexes block groups. $InCZ_b$ is the Coastal Zone area share for block group b. ADU_b is either cumulative ADU permits from 2018 to 2024 for block group b or the cumulative ADU permits from 2018 to 2024 per 1,000 housing units in 2016 for block group b. $\%\Delta Rent_b$ is the percentage change in mean gross rent between 2016 and 2024 for block group b. We include a control set, X_b',

[103] We work with 2020 block groups. We crosswalk 2010 block groups to 2020 geometries using IPUMS NHGIS (2026).

[104] All monetary values are in 2025 USD (U.S. Bureau of Economic Analysis, 2025).

[105] Due to the high frequency at which ADUs are in tracts with high ratios of owners to renters, we weight by all housing units rather than by rented housing units.

meant to both add precision to our estimates and to address concerns that the Coastal Zone is correlated with other drivers of rent growth. The controls come in 5 groups: terrain and density,[106] age structure,[107] job access,[108] non-ADU permitting,[109] and amenities.[110] σ_s is a Coastal Zone segment fixed effect that captures heterogeneous developments in the real estate market across space over our sample. We present all estimates with spatial correlation robust standard errors with a 1,000 meter cutoff (Conley, 1999).

Our instrumental variables estimates rest on two assumptions (Cunningham, 2021). First, the share that a block group overlaps with the Coastal Zone only impacts changes in rents through its impacts on permitting for ADUs. Given the tenuous nature of this assumption, our control set tries to partially account for the other ways that the Coastal Zone may influence rental price changes. This approach is imperfect, but local to the boundary it is more reasonable than on a sample that extends far inland. Second, the block group Coastal Zone overlap share needs to be a strong predictor of ADU permitting. We can provide some evidence in support of our second assumption below, but the exclusion restriction we cannot test directly in our just identified setting.[111]

We show our first stage estimates from Equation F1a in Table F.1. In Columns 1 and 2 we show our unscaled endogenous variable, and in Columns 3 and 4 we show our scaled endogenous variable. Odd columns don't include controls and even Columns add the full control set. Our preferred estimates are in Column 4. The Coastal Zone overlap instrument is a strong predictor of decreased ADU permitting, with block groups completely inside of the Coastal Zone permitting 6 fewer ADUs per 1,000 housing units than block groups completely outside the Coastal Zone. The F-statistic for the excluded instrument is 17. These estimates are large in magnitude, as the average block group in our sample has only 12 ADU permits per 1,000 housing units.

With our first stage estimates in hand, we proceed to estimate our full 2SLS model. We present estimates of Equations F1b in Table F.2. In Columns 1 and 2 we show our unscaled endogenous variable, and in Columns 3 and 4 we show our scaled endogenous variable. Odd columns don't include controls and even Columns add the full control set. Our preferred estimates are in Column 4. Our point estimates in Column 4 suggest that 1 additional ADU permit per 1,000 housing units in a block group reduced rents by about 0.076 percentage points between 2016 and 2024. So, a roughly 1 standard deviation increase in ADU permitting lowers rents by 1.2 percentage points, in a setting where rents grew 28% over the sample period. However, these estimates are not significant at traditional significance levels. We show a companion set of estimates that add each control set

[106] Elevation, block group population per square kilometer.

[107] 2016 population share over age 65.

[108] Distance to the nearest CBD, change in jobs within 10 miles between 2016 and 2024, average commute distance for the block group.

[109] Cumulative non-ADU permits from 2018 to 2024 (when the endogenous variable is scaled by housing units we scale this variable as well.).

[110] A dummy variable that take a value of 1 when the closest shoreline category is a sandy beach and a value of 0 otherwise, 2016 to 2024 change in service sector jobs within 3 miles.

[111] We also need a monotonicity assumption on the relationship between our instrument and the endogenous variable: more overlap with the Coastal Zone needs to weakly decrease permitting activity. Controlling for coastal access helps address this.

one at a time in Table F.3.

These estimates are at best suggestive evidence of the relationship between ADU permitting and rent changes. First, these estimates are implausibly large, implying an elasticity between ADUs and rental prices of -0.76 within a block group. For example, Li (2022) finds an elasticity for new buildings in New York City of -0.1, which is within our confidence interval. They are too imprecise to draw strong conclusions about the impacts on ADUs and rents. However, these estimates are broadly consistent with findings from Bhatt (2023). Bhatt (2023) uses a difference in differences approach with census tract data from the ACS and permits from San Francisco, Los Angeles, and San Diego. He compares tracts with different shares of land zoned for single-family and finds imprecise estimates of the impacts of ADUs on rents that could be consistent with a range of elasticities from the literature. Both of these results are likely reflective of the low levels of ADU permitting as a share of housing units and the imprecision of the ACS rental data.

Table F.1: First Stage Estimates of Block Group Coastal Zone Overlap on Cumulative Change in ADU Permitting (2018-2024)

	Dep. Var.: Cumulative ADU Permits, 2018–2024			
	Count (units)		Per 1,000 Housing Units	
	(1)	(2)	(3)	(4)
Coastal Zone Area Share	−3.243***	−3.233***	−6.182***	−6.143***
	(0.753)	(0.765)	(1.470)	(1.488)
Controls		Y		Y
CZ Segment FE	Y	Y	Y	Y
N	2186	2185	2186	2185
R^2	0.302	0.340	0.257	0.300
First-stage F	18.54	17.86	17.68	17.04
Mean	7.002	7.002	11.69	11.69
SD	8.995	8.995	16.53	16.53
In-CZ Mean	6.142	6.142	9.336	9.336
In-CZ SD	8.015	8.015	12.77	12.77

Notes: This table uses data from U.S. Census Bureau (2016), U.S. Census Bureau (2024), California Department of Housing and Community Development (2025), California Department of Fish and Wildlife (2026), California Coastal Commission (2014), US Census Bureau (2023), U.S. Geological Survey (2026), and Holian (2019) to estimate Equation F1a. In Columns 1 and 2 we show our unscaled endogenous variable, and in Columns 3 and 4 we show our scaled endogenous variable. Odd columns don't include controls and even Columns add the full control set. Our preferred estimates are in Column 4. We show first stage F-statistics on the excluded instrument, dependent variable means and standard deviations for the analysis sample, and dependent variable means and standard deviations calculated on the share of block groups with more than half of their area within the Coastal Zone. Standard errors follow Conley (1999), computed on the block-group centroid with a distance cutoff of 1,000 m. We weight estimates by 2016 ACS housing units. *$p < .1$, ** $p < .05$, *** $p < .01$

Table F.2: 2SLS Estimates of ADU Permitting Effects on Percent Changes in Block Group Gross Rent (2016-2024)

	Per. Ch. Mean Gross Rent (2016–2024)			
	Count		Per 1,000 Units	
	(1)	(2)	(3)	(4)
Cumulative ADU Permits	−0.280	−0.166	−0.146	−0.076
	(0.726)	(0.735)	(0.380)	(0.382)
Controls		Y		Y
CZ Segment FE	Y	Y	Y	Y
N	2126	2126	2126	2126
R^2	0.082	0.096	0.075	0.093
First-stage F	18.54	17.86	17.68	17.04
Mean	27.72	27.72	27.72	27.72
SD	35.83	35.83	35.83	35.83
In-CZ Mean	29.66	29.66	29.66	29.66
In-CZ SD	34.57	34.57	34.57	34.57
ADU Permits Mean	7.002	7.002	11.69	11.69
ADU Permits SD	8.995	8.995	16.53	16.53

Notes: This table uses data from U.S. Census Bureau (2016), U.S. Census Bureau (2024), California Department of Housing and Community Development (2025), California Department of Fish and Wildlife (2026), California Coastal Commission (2014), US Census Bureau (2023), U.S. Geological Survey (2026), and Holian (2019) to estimate Equation F1b. In Columns 1 and 2 we show our unscaled endogenous variable, and in Columns 3 and 4 we show our scaled endogenous variable. Odd columns don't include controls and even Columns add the full control set. Our preferred estimates are in Column 4. We show first stage F-statistics on the excluded instrument, dependent variable means and standard deviations for the analysis sample, dependent variable means and standard deviations calculated on the share of block groups with more than half of their area within the Coastal Zone, and mean and standard deviations for our endogenous regressors. Standard errors follow Conley (1999), computed on the block-group centroid with a distance cutoff of 1,000 m. We weight estimates by 2016 ACS housing units. *$p < .1$, ** $p < .05$, *** $p < .01$

Table F.3: Sensitivity of 2SLS Estimates of ADU Permitting Effects on Percent Changes in Block Group Gross Rent (2016-2024)

	Dep.Var.: Change in Mean Gross Rent, 2016–2024 (%)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Cumulative ADU Permits	−0.146	−0.109	−0.065	−0.211	−0.153	−0.114	−0.076
	(0.380)	(0.351)	(0.361)	(0.420)	(0.382)	(0.392)	(0.382)
Terrain & Density		Y					Y
Age Structure			Y				Y
Job Access				Y			Y
Non-ADU Permitting					Y		Y
Coastal Amenity						Y	Y
CZ Segment FE	Y	Y	Y	Y	Y	Y	Y
N	2126	2126	2126	2126	2126	2126	2126
R^2	0.075	0.083	0.090	0.069	0.074	0.078	0.093
First-stage F	17.68	20.78	18.51	14.96	17.73	16.74	17.04
Mean	27.72	27.72	27.72	27.72	27.72	27.72	27.72
SD	35.83	35.83	35.83	35.83	35.83	35.83	35.83
In-CZ Mean	29.66	29.66	29.66	29.66	29.66	29.66	29.66
In-CZ SD	34.57	34.57	34.57	34.57	34.57	34.57	34.57
ADU Permits Mean	11.69	11.69	11.69	11.69	11.69	11.69	11.69
ADU Permits SD	16.53	16.53	16.53	16.53	16.53	16.53	16.53

Notes: This table uses data from U.S. Census Bureau (2016), U.S. Census Bureau (2024), California Department of Housing and Community Development (2025), California Department of Fish and Wildlife (2026), California Coastal Commission (2014), US Census Bureau (2023), U.S. Geological Survey (2026), and Holian (2019) to estimate Equation F1b. In Columns 1 and 2 we show our unscaled endogenous variable, and in Columns 3 and 4 we show our scaled endogenous variable. Odd columns don't include controls and even Columns add the full control set. Our preferred estimates are in Column 4. We show first stage F-statistics on the excluded instrument, dependent variable means and standard deviations for the analysis sample, dependent variable means and standard deviations calculated on the share of block groups with more than half of their area within the Coastal Zone, and mean and standard deviations for our endogenous regressors. Standard errors follow Conley (1999), computed on the block-group centroid with a distance cutoff of 1,000 m. We weight estimates by 2016 ACS housing units. *$p < .1$, ** $p < .05$, *** $p < .01$